

Chapter 1: The Internet at Scale

How routers, ISPs, and shared paths make the Internet scale

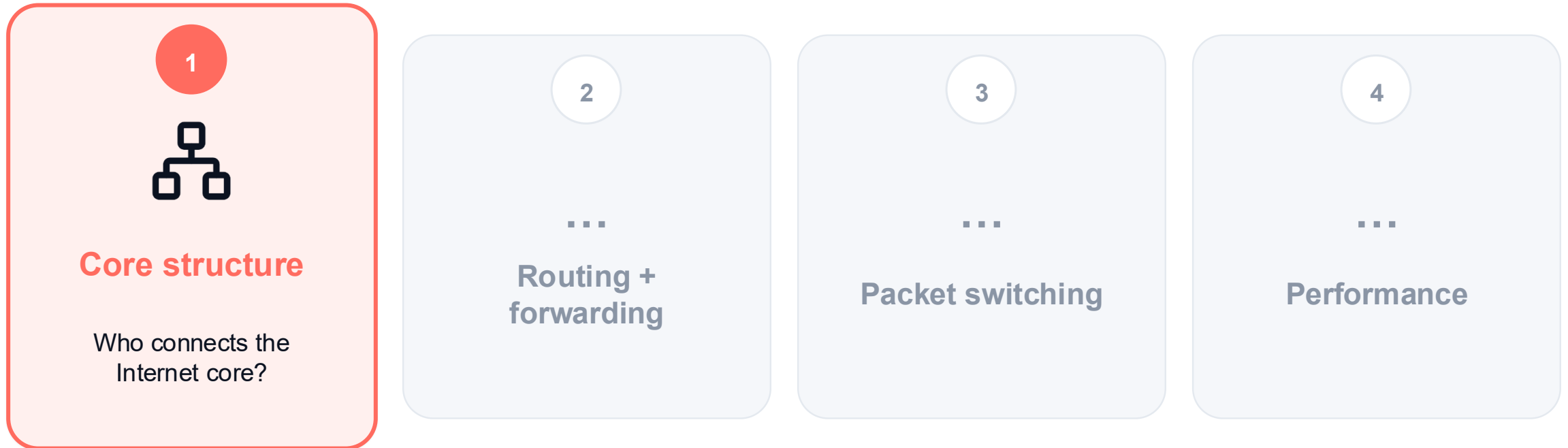
Yaxiong Xie

CSE589 · Modern Network Concepts · Fall 2026



Four questions organize the Internet core

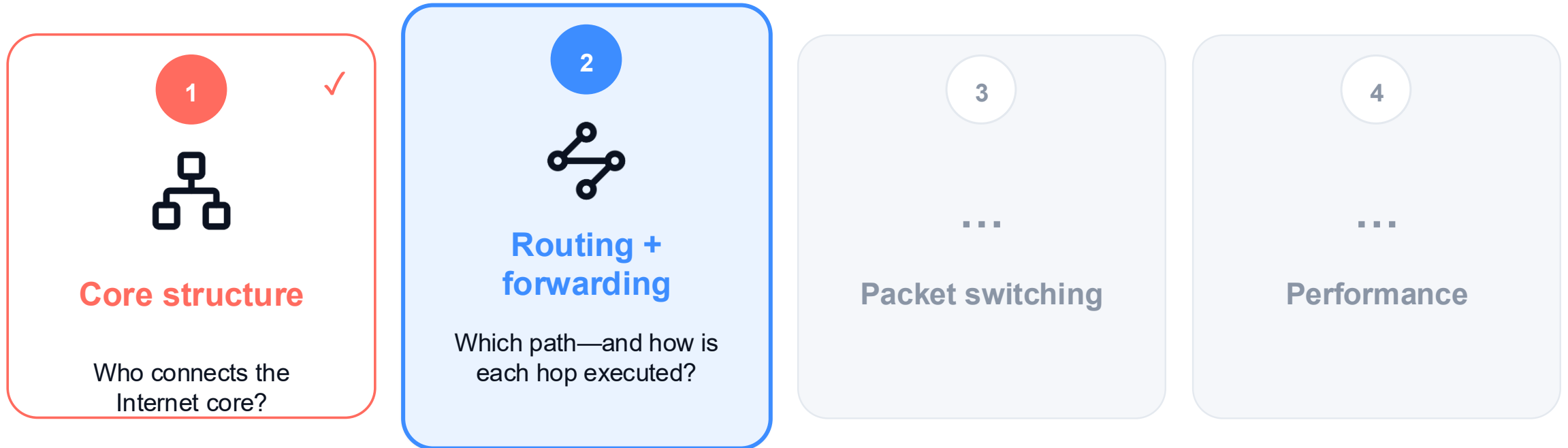
By the end of this part, you should be able to answer four questions.



See how routers, ISPs, content networks, and IXPs form the Internet core.

Four questions organize the Internet core

By the end of this part, you should be able to answer four questions.



Choose an end-to-end route, then execute it one hop at a time.

Four questions organize the Internet core

By the end of this part, you should be able to answer four questions.

1



Core structure

Who connects the Internet core?

2



Routing + forwarding

Which path—and how—is each hop executed?

3



Packet switching

How do packets share link capacity?

4



Performance

Explain how packet and circuit switching share capacity.

Four questions organize the Internet core

By the end of this part, you should be able to answer four questions.

1



Core structure

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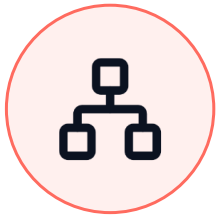
Performance

What determines delay, loss, and throughput?

Connect link properties and contention to delay, loss, and throughput.

Part 1 · Core structure

01



From one router to a network of networks.

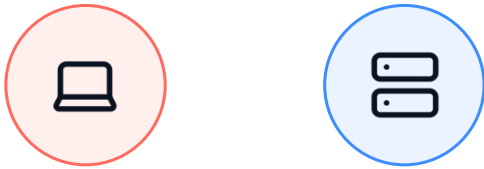
Core structure

Routing + forwarding

Packet switching

Performance

How can every endpoint reach every other endpoint?



Two endpoints

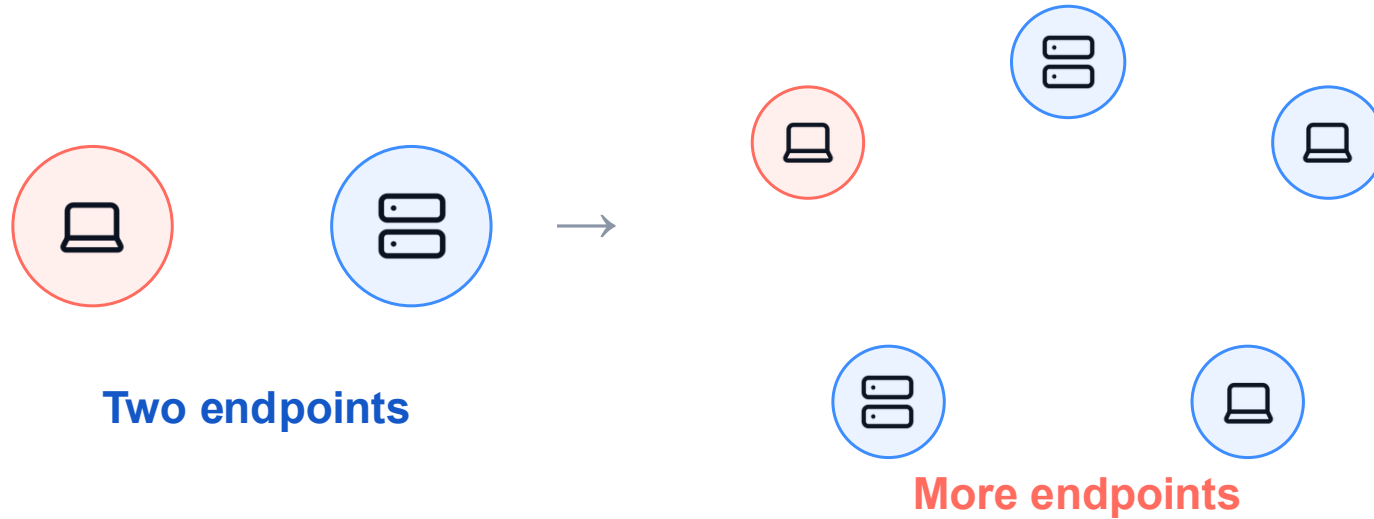
1 Direct links

2 Scaling problem

3 Shared routers

4 Network core

How can every endpoint reach every other endpoint?



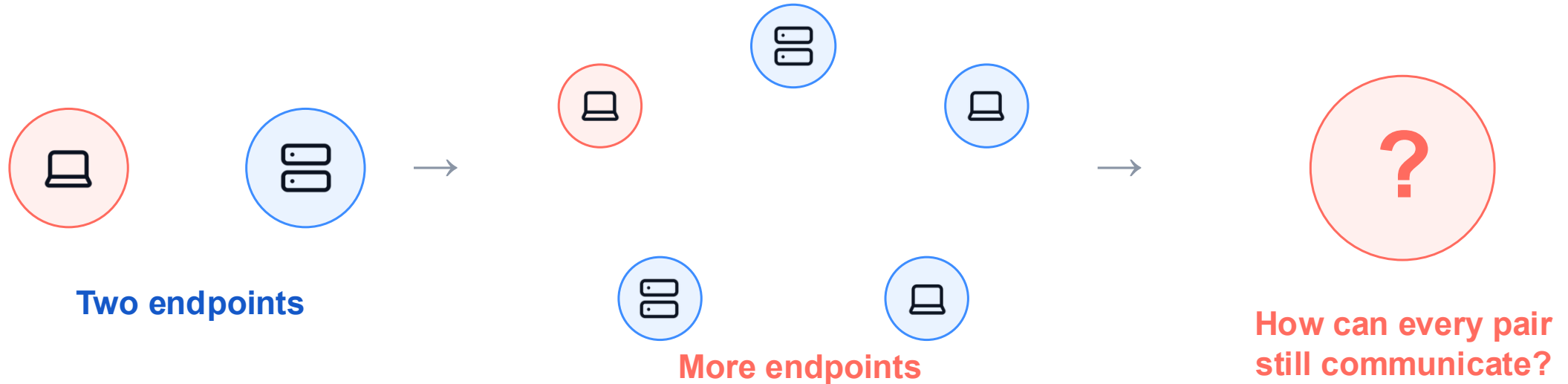
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How can every endpoint reach every other endpoint?



1 Direct links

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Start simple: two endpoints need one link

1 link



Direct connection is simple at tiny scale.

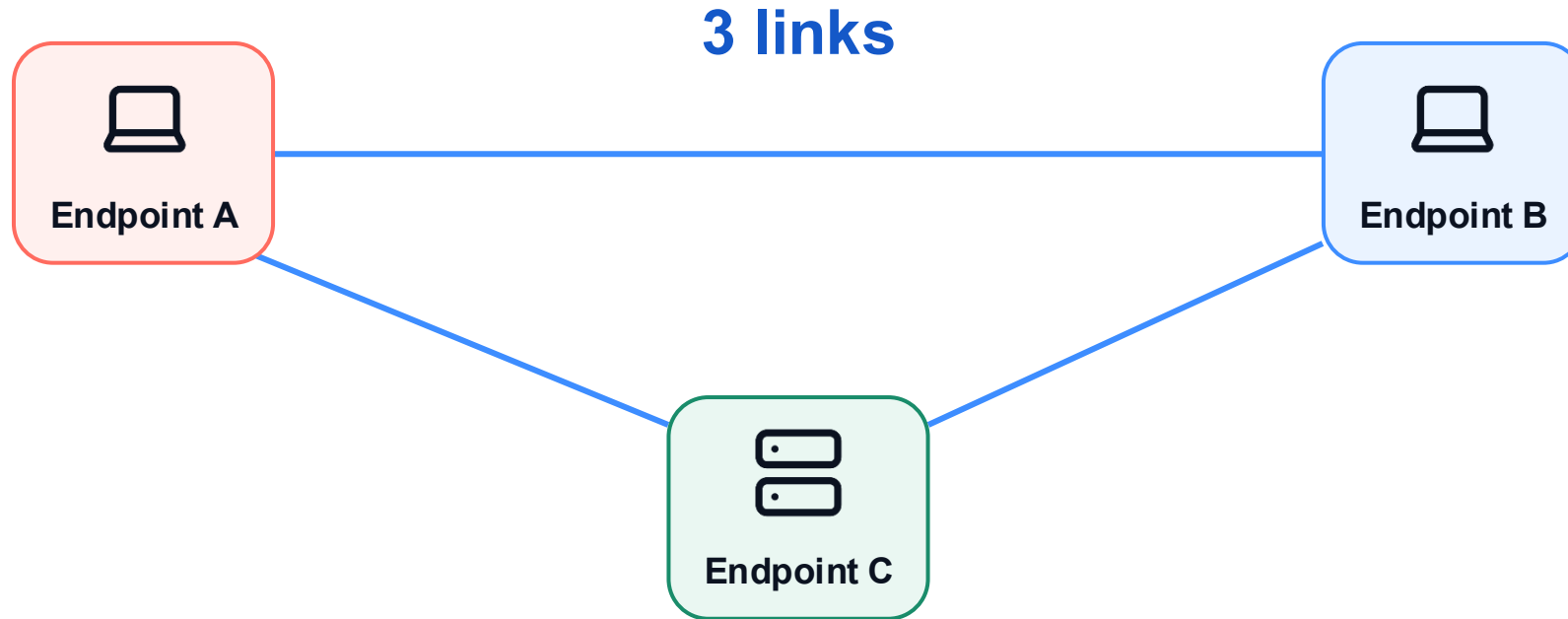
1 Direct links

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4 Network core

Add one endpoint: now three links are needed



Every new endpoint creates more pairwise connections.

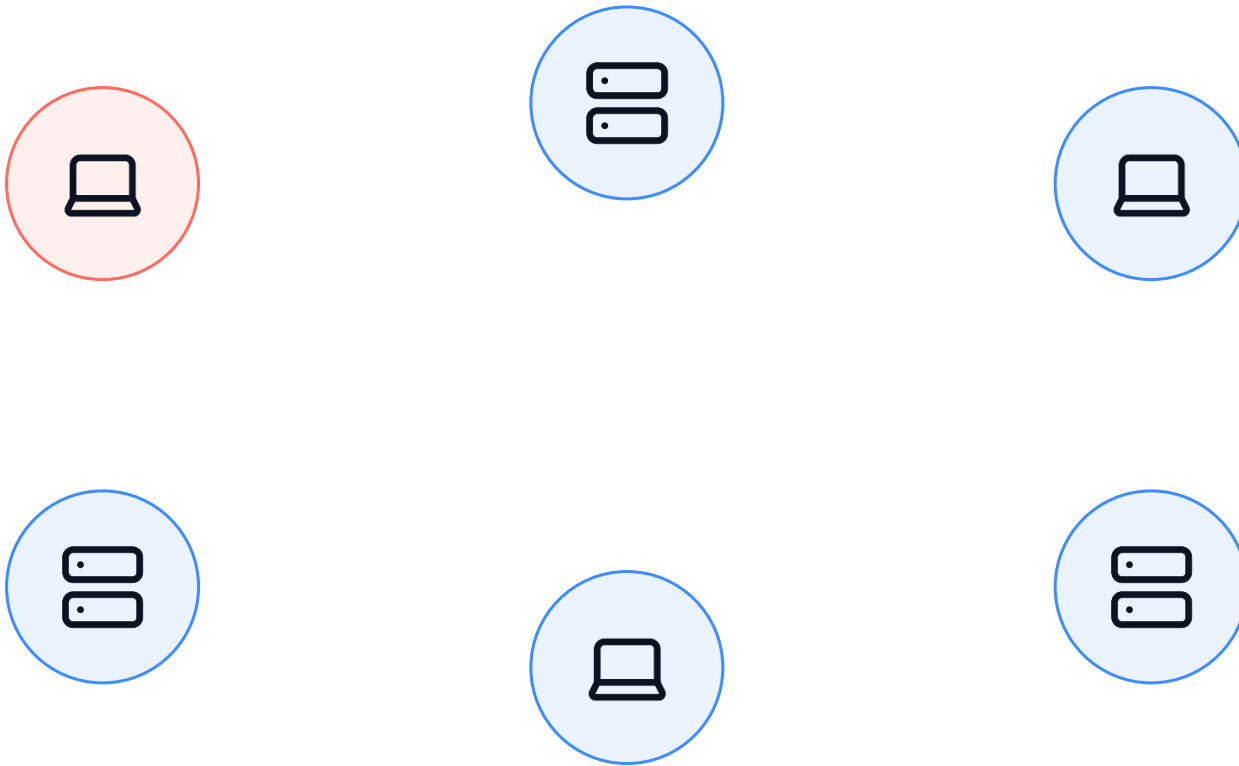
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Direct connections grow quadratically



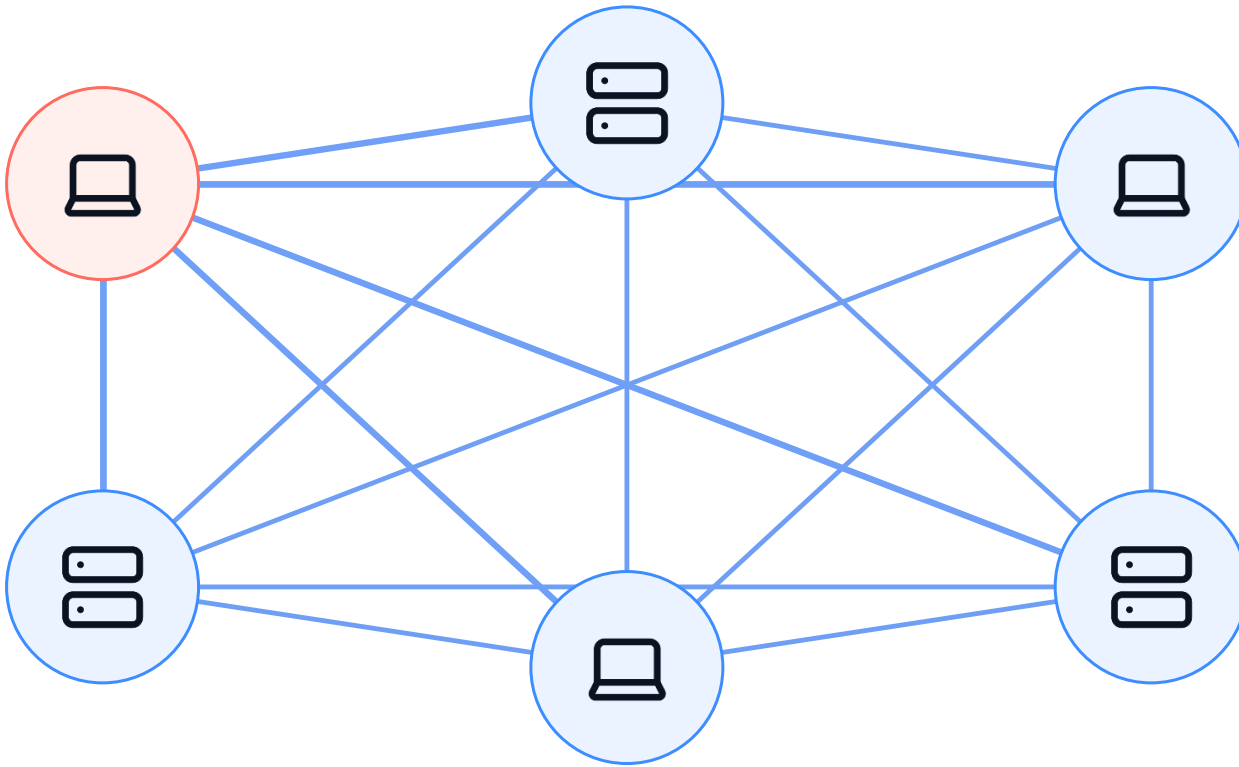
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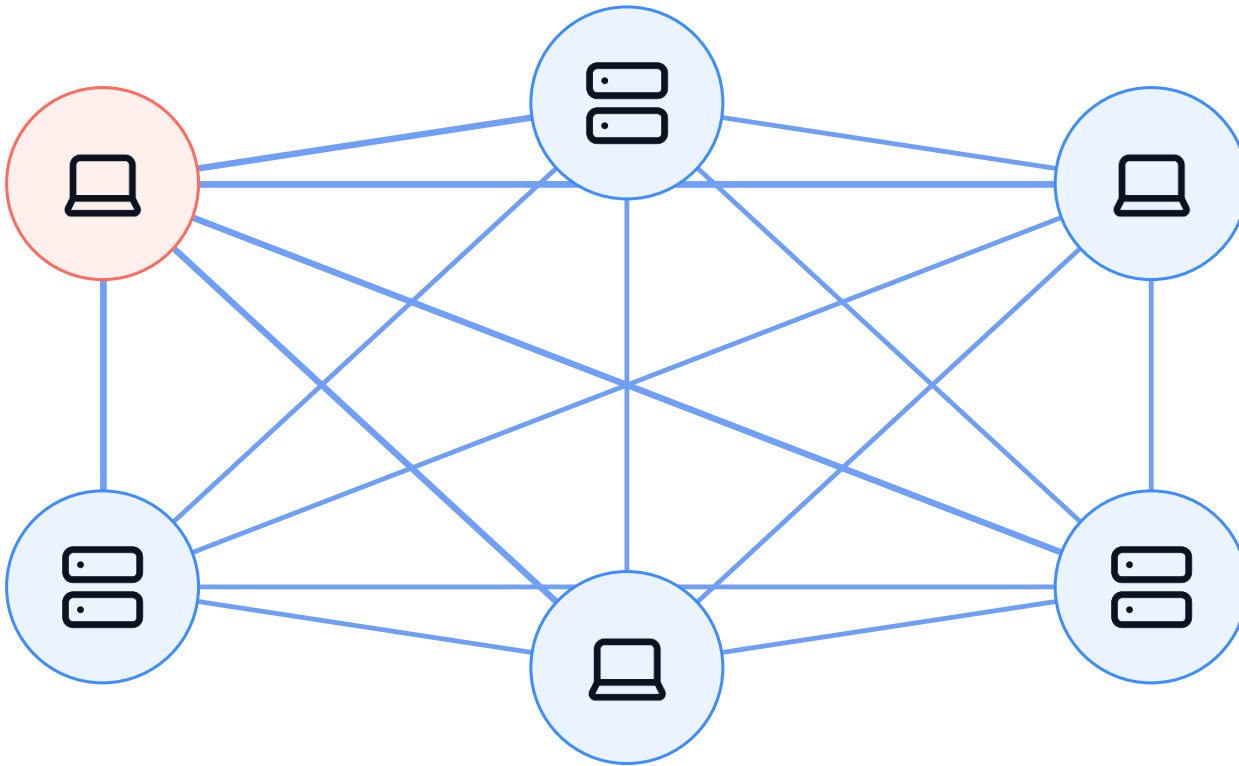
1 Direct links

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Direct connections grow quadratically



Every pair needs one link

$$E = \frac{N(N - 1)}{2}$$

E = number of direct links

The number of direct links grows quadratically with N.

1 Direct links

2 Scaling problem

3 Shared routers

4 Network core

One router connects several devices through separate ports



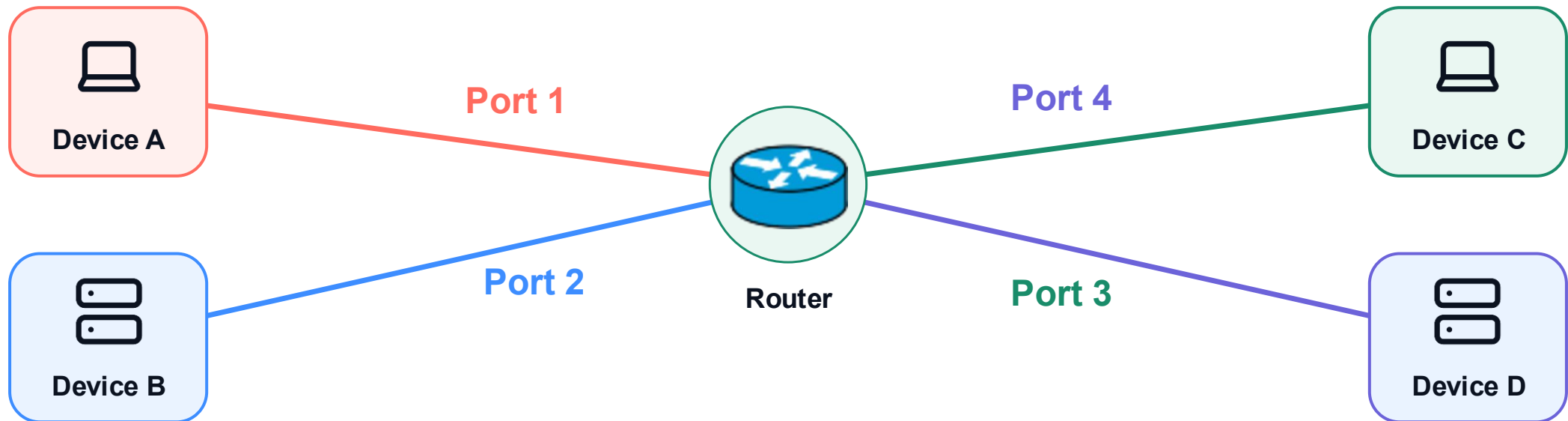
1 Direct links

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One router connects several devices through separate ports



But a router has a finite number of ports—and a network must serve far more devices.

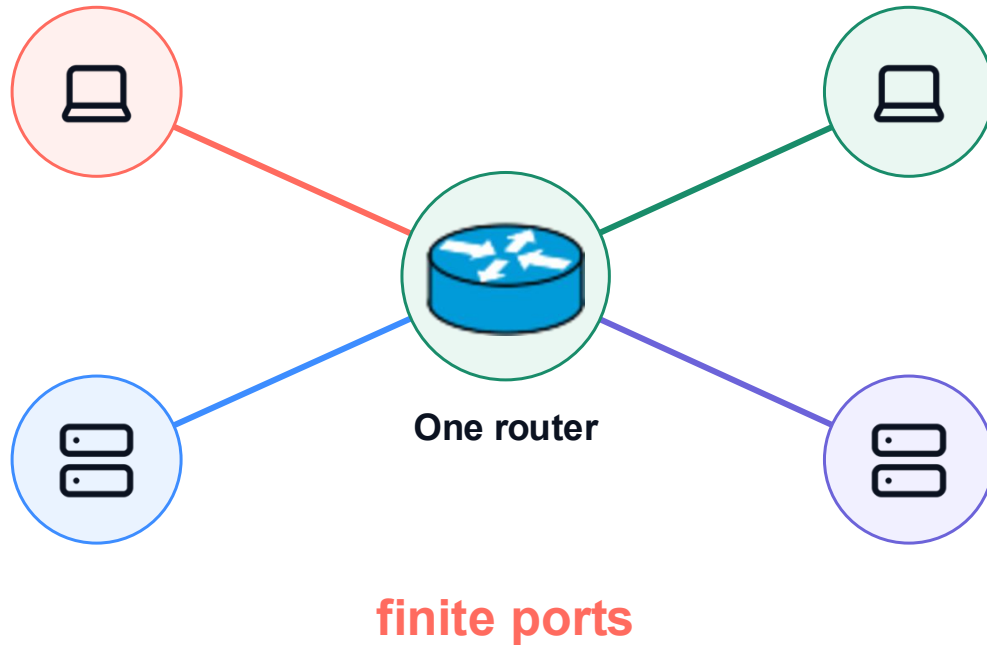
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One router is not enough for a large network



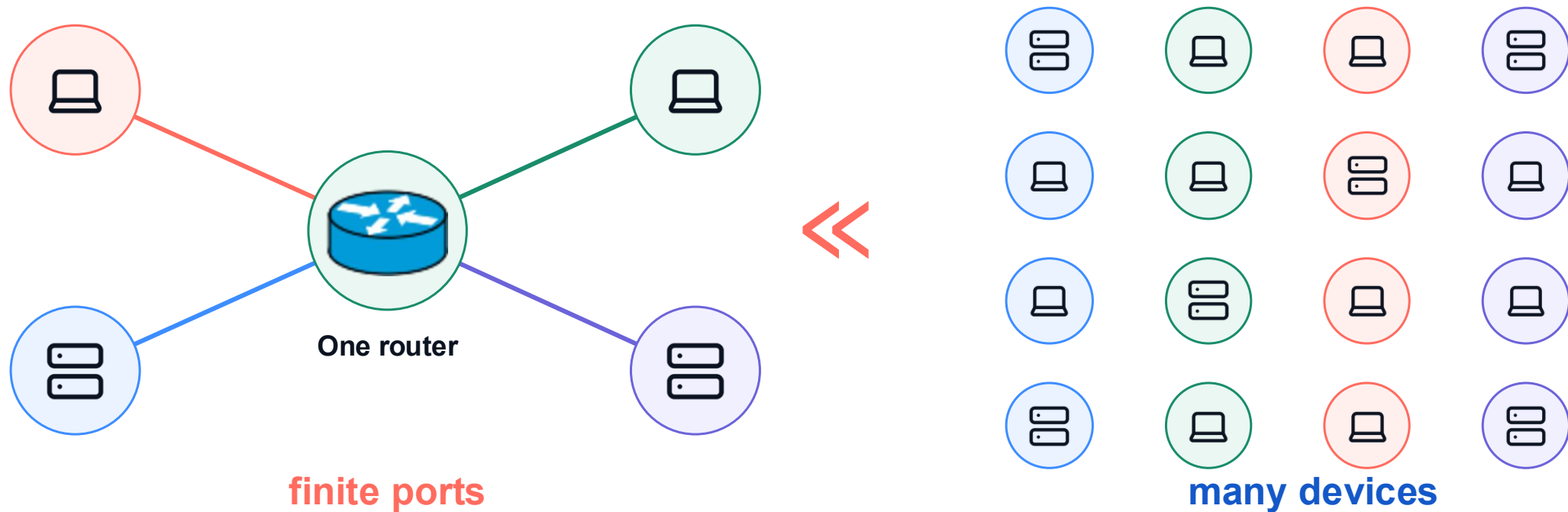
1 Direct links

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One router is not enough for a large network



To scale, routers must connect to other routers.

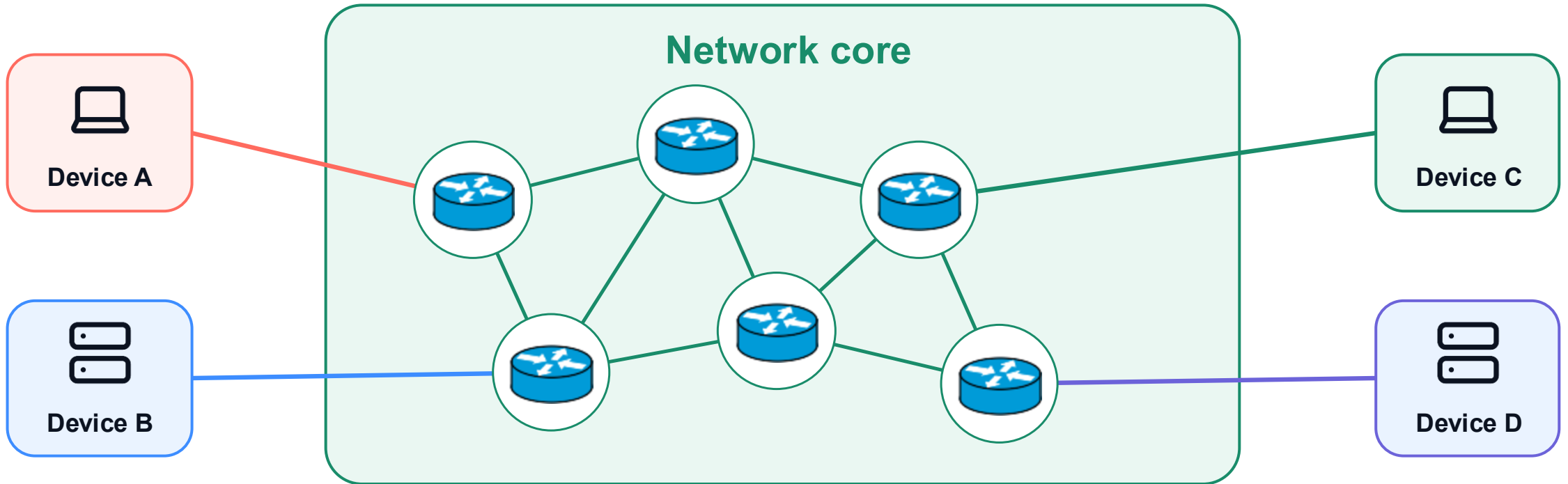
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Multiple routers interconnect to form the network core



Edge devices attach to routers; interconnected routers form the network core.

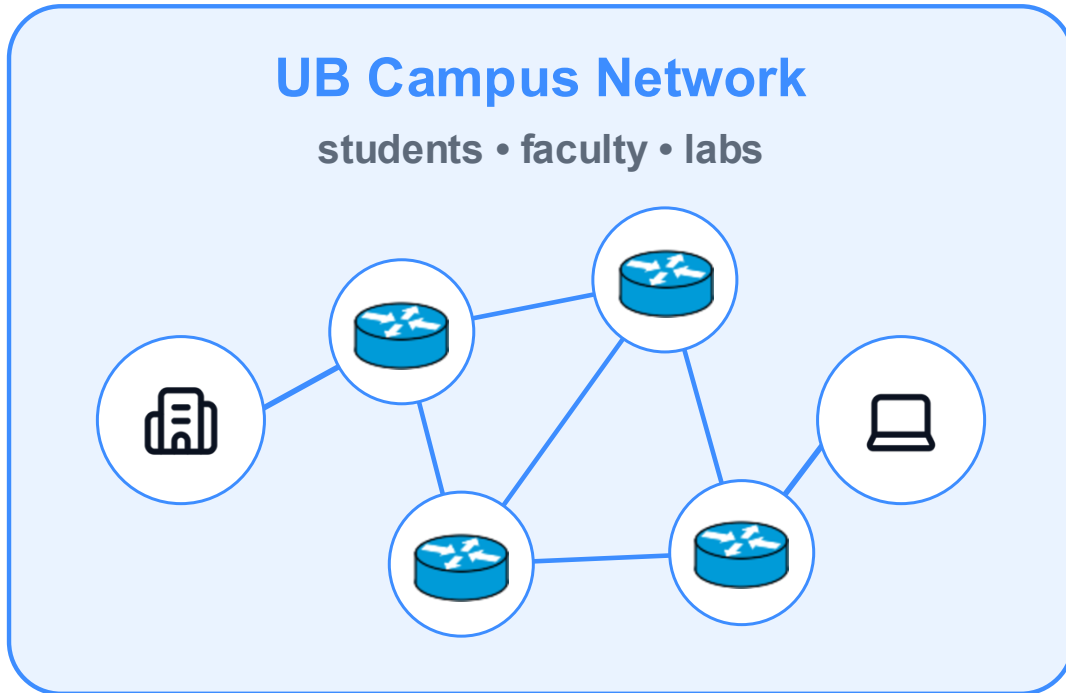
Core structure

Routing + forwarding

Packet switching

Performance

A network or ISP serves only part of the users



Each network covers a limited community or region—not everyone.

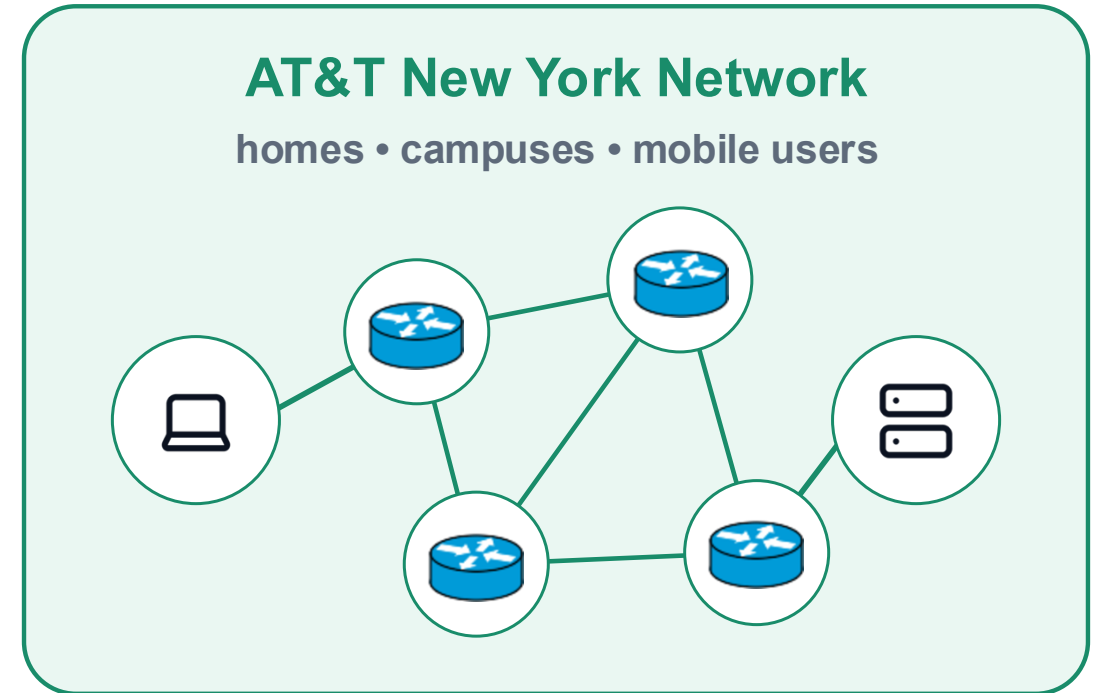
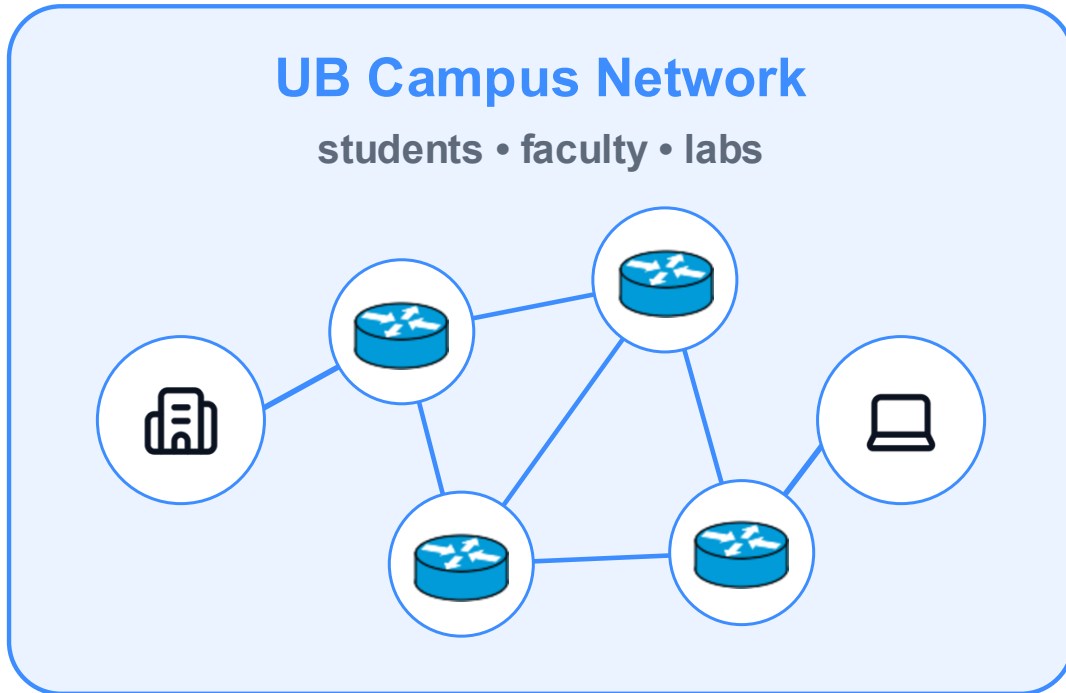
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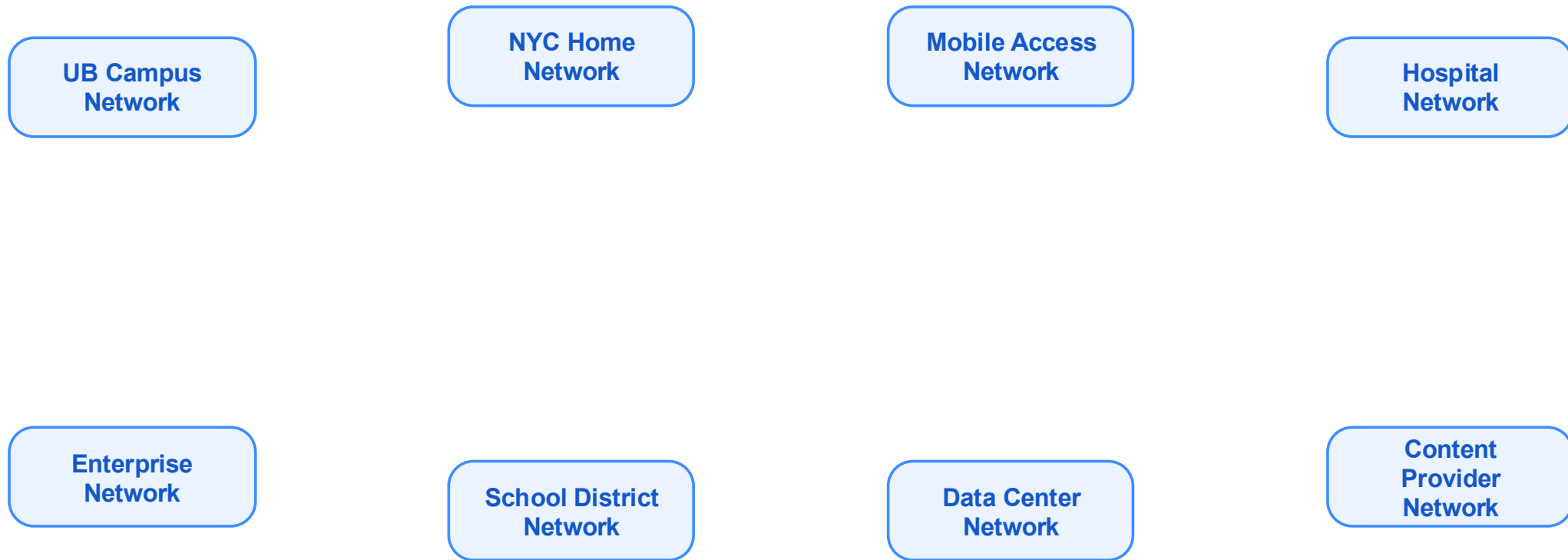
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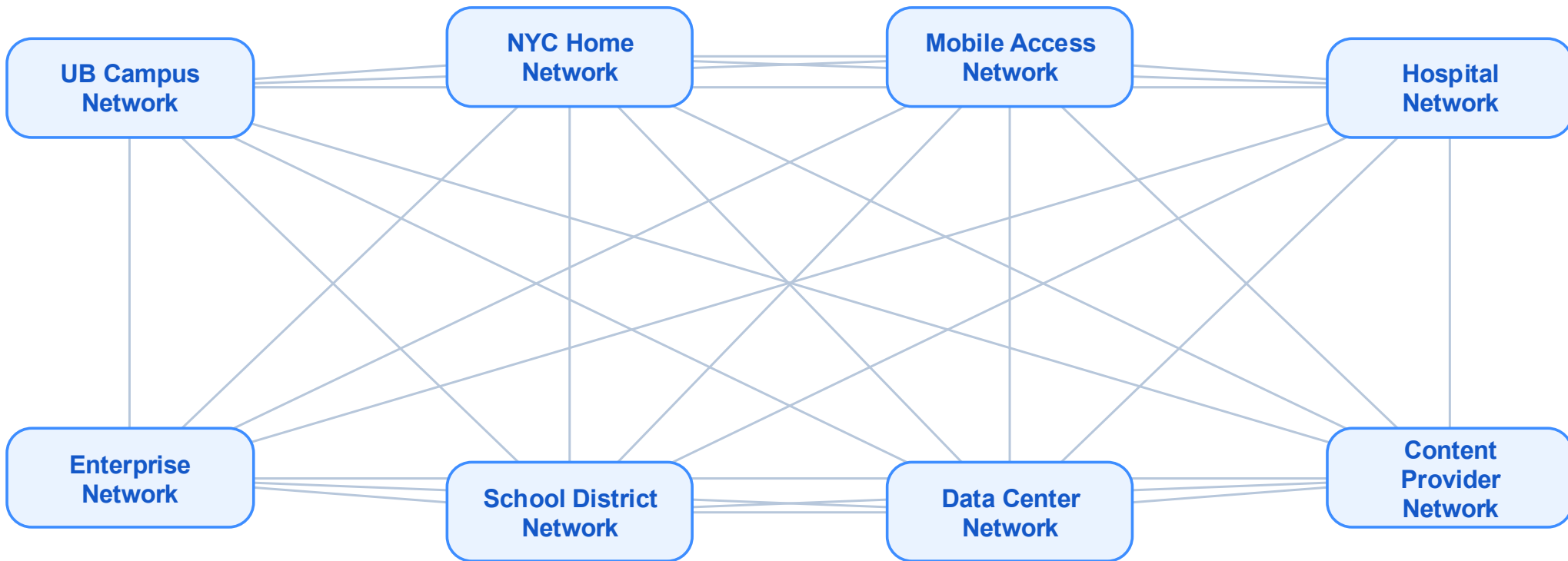
Directly connecting every small network does not scale



The same scaling problem appears again—now between networks.



Directly connecting every small network does not scale



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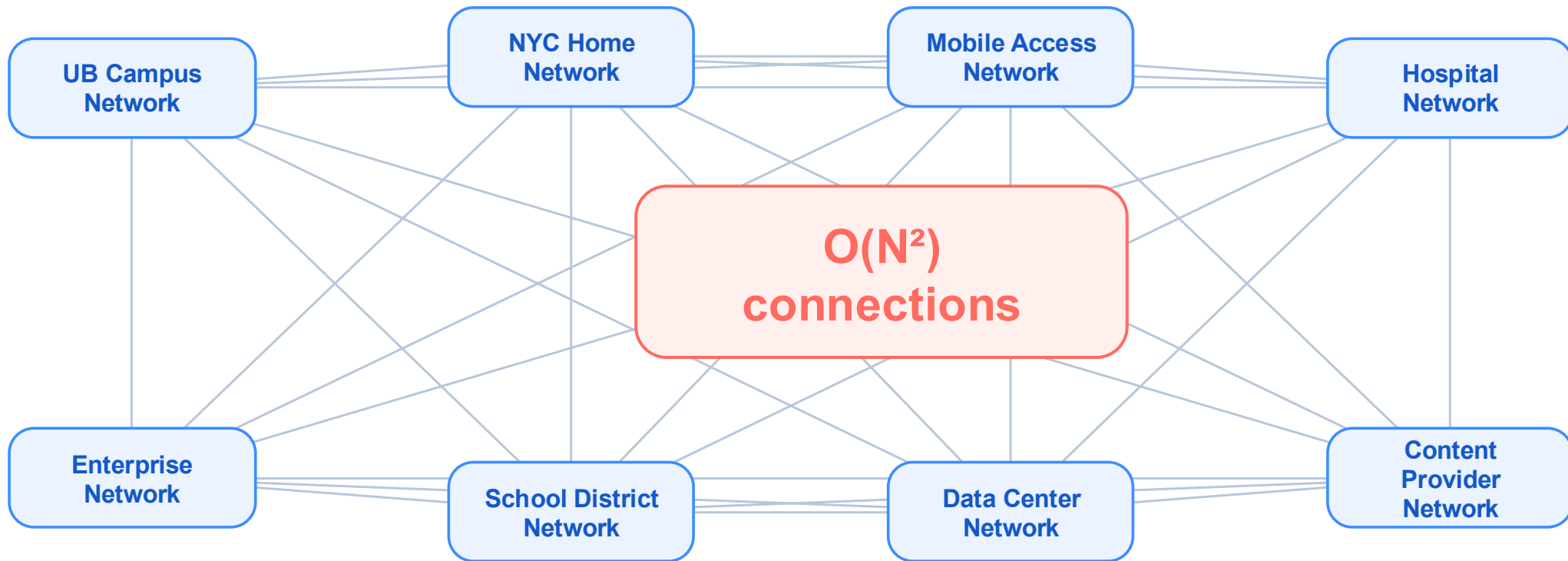
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Directly connecting every small network does not scale



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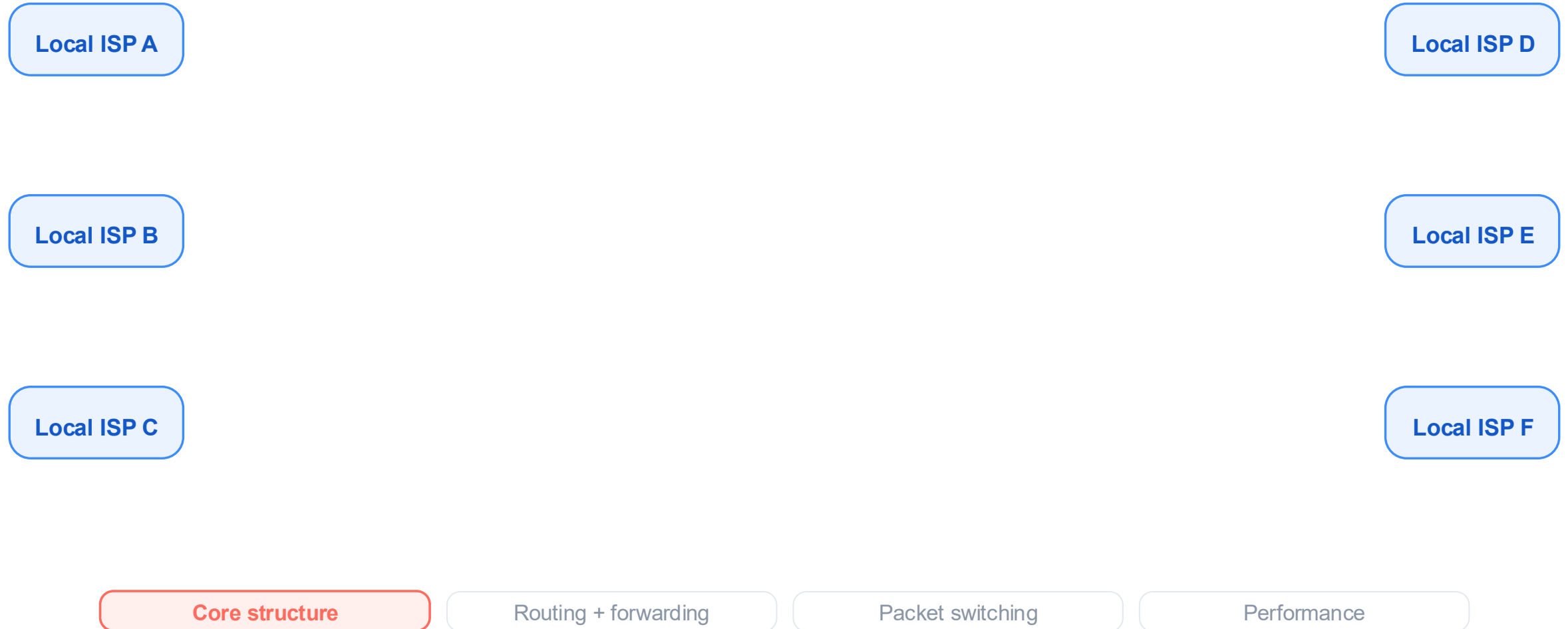
Core structure

Routing + forwarding

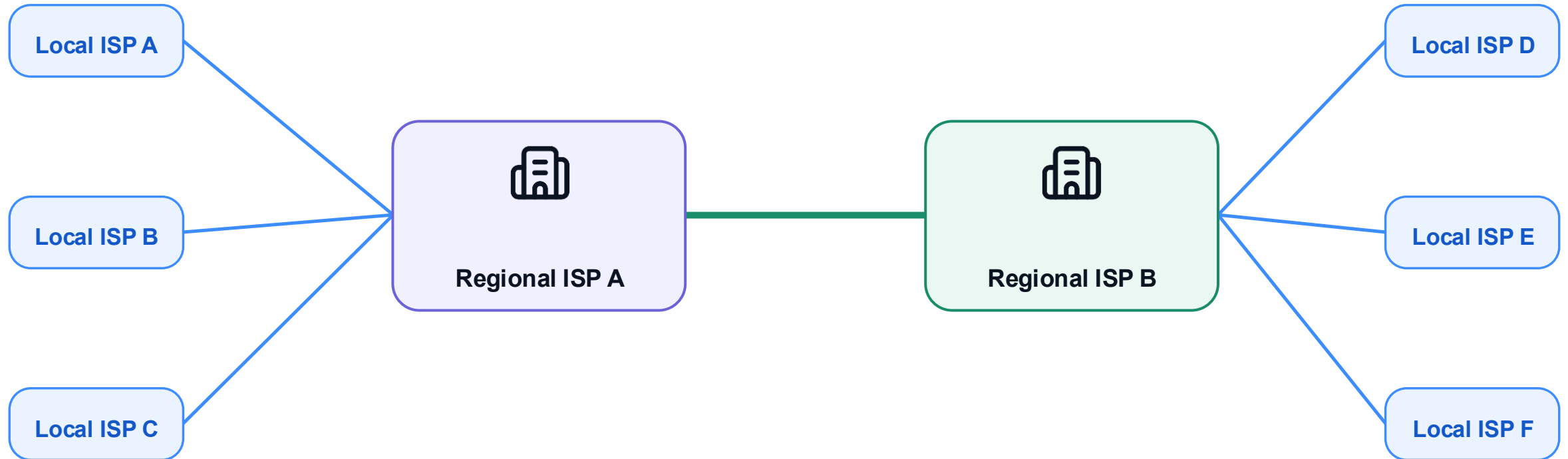
Packet switching

Performance

Local ISPs begin as separate access networks



Regional ISPs aggregate many local ISPs



Hierarchy begins when regional ISPs aggregate many local ISPs.

Core structure

Routing + forwarding

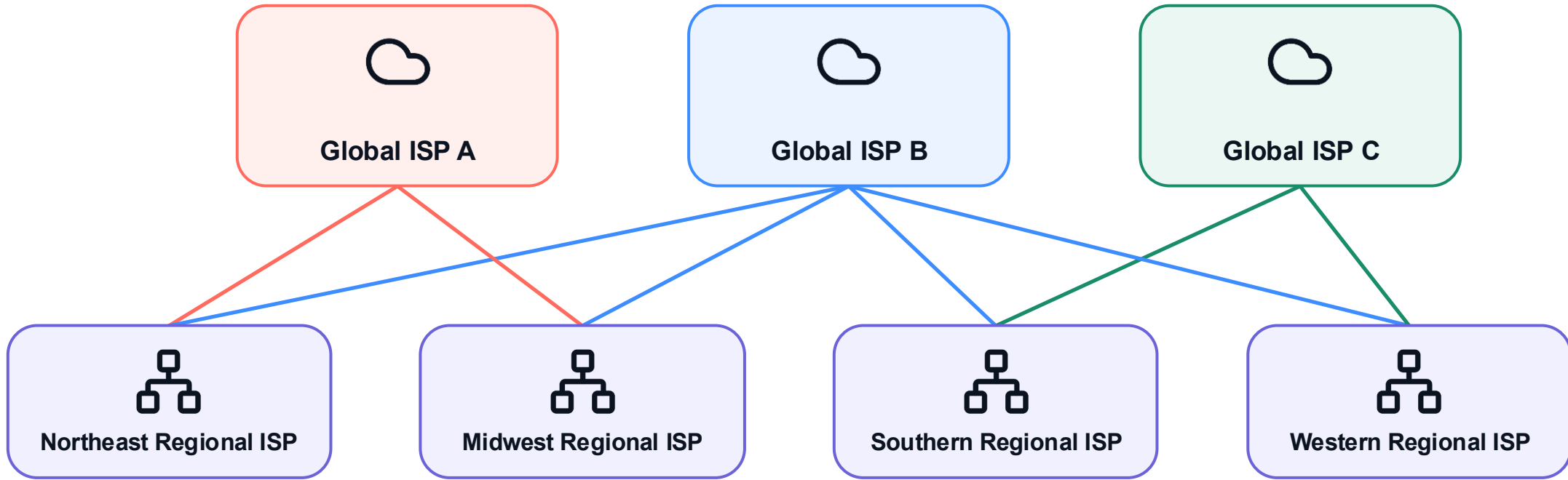
Packet switching

Performance

Regional ISPs first serve separate geographic regions



Regional ISPs connect through global providers



Regional ISPs connect upward to multiple global providers.

Core structure

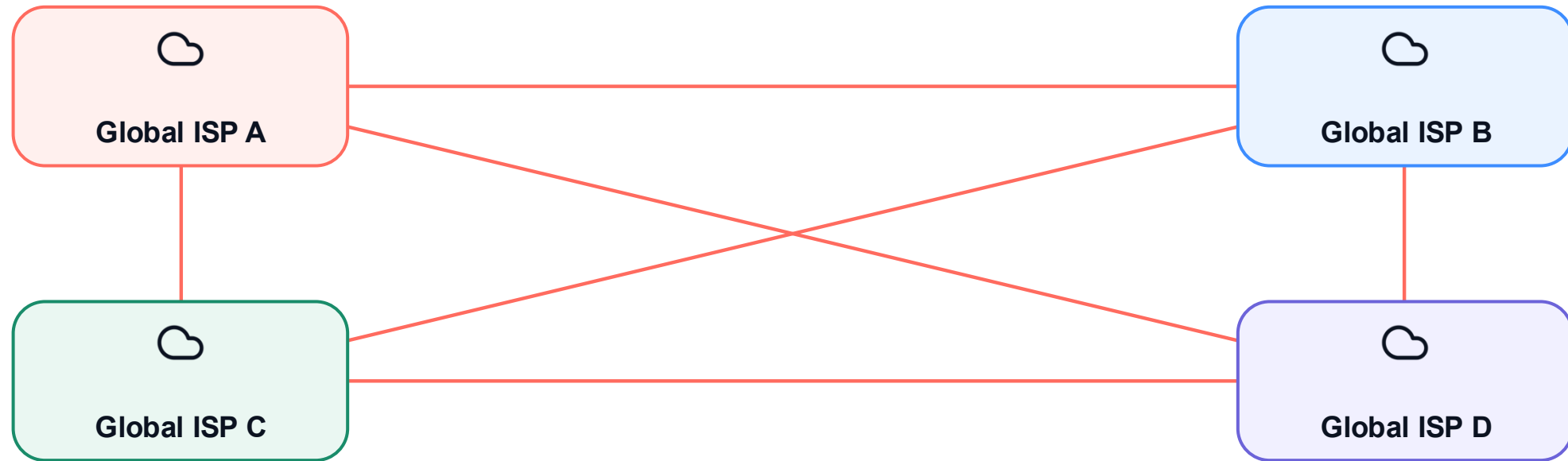
Routing + forwarding

Packet switching

Performance

But global ISPs also need to exchange traffic

If they rely only on bilateral private links...



Does every pair need its own private link?

The same scaling problem appears whenever many independent networks need to exchange traffic.

Core structure

Routing + forwarding

Packet switching

Performance

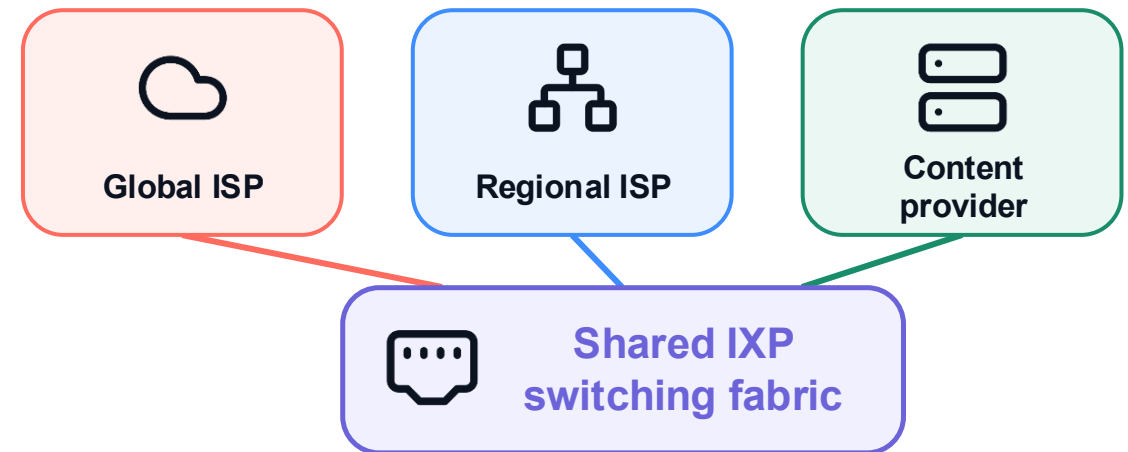
An IXP is where independent networks exchange traffic

IXP = Internet Exchange Point

A physical facility where independently operated networks connect and exchange traffic.



Example: an IXP facility in London



IXP is the meeting place; peering is the traffic-exchange relationship.

Core structure

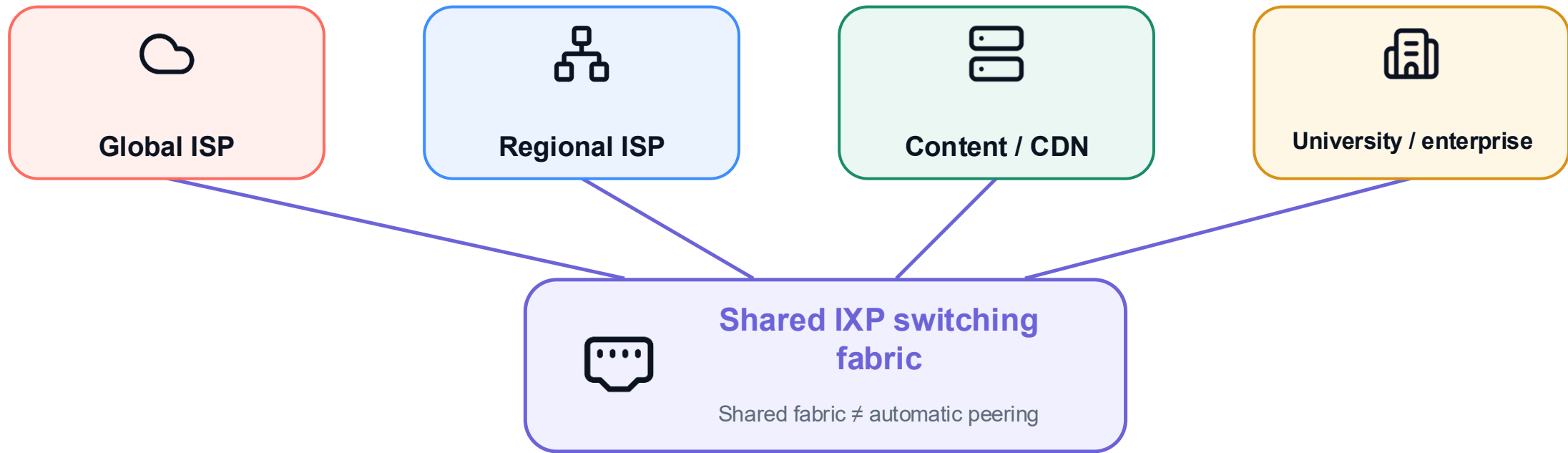
Routing + forwarding

Packet switching

Performance

An IXP connects networks—not only ISPs

Participants are independently operated IP networks



Global and regional ISPs can peer here—if both connect and their policies allow it.

Core structure

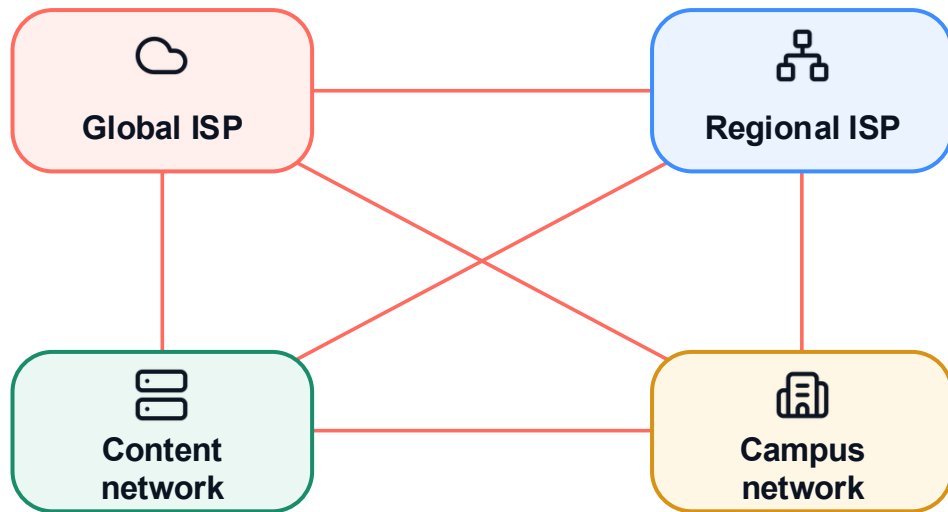
Routing + forwarding

Packet switching

Performance

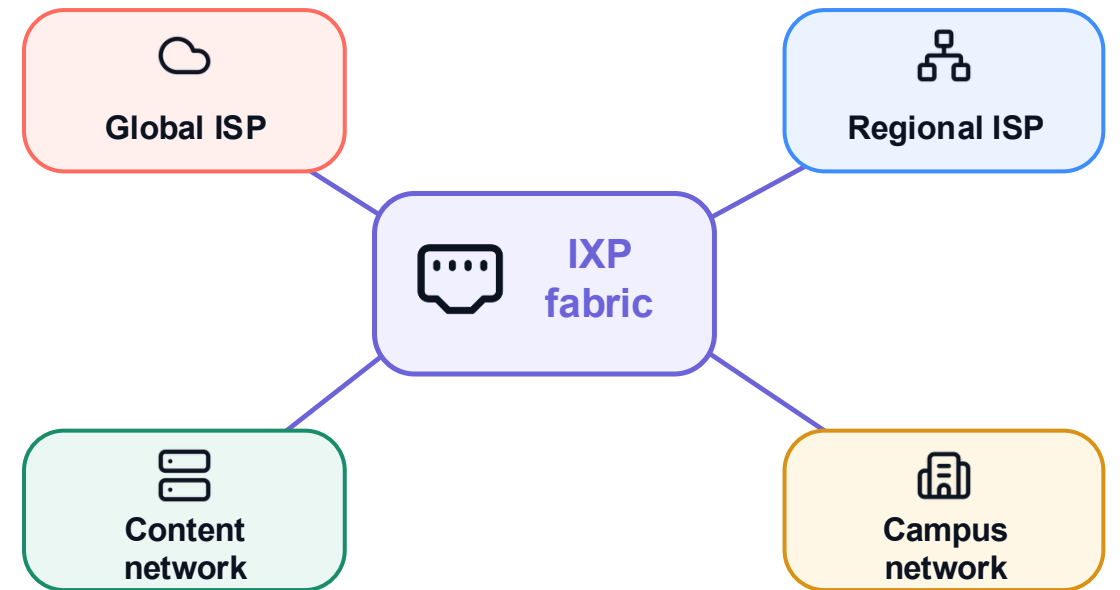
A shared IXP fabric scales better than pairwise links

Private cross-connects



4 networks → 6 private links
Pairwise links grow as $N(N-1)/2$

Shared IXP fabric



4 networks → 4 IXP ports
One connection can support many peerings

Private links still help heavy pairs; an IXP scales many possible peerings.

Core structure

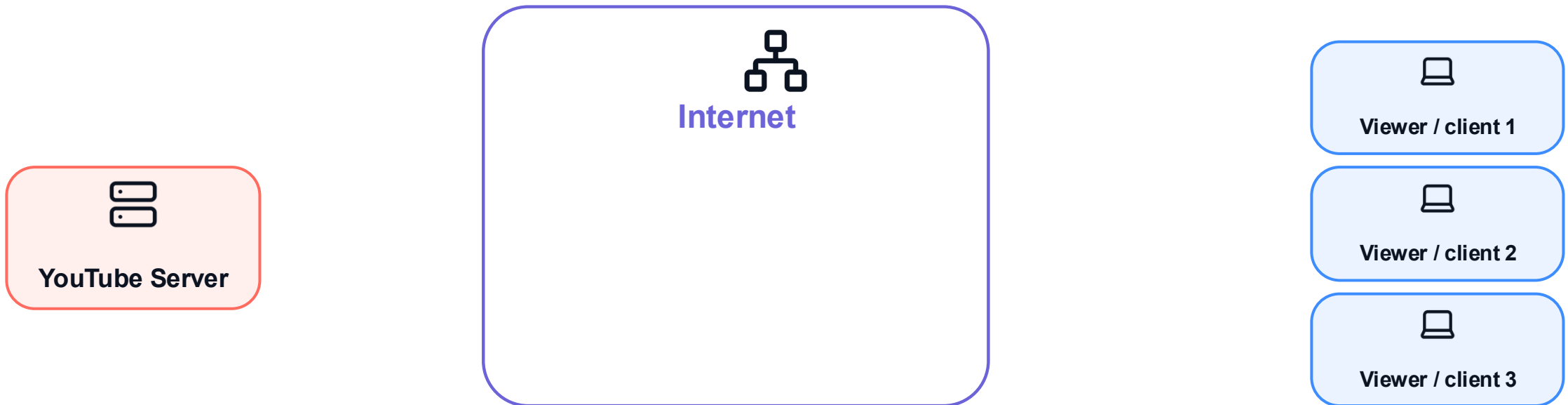
Routing + forwarding

Packet switching

Performance

The same video crosses the Internet once per viewer

Ordinary delivery is unicast: one server-to-client stream per viewer



Three viewers request the same video. How many streams cross the Internet?

Core structure

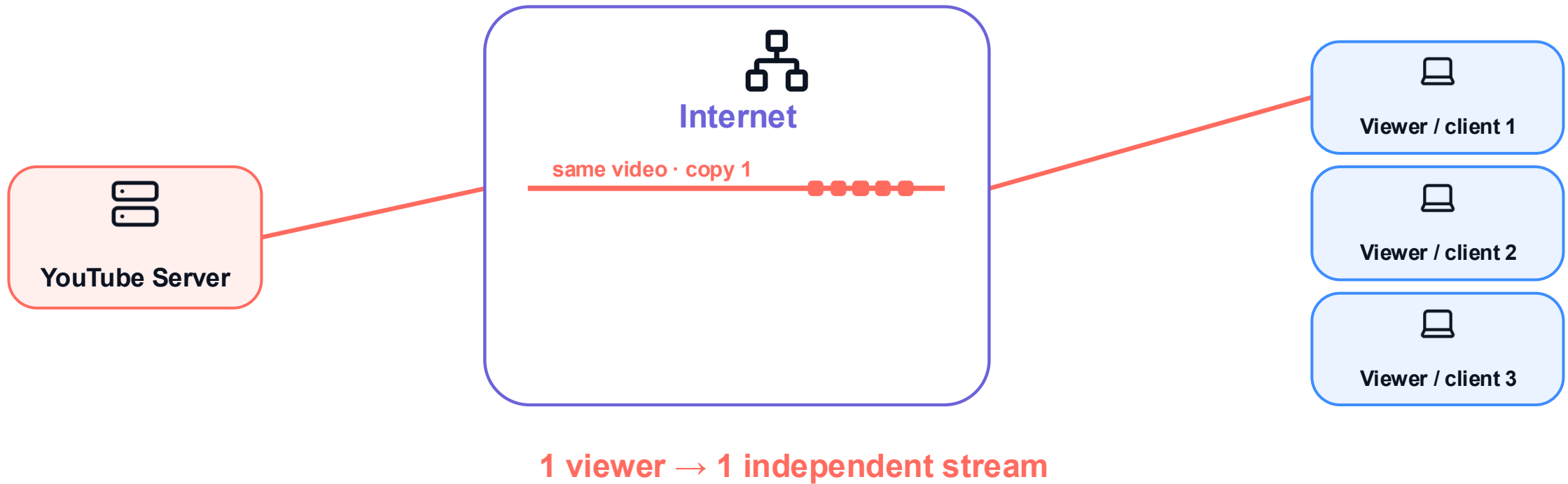
Routing + forwarding

Packet switching

Performance

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Core structure

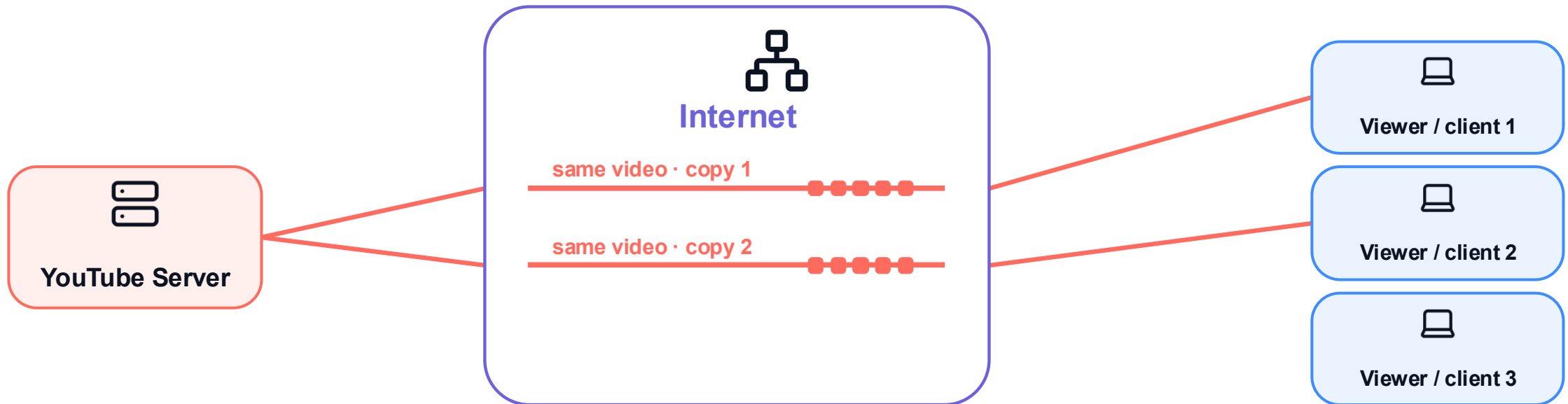
Routing + forwarding

Packet switching

Performance

The same video crosses the Internet once per viewer

Ordinary delivery is unicast: one server-to-client stream per viewer



2 viewers → 2 independent streams

Core structure

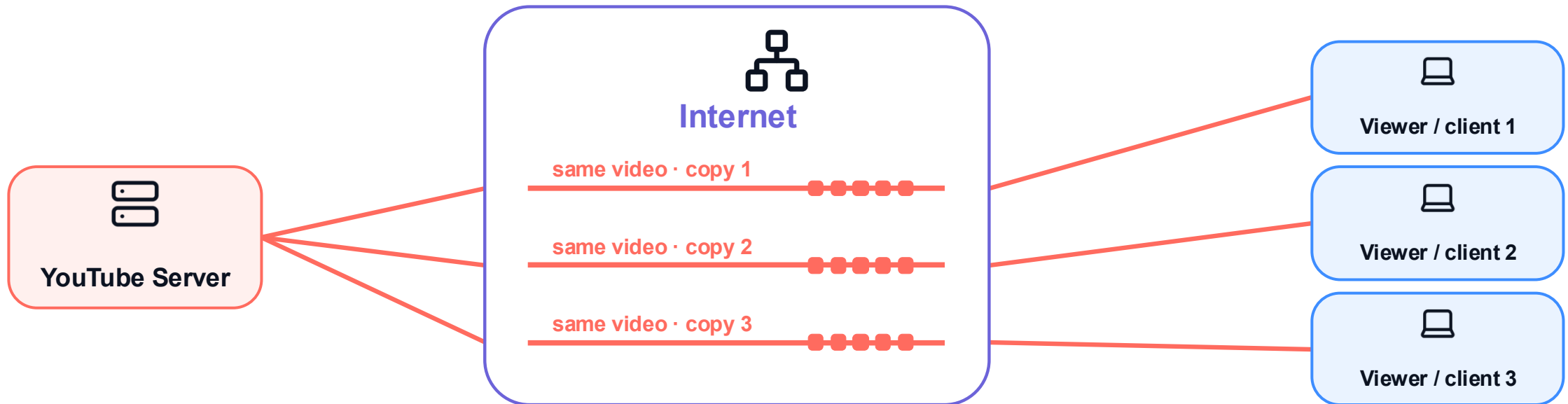
Routing + forwarding

Packet switching

Performance

The same video crosses the Internet once per viewer

Ordinary delivery is unicast: one server-to-client stream per viewer



3 viewers → 3 independent streams

Same content × 3 unicast streams = the video crosses shared Internet paths three times.

Core structure

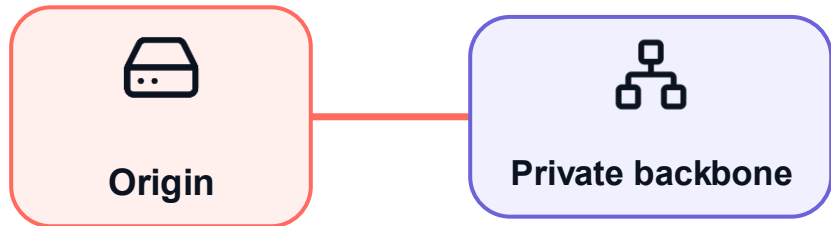
Routing + forwarding

Packet switching

Performance

A content network moves delivery capacity closer to users

Step 1 · Carry content from the origin on a private backbone

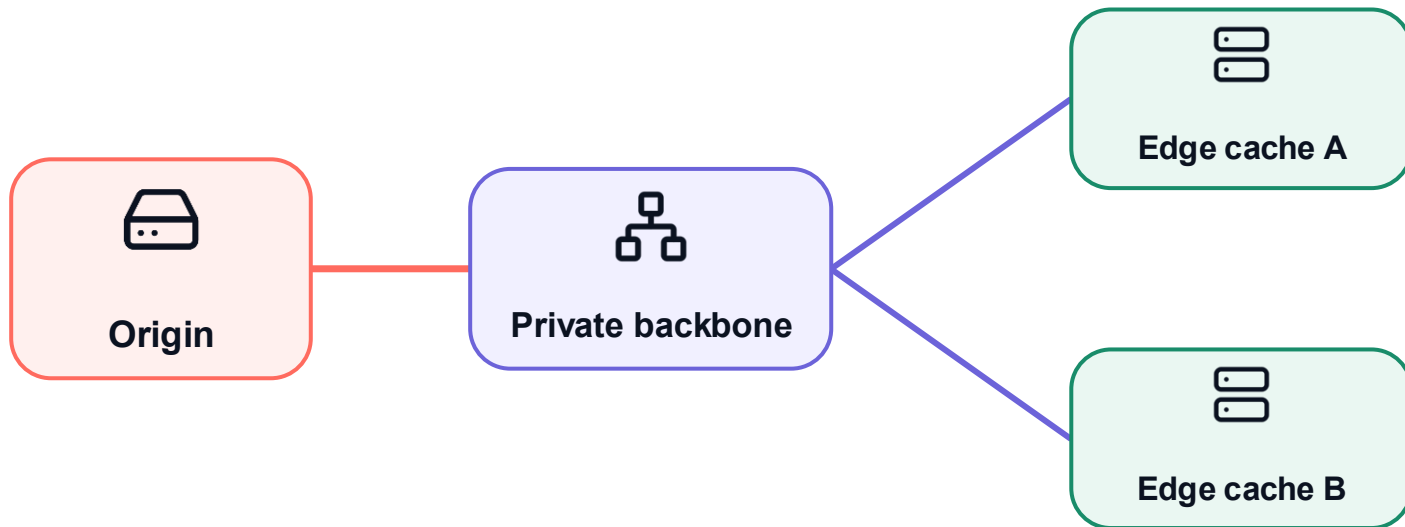


One origin still needs a way to reach many regions.



A content network moves delivery capacity closer to users

Step 2 · Place copies at distributed edge caches



Popular video objects can be copied to multiple edge locations.

Core structure

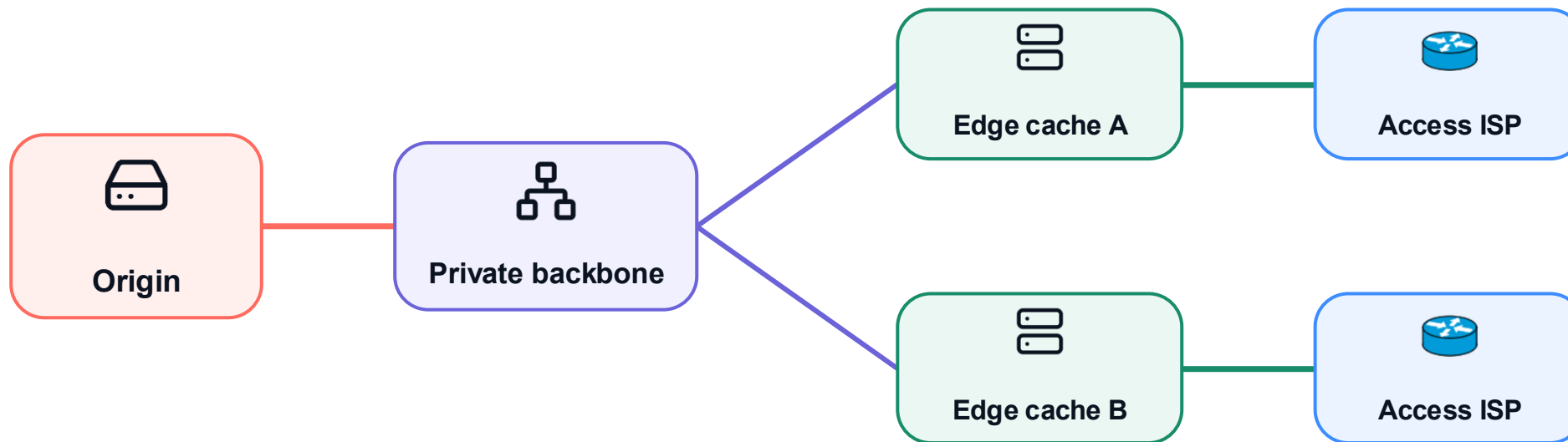
Routing + forwarding

Packet switching

Performance

A content network moves delivery capacity closer to users

Step 3 · Interconnect those caches near access ISPs



Each edge cache connects where users enter the Internet.

Core structure

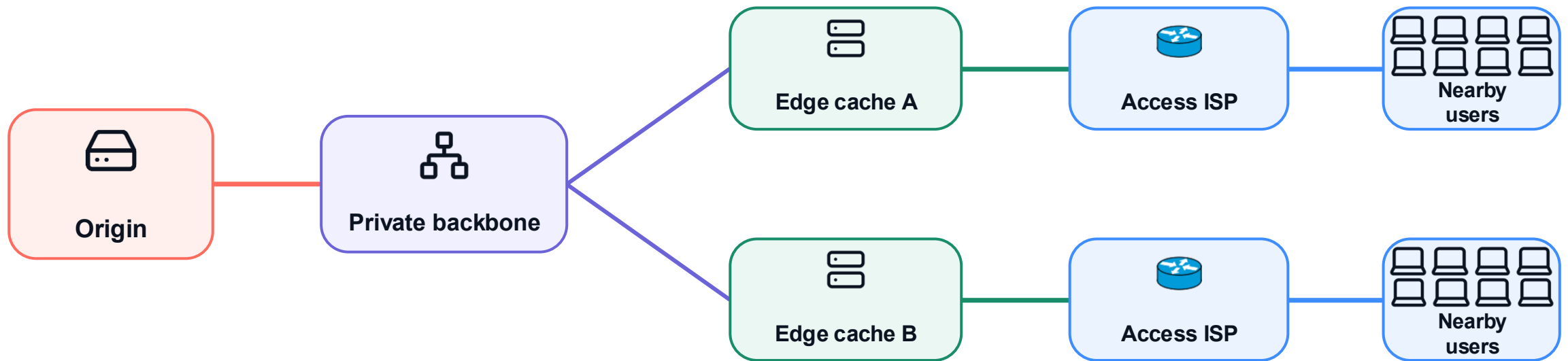
Routing + forwarding

Packet switching

Performance

A content network moves delivery capacity closer to users

Step 4 · Serve viewers from a nearby copy whenever possible



Cache hit: serve nearby · Cache miss: fetch or refill from the origin

Nearby copies shorten delivery paths and reduce repeated long-haul traffic.

Core structure

Routing + forwarding

Packet switching

Performance

Content networks can be service-owned or shared CDNs

Service-owned delivery networks

- **Google / YouTube**
video, search, apps, and popular cached objects
- **Netflix Open Connect**
a purpose-built network for streaming delivery
- **Meta**
feeds, photos, video, messaging, and application traffic

Shared CDN / cloud-edge networks

- **Amazon CloudFront**
global edge locations and regional edge caches
- **Microsoft Azure Front Door**
cloud CDN and global application delivery
- **Akamai**
distributed delivery for many content owners
- **Cloudflare**
CDN, edge delivery, performance, and security

Different owners; the same design goal: put delivery capacity close to demand.

A content network is a network role—not another ISP tier.

Core structure

Routing + forwarding

Packet switching

Performance

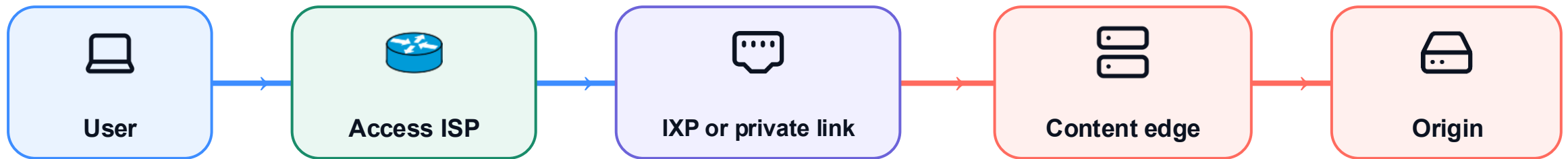
ISPs provide reach; content networks deliver the service

ISP: access + reach

Connects users to other networks

Content network: service + delivery

Operates the service, edge sites, and origin



1 ISP carries the request

2 Networks interconnect

3 Content network answers

Connectivity and service are different roles—even when both networks meet at the same IXP.

The ISP supplies connectivity; the content network supplies the service.

Core structure

Routing + forwarding

Packet switching

Performance

Millions of edge networks sit at the bottom

MILLIONS AT
EDGE
18 shown



home • campus • mobile networks

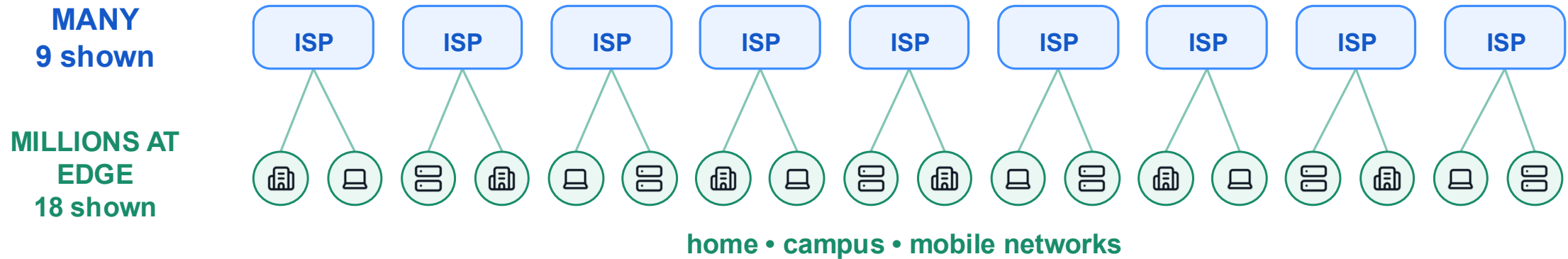
Core structure

Routing + forwarding

Packet switching

Performance

Access ISPs aggregate many edge networks



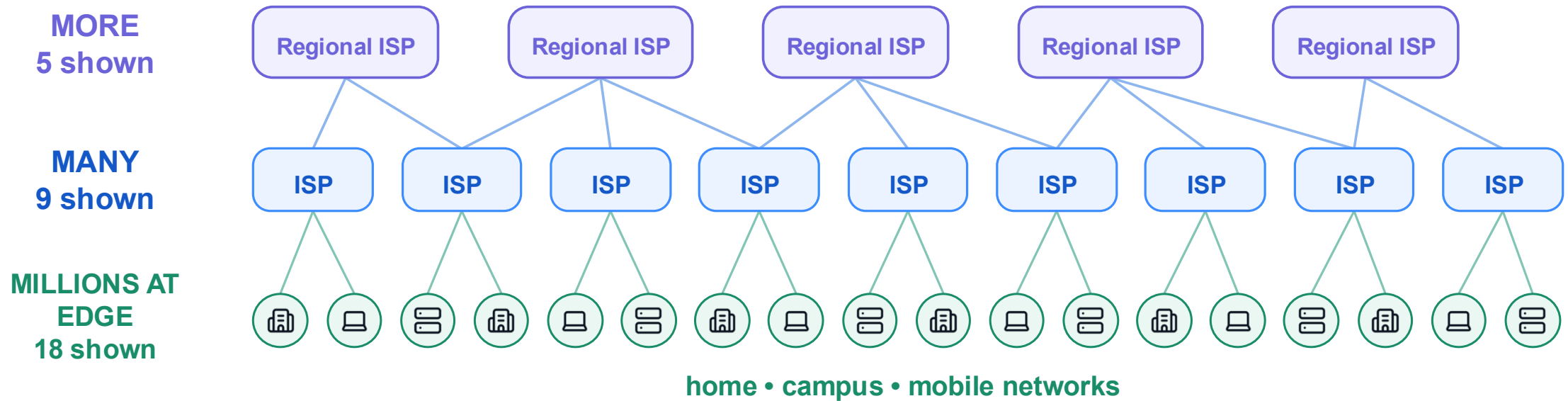
Core structure

Routing + forwarding

Packet switching

Performance

Regional ISPs aggregate many access ISPs



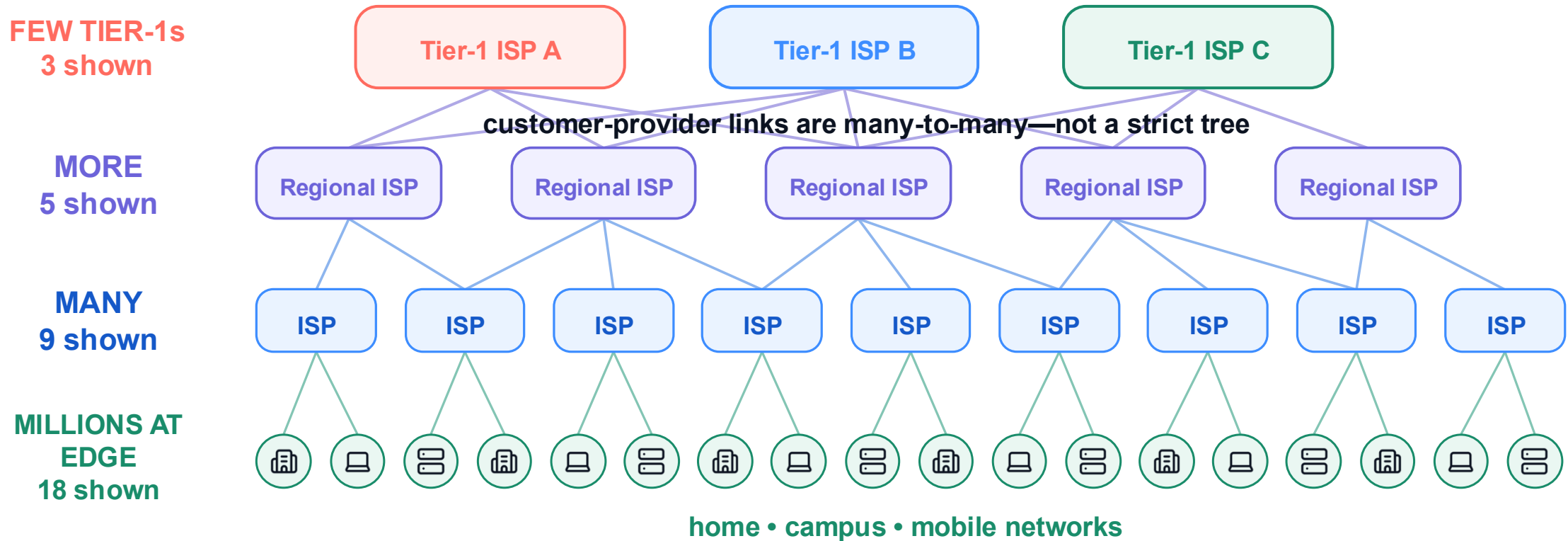
Core structure

Routing + forwarding

Packet switching

Performance

The ISP hierarchy is many-to-many, not a strict tree



One Tier-1 serves many regional ISPs; a regional ISP can also connect to multiple upstreams.

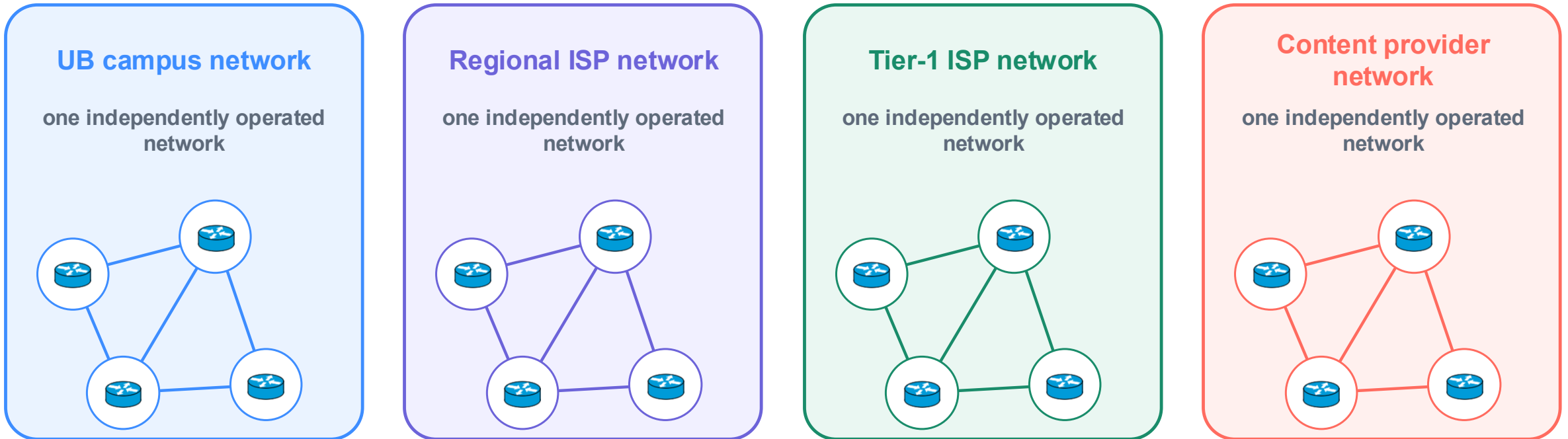
Core structure

Routing + forwarding

Packet switching

Performance

Independent networks can operate on their own



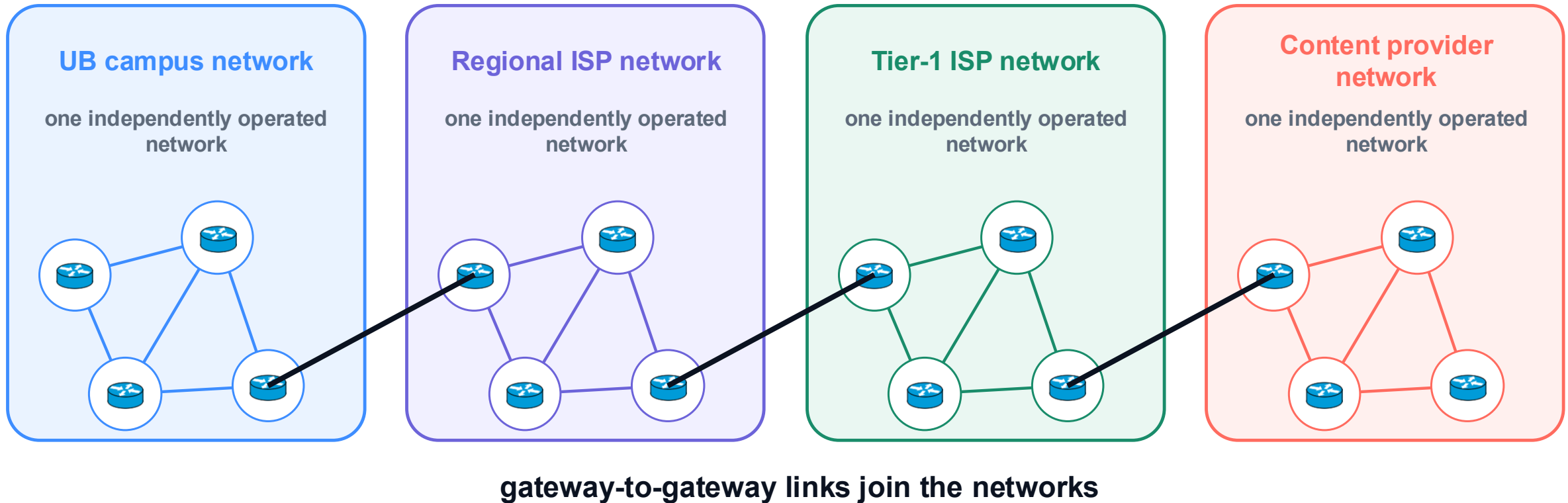
Core structure

Routing + forwarding

Packet switching

Performance

The Internet is many complete networks joined at gateways



The Internet is not one network—it is many complete networks joined at their gateways.

Core structure

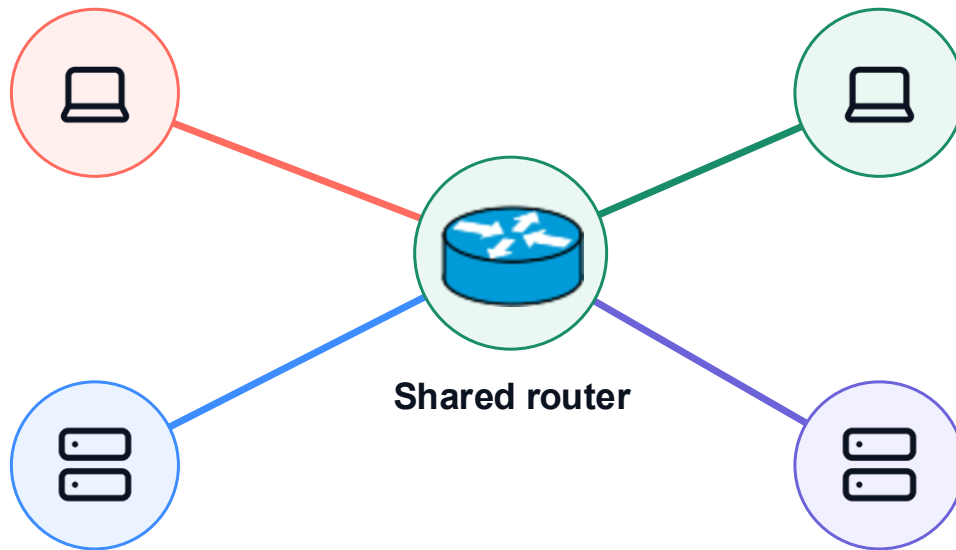
Routing + forwarding

Packet switching

Performance

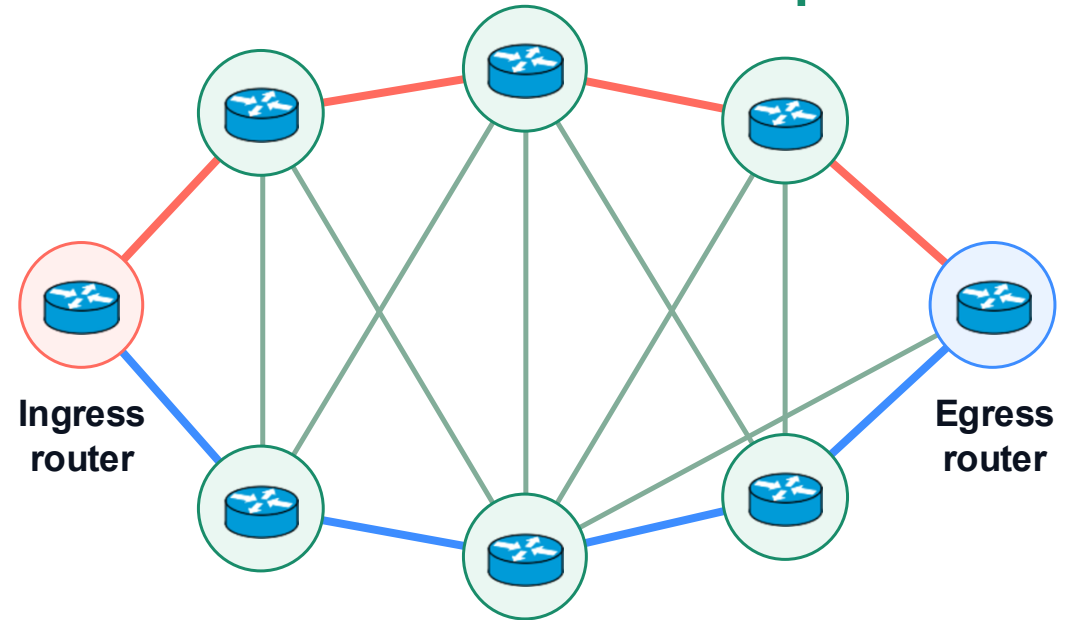
Shared routers reduce links; a meshed core adds paths

1 Fewer endpoint links



4 router links < 6 pairwise links

2 More router-to-router paths



Edge links are reduced; links among core routers are added intentionally for path diversity.

Core structure

Routing + forwarding

Packet switching

Performance

Part 2 · Routing + forwarding

02



Routing chooses. Forwarding executes.

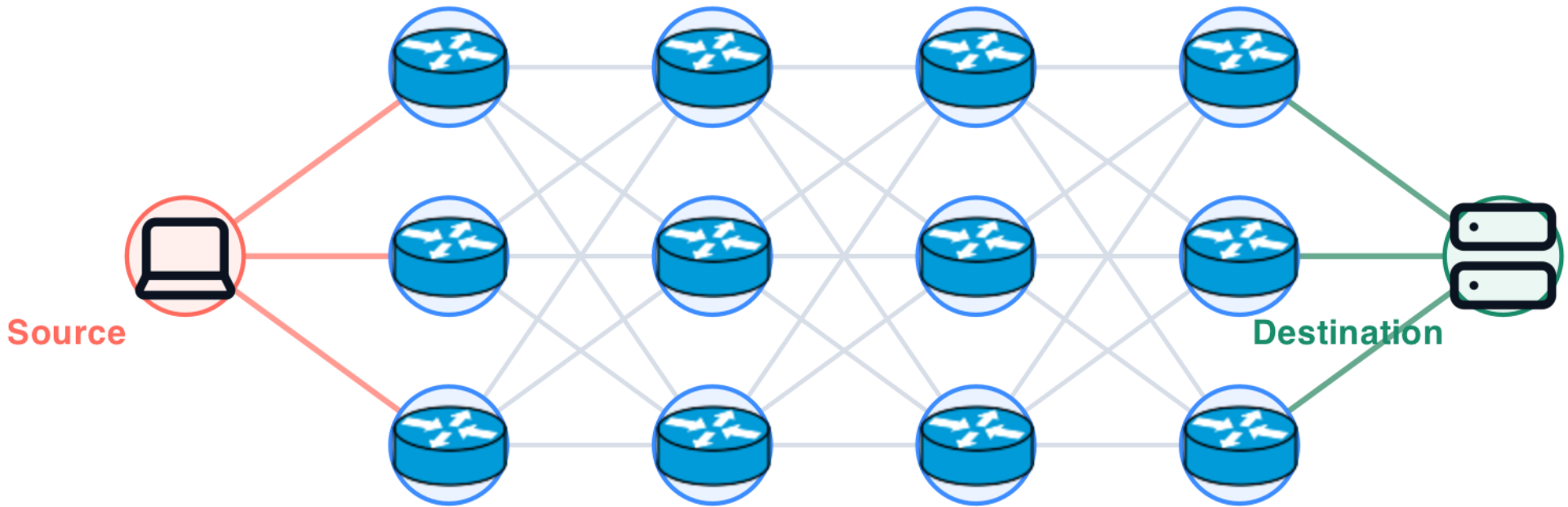
Core structure

Routing + forwarding

Packet switching

Performance

Many possible paths make routing a difficult choice



Which route is best?

More possible paths make the routing decision harder.

Core structure

Routing + forwarding

Packet switching

Performance

Optimal depends on the metric



Shortest

Fewest hops or least distance



Lowest cost

Cheapest links or agreements



Policy compliant

Paths allowed by operators

A route is optimal only relative to a stated objective and policy.

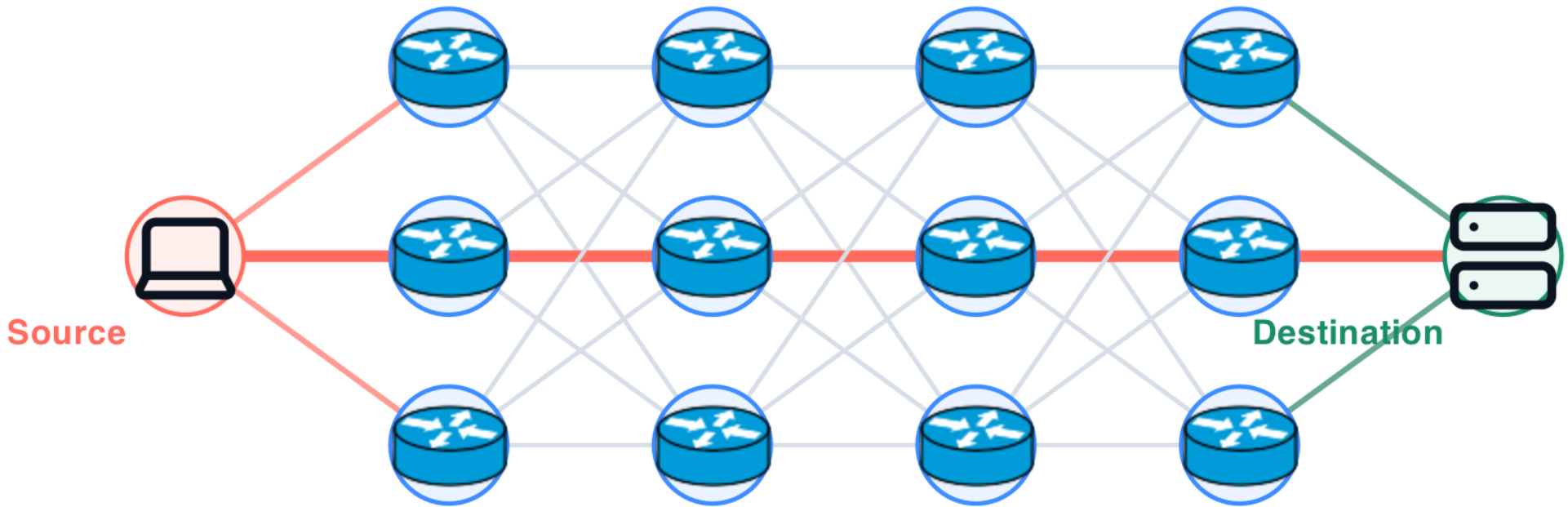
Core structure

Routing + forwarding

Packet switching

Performance

Routing selects one path through a complex topology



Routing is global: it chooses one path through the full topology.

Core structure

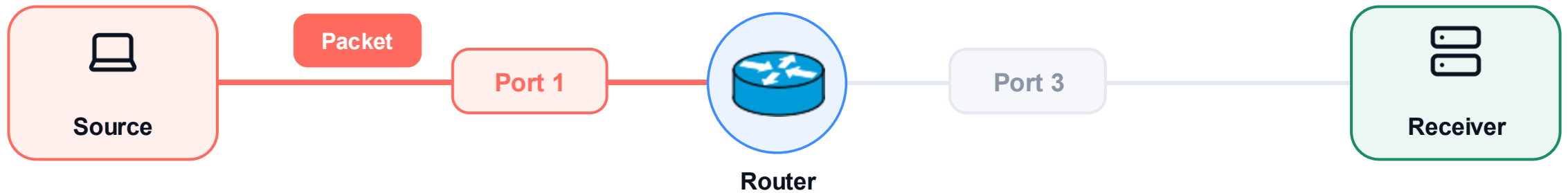
Routing + forwarding

Packet switching

Performance

Step 1 · The packet arrives on an input port

FORWARDING = **receive** → choose → send

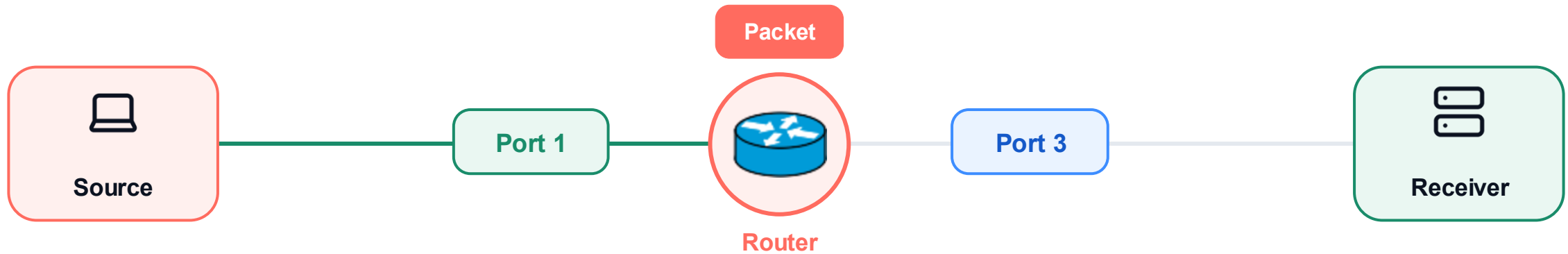


The router first receives one packet on Port 1.



Step 2 · The router chooses one output

FORWARDING = **receive** → **choose** → send



The router uses its forwarding information to choose Port 3.

Core structure

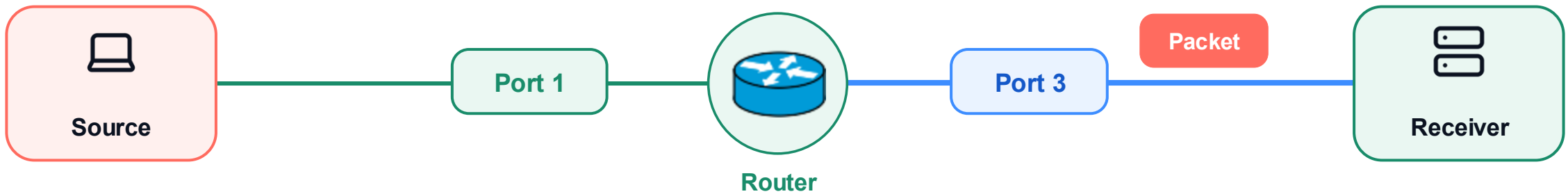
Routing + forwarding

Packet switching

Performance

Step 3 · The packet leaves through the selected output

FORWARDING = receive → choose → send



Forwarding is local: one router moves one packet from Port 1 to Port 3.

Core structure

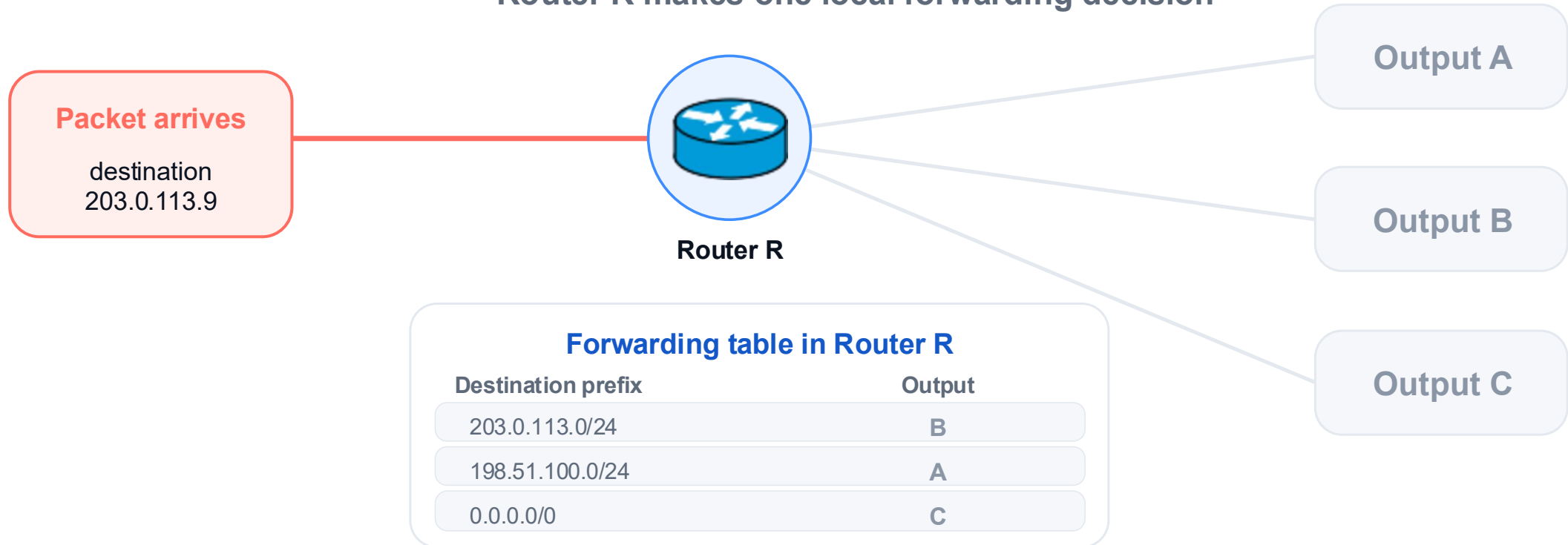
Routing + forwarding

Packet switching

Performance

Step 1 · Read the packet's destination

Router R makes one local forwarding decision



The destination address is the lookup key.

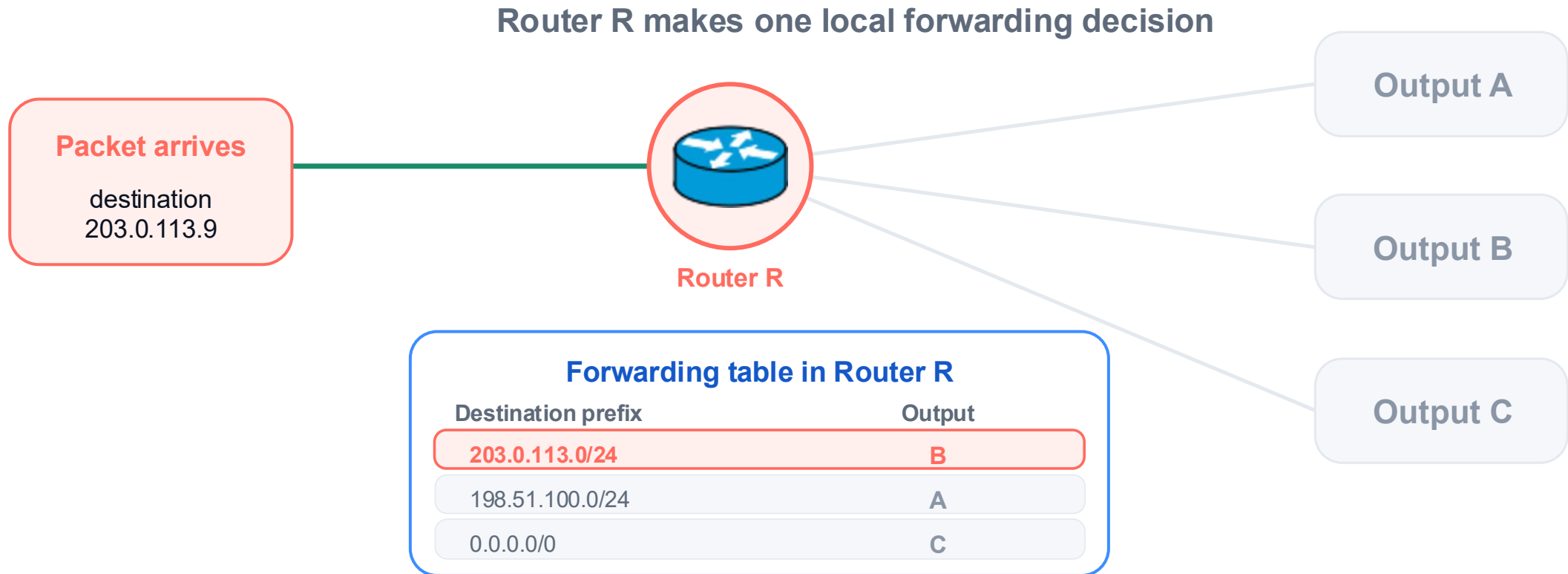
Core structure

Routing + forwarding

Packet switching

Performance

Step 2 · Match the destination to a table entry



203.0.113.9 matches 203.0.113.0/24, so the table selects B.

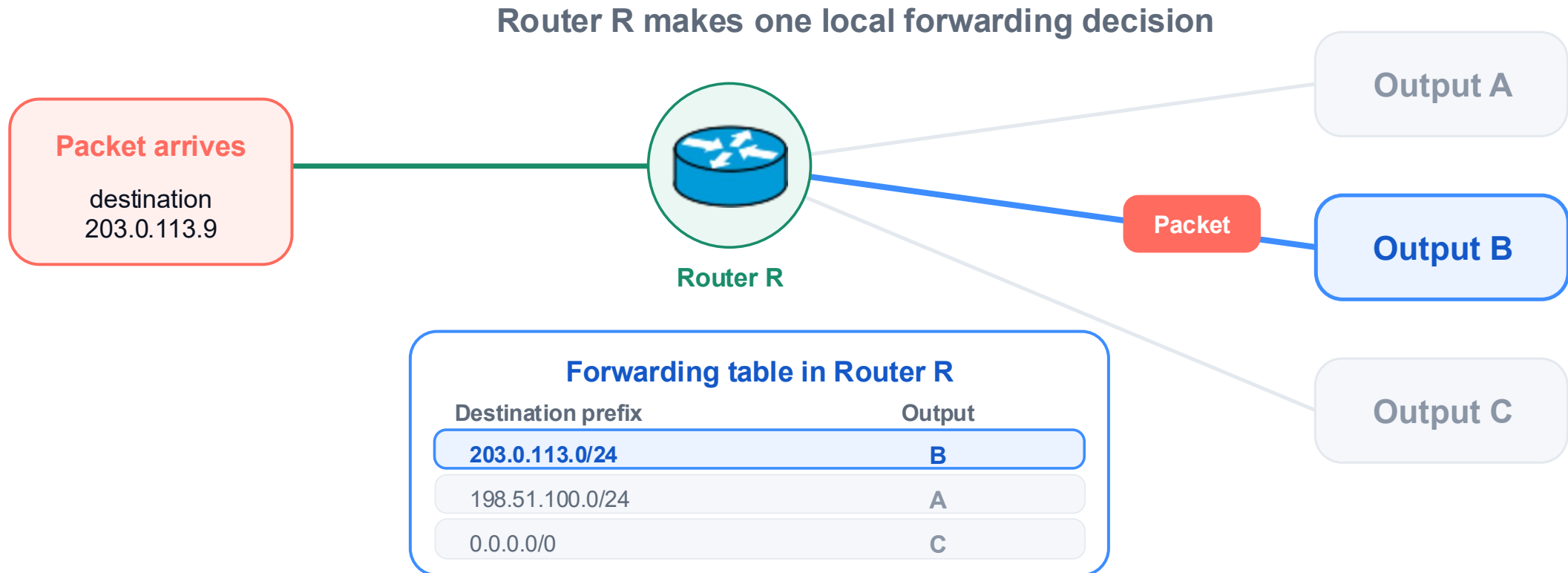
Core structure

Routing + forwarding

Packet switching

Performance

Step 3 · Send the packet through output B



Forwarding reads one local rule and activates one output.

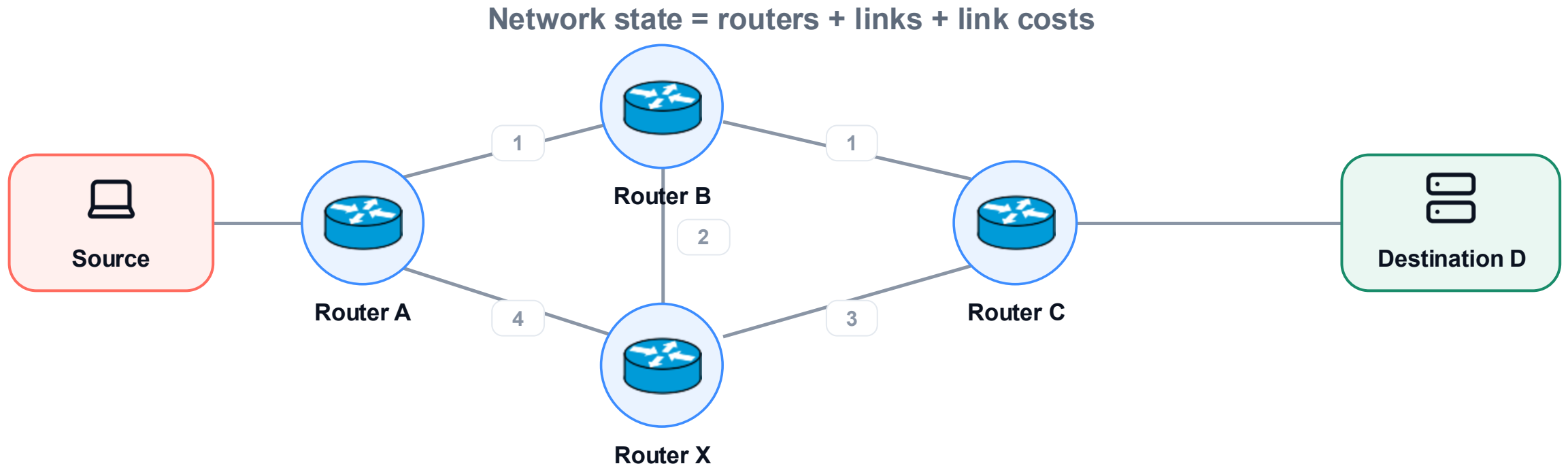
Core structure

Routing + forwarding

Packet switching

Performance

Routing starts with a network-wide topology



Routing reasons about the network as a whole.

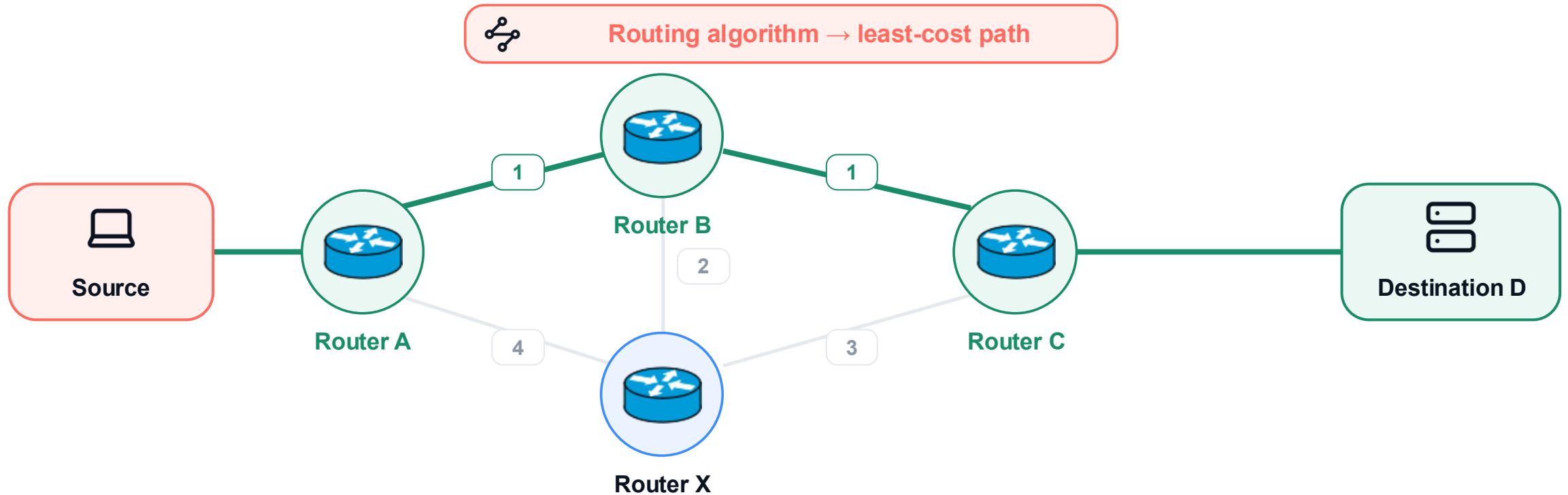
Core structure

Routing + forwarding

Packet switching

Performance

The routing algorithm selects one path through the topology



For this example, the least-cost route is **Source → A → B → C → Destination**.

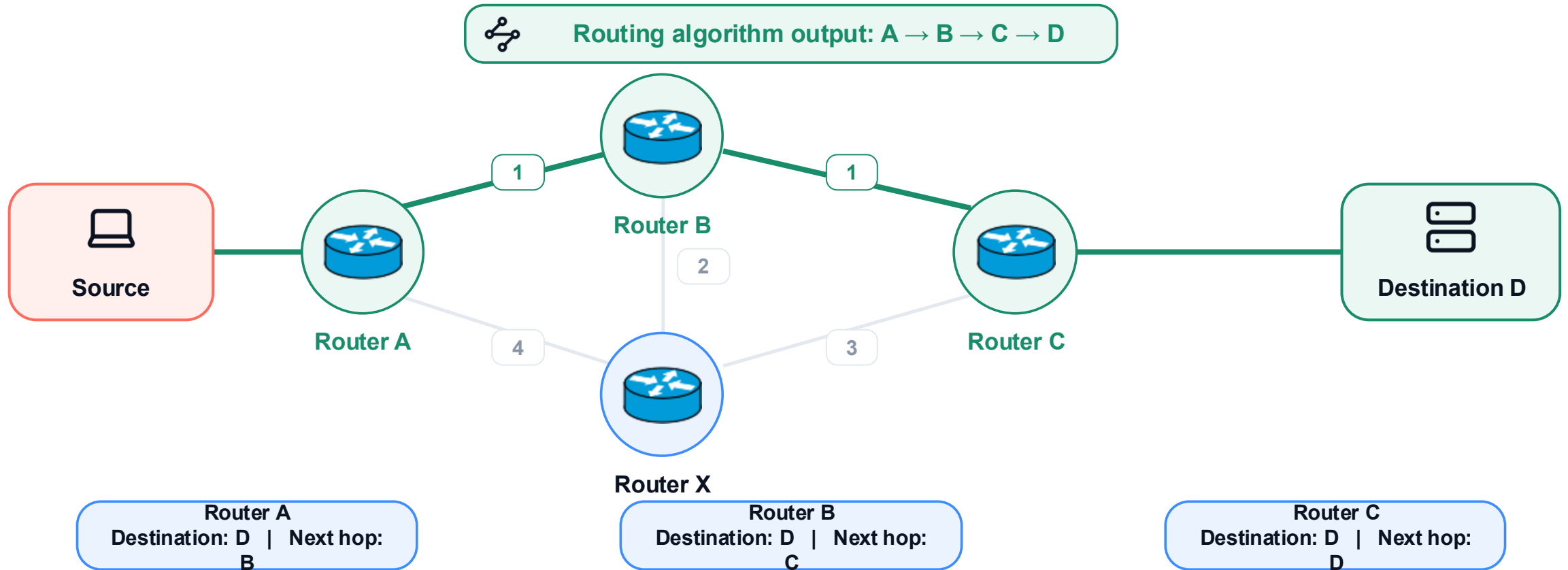
Core structure

Routing + forwarding

Packet switching

Performance

For destination D, each router stores its next hop



The algorithm computed three next-hop rules. Next, install one rule in each router.

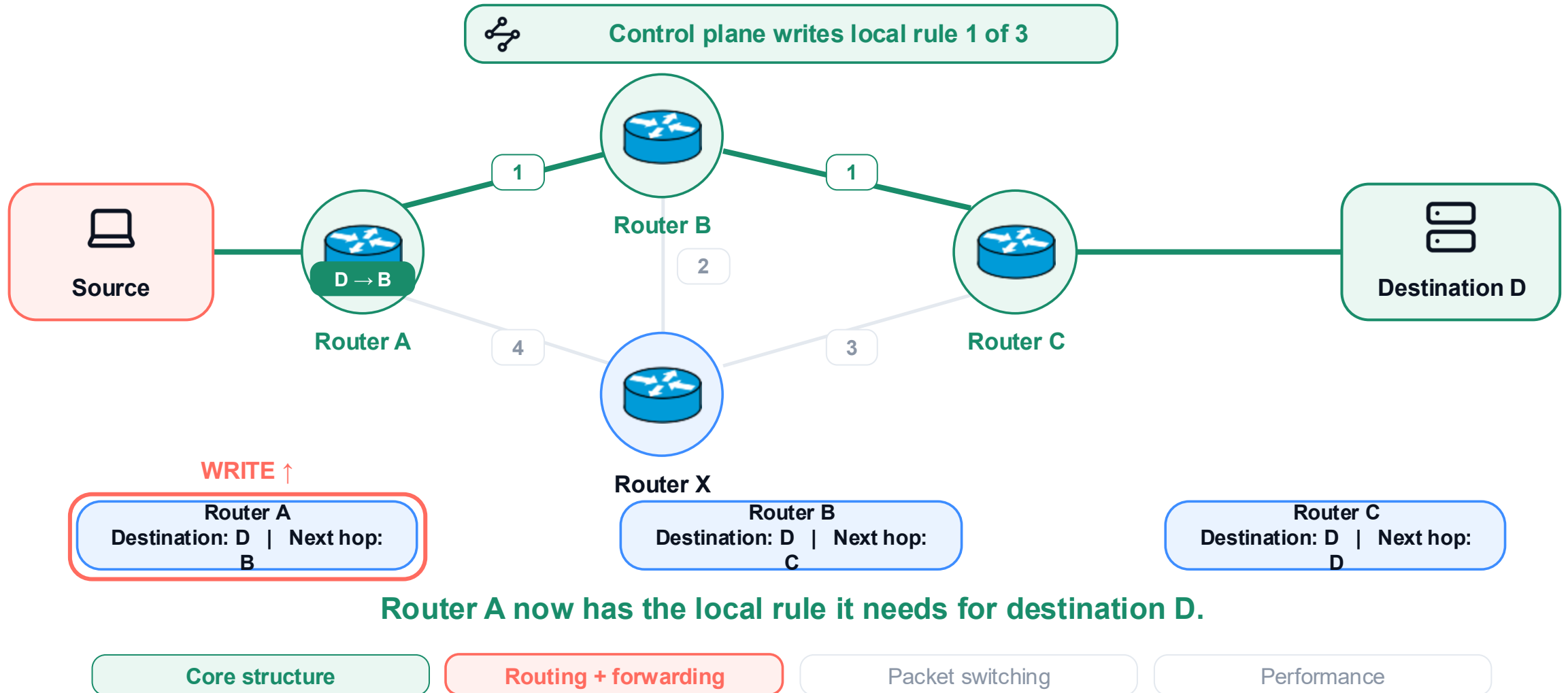
Core structure

Routing + forwarding

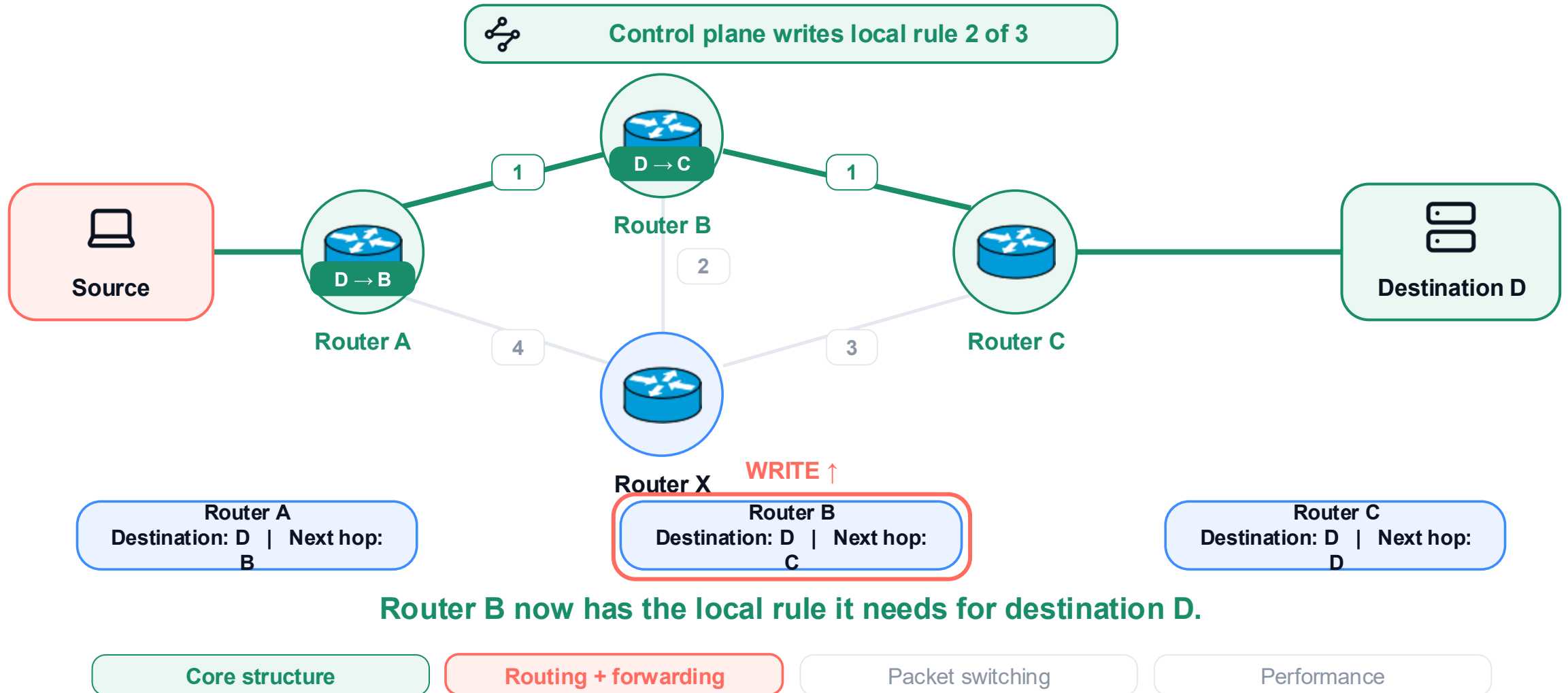
Packet switching

Performance

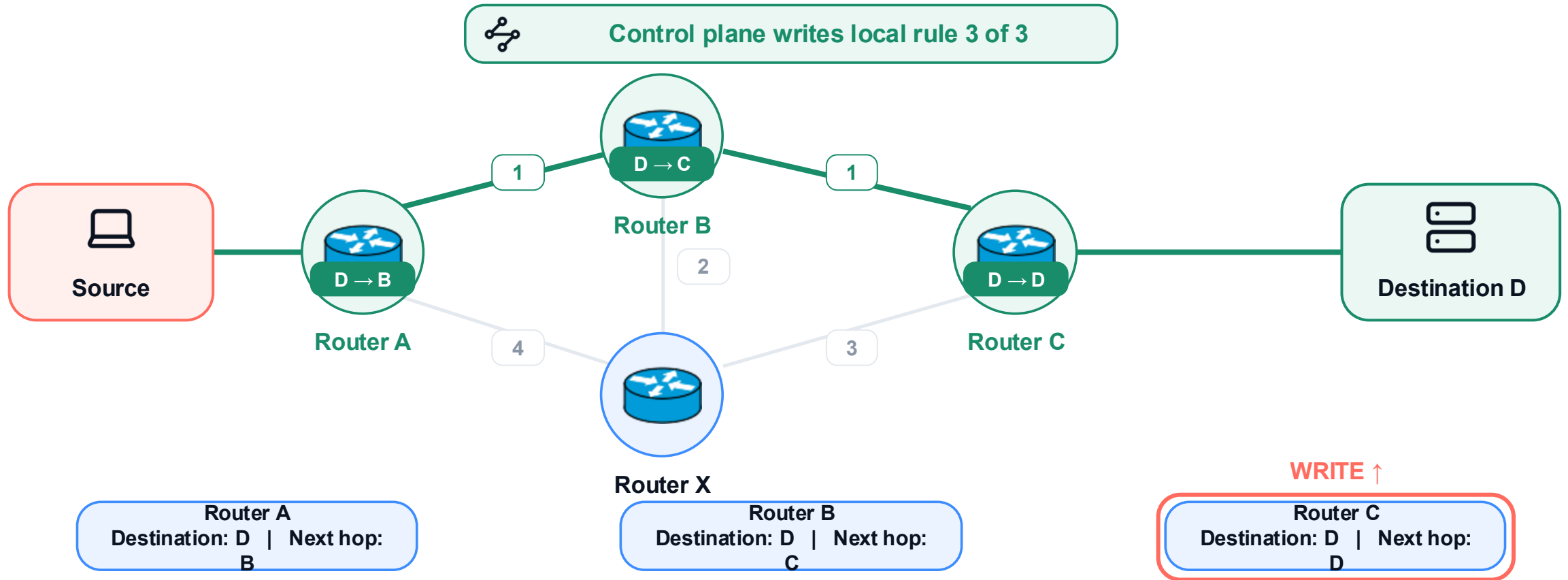
Install Router A's rule: destination D → next hop B



Install Router B's rule: destination D → next hop C



Install Router C's rule: destination D → next hop D



The end-to-end path is now encoded as three local forwarding rules.

Core structure

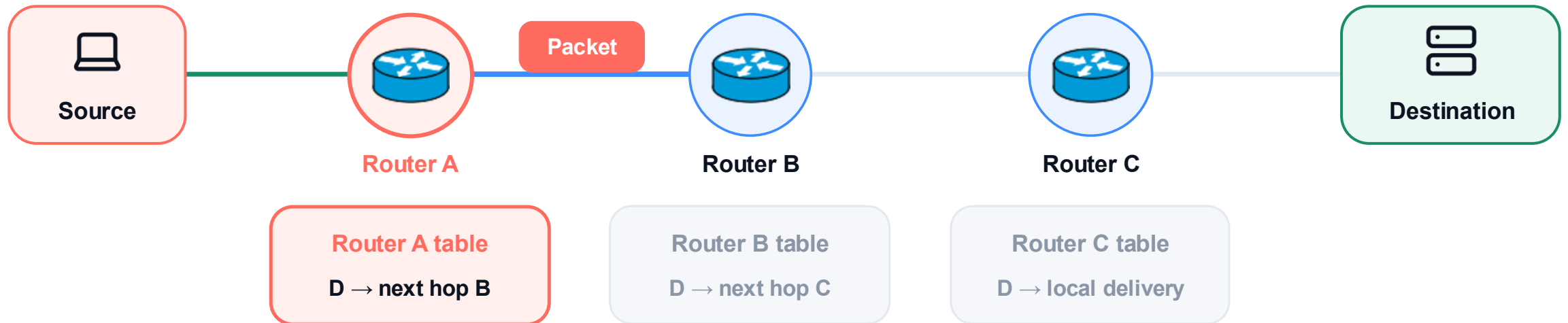
Routing + forwarding

Packet switching

Performance

Router A forwards the packet to Router B

Each router consults only its own forwarding table



One table lookup changes only Router A's local action.

Core structure

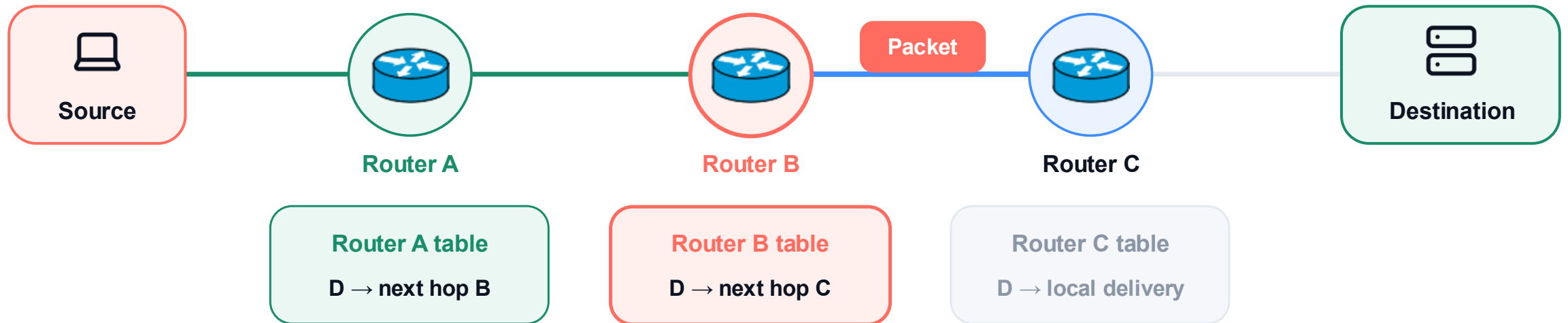
Routing + forwarding

Packet switching

Performance

Router B forwards the packet to Router C

Each router consults only its own forwarding table



The packet reaches Router B, which makes a new local decision.

Core structure

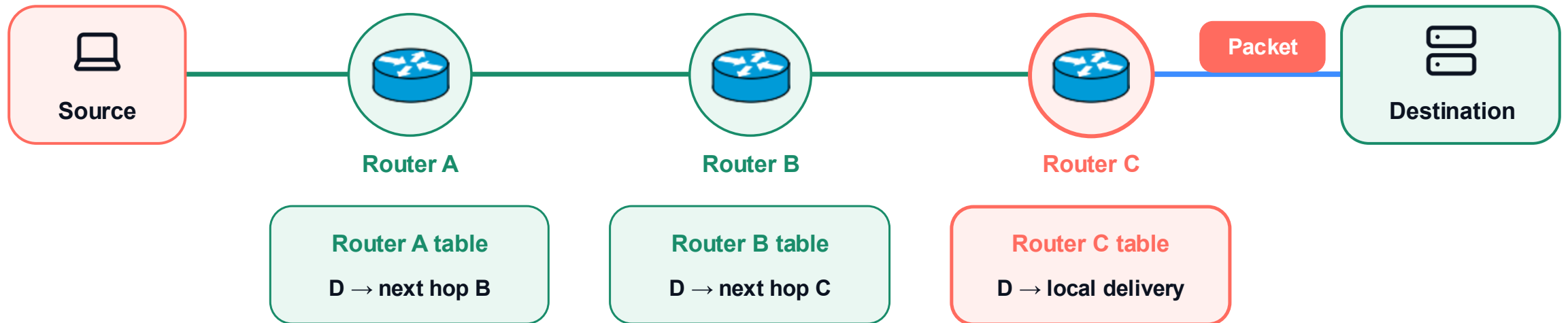
Routing + forwarding

Packet switching

Performance

Router C forwards the packet to the destination

Each router consults only its own forwarding table



Repeated local decisions carry the packet across the selected route.

Core structure

Routing + forwarding

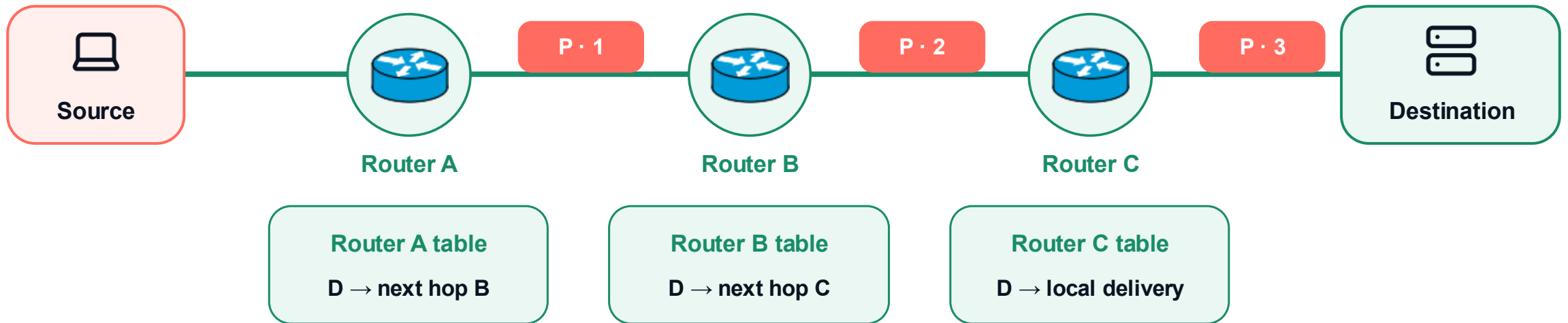
Packet switching

Performance

Routing chooses the path; forwarding moves each packet

routing (global) → forwarding tables (local state) → forwarding (per packet)

the same packet shown at three successive hops



Routing selected the path; repeated local forwarding actions realize the end-to-end route.

Core structure

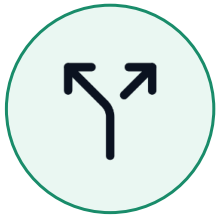
Routing + forwarding

Packet switching

Performance

Part 3 · Packet switching

03



Packets share links on demand.

Core structure

Routing + forwarding

Packet switching

Performance

Two switching models allocate resources differently



Packet switching

Share links on demand



Circuit switching

Reserve a path before sending

Both models move data, but they allocate link capacity differently.

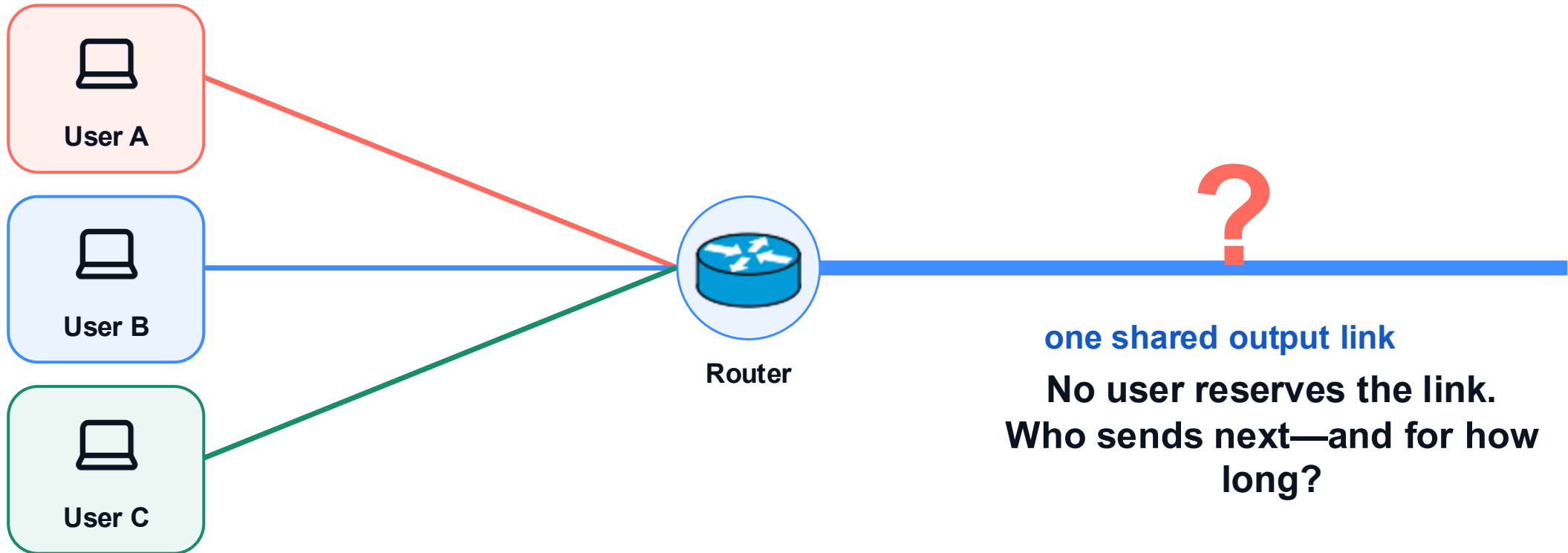
Core structure

Routing + forwarding

Packet switching

Performance

How can several users share one outgoing link?



Question: can the router interleave packets from all three users?

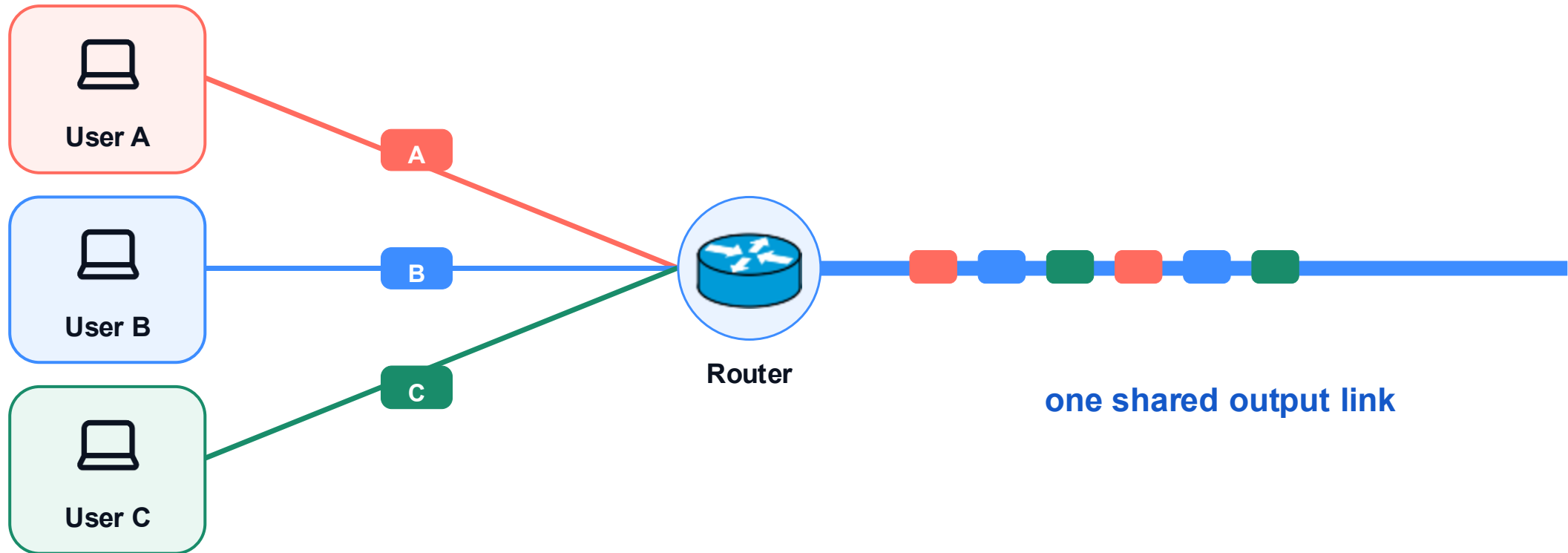
Core structure

Routing + forwarding

Packet switching

Performance

Packets from several users share one outgoing link



Packets from different users take turns on the same link.

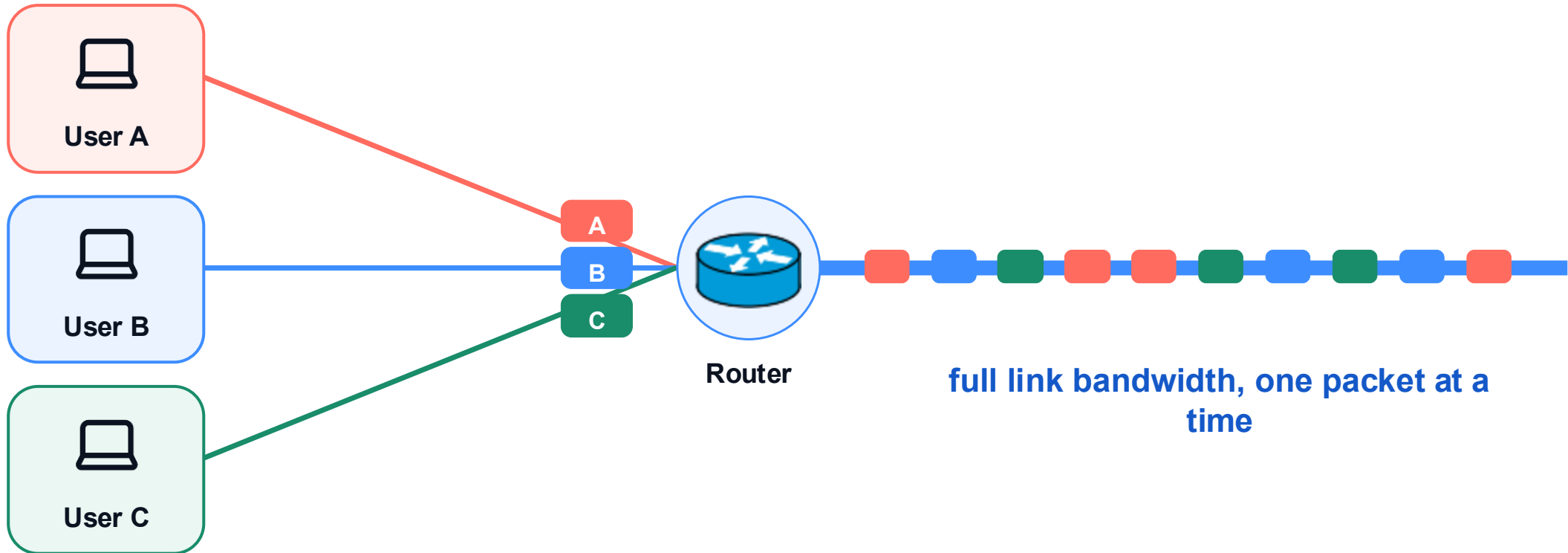
Core structure

Routing + forwarding

Packet switching

Performance

Packets from many users share the same links



Packet switching interleaves users without reserving fixed slices of the link.

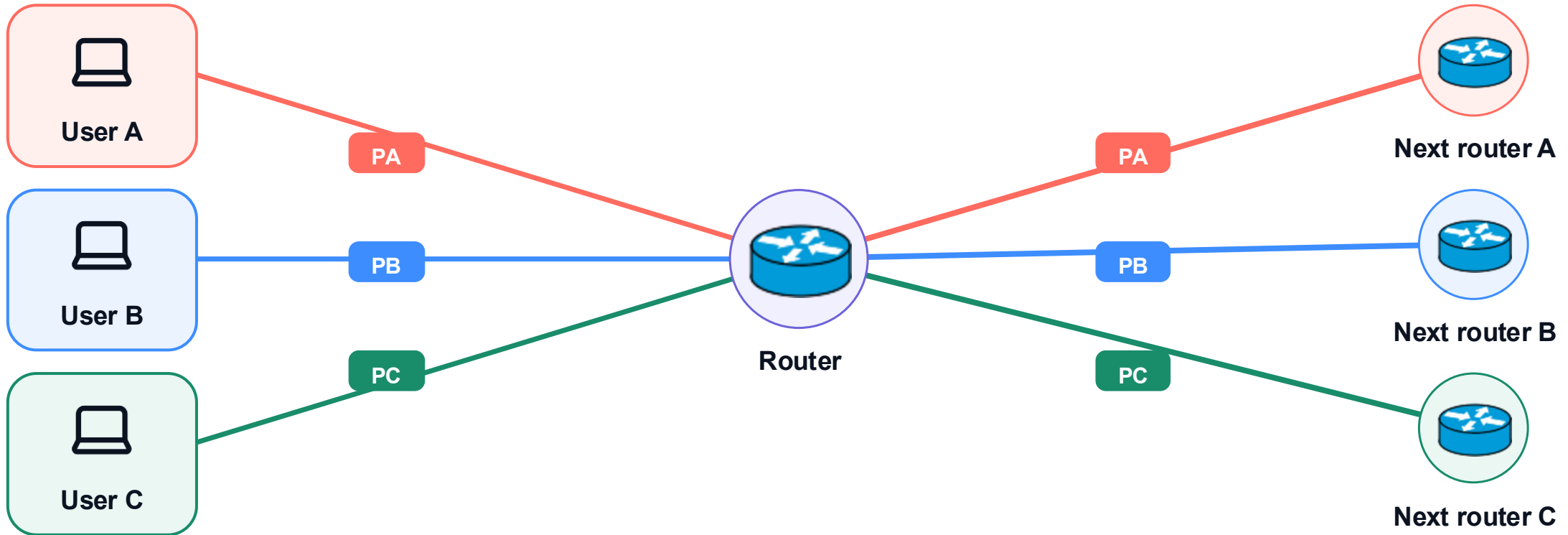
Core structure

Routing + forwarding

Packet switching

Performance

Three packets can take three different next hops



The router reads each destination and can send different packets to different next routers.

Core structure

Routing + forwarding

Packet switching

Performance

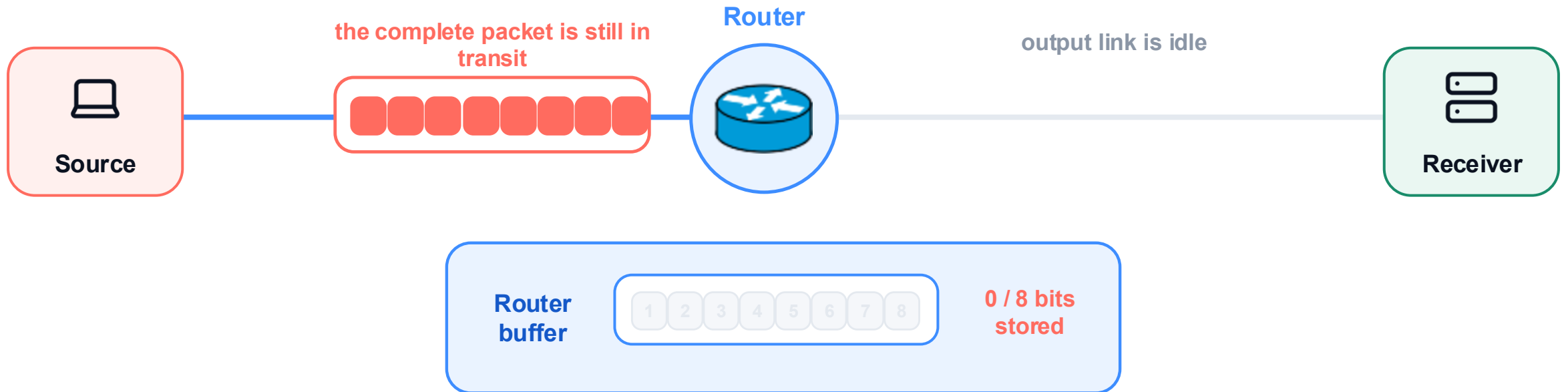
The packet is still traveling toward the router

1 RECEIVE + STORE ALL 8 BITS



2 TRANSMIT

one square = one bit · eight bits = one packet



The router has received 0 / 8 bits, so it cannot transmit.

Core structure

Routing + forwarding

Packet switching

Performance

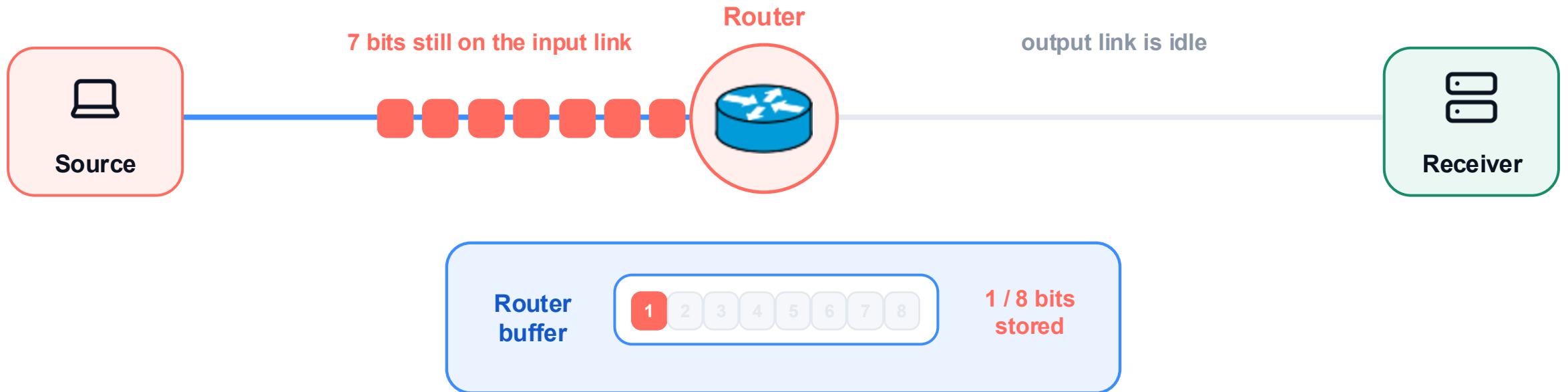
Bit 1 arrives and is stored in the router buffer

1 RECEIVE + STORE ALL 8 BITS



2 TRANSMIT

one square = one bit · eight bits = one packet



The output stays empty: store-and-forward is still waiting for bit 8.

Core structure

Routing + forwarding

Packet switching

Performance

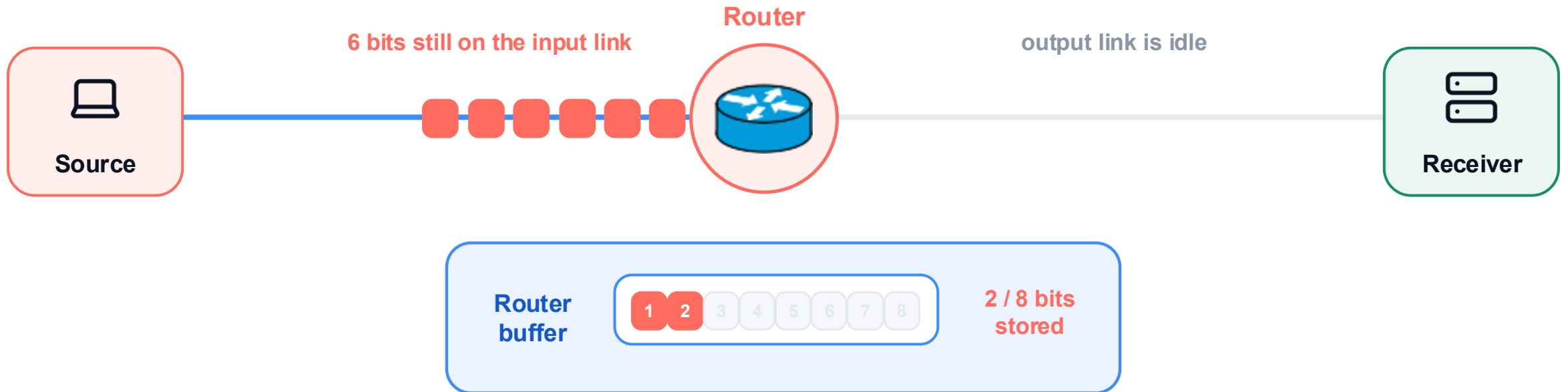
Bit 2 arrives and is stored in the router buffer

1 RECEIVE + STORE ALL 8 BITS



2 TRANSMIT

one square = one bit · eight bits = one packet



The output stays empty: store-and-forward is still waiting for bit 8.

Core structure

Routing + forwarding

Packet switching

Performance

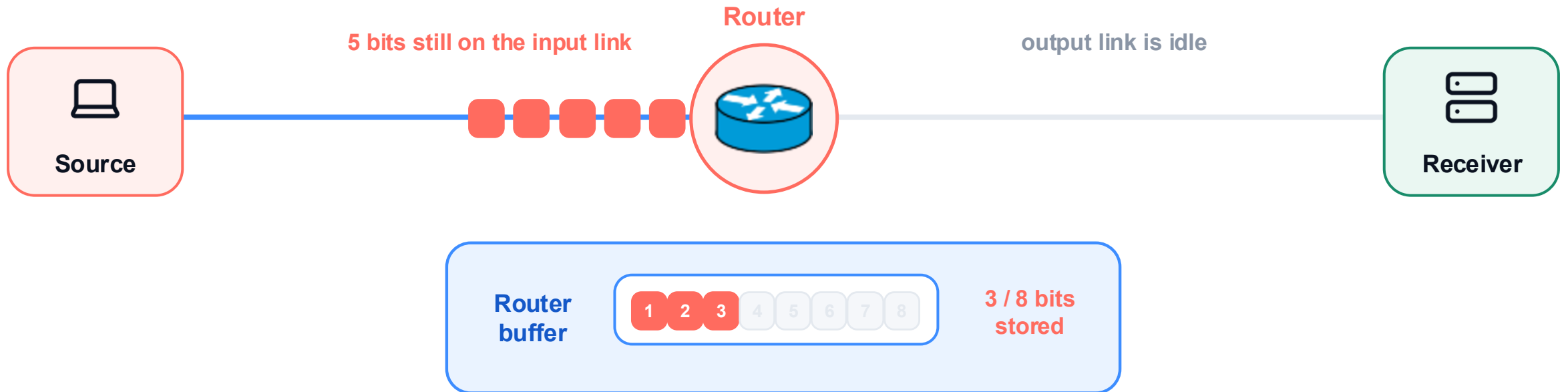
Bit 3 arrives and is stored in the router buffer

1 RECEIVE + STORE ALL 8 BITS



2 TRANSMIT

one square = one bit · eight bits = one packet



The output stays empty: store-and-forward is still waiting for bit 8.

Core structure

Routing + forwarding

Packet switching

Performance

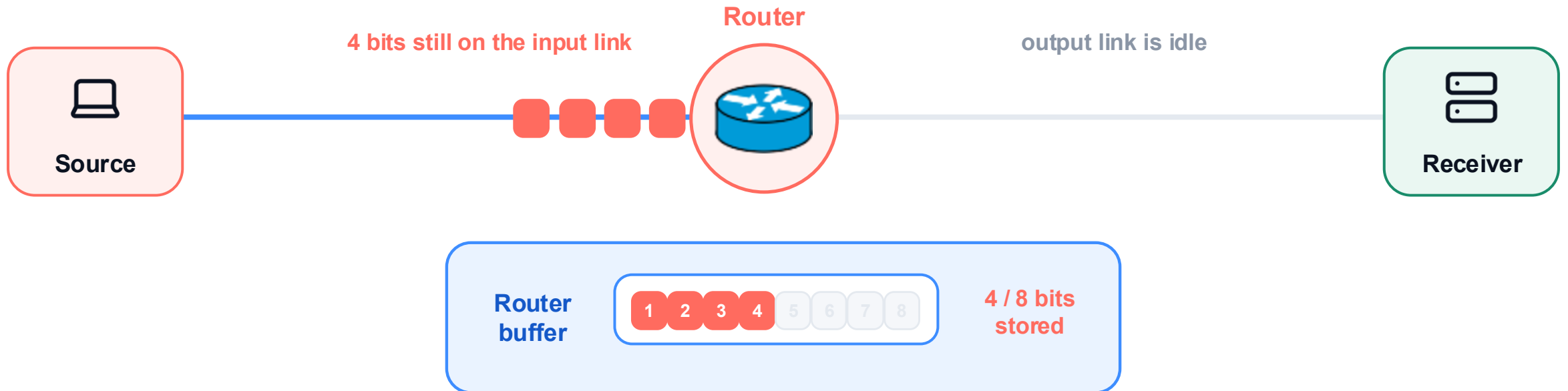
Bit 4 arrives and is stored in the router buffer

1 RECEIVE + STORE ALL 8 BITS



2 TRANSMIT

one square = one bit · eight bits = one packet



The output stays empty: store-and-forward is still waiting for bit 8.

Core structure

Routing + forwarding

Packet switching

Performance

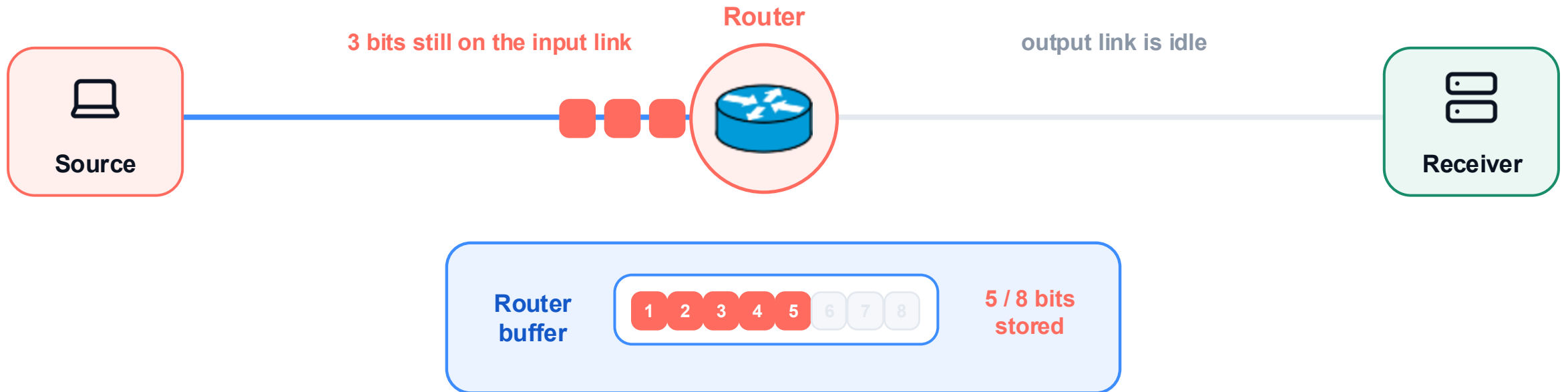
Bit 5 arrives and is stored in the router buffer

1 RECEIVE + STORE ALL 8 BITS



2 TRANSMIT

one square = one bit · eight bits = one packet



The output stays empty: store-and-forward is still waiting for bit 8.

Core structure

Routing + forwarding

Packet switching

Performance

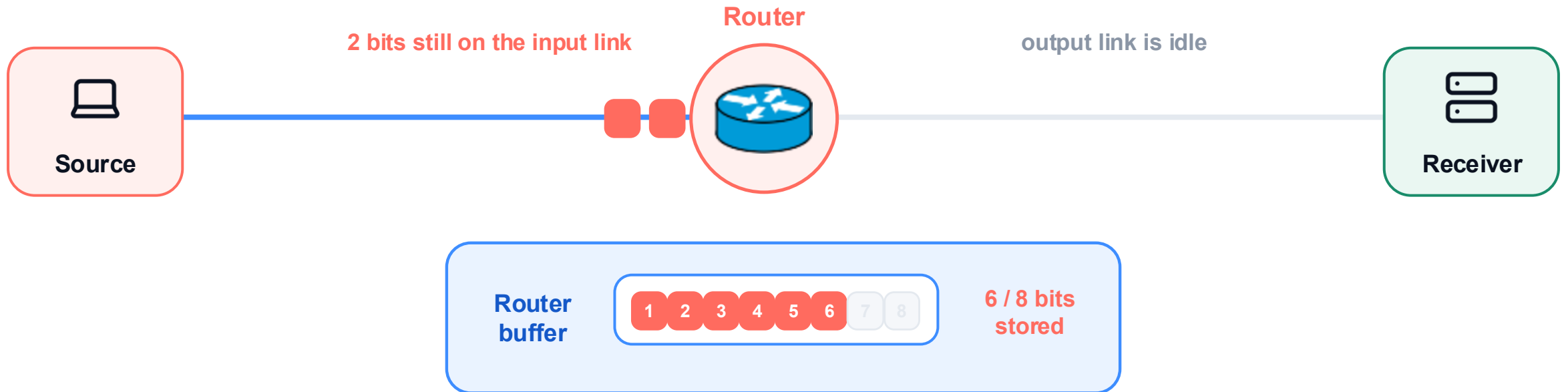
Bit 6 arrives and is stored in the router buffer

1 RECEIVE + STORE ALL 8 BITS



2 TRANSMIT

one square = one bit · eight bits = one packet



The output stays empty: store-and-forward is still waiting for bit 8.

Core structure

Routing + forwarding

Packet switching

Performance

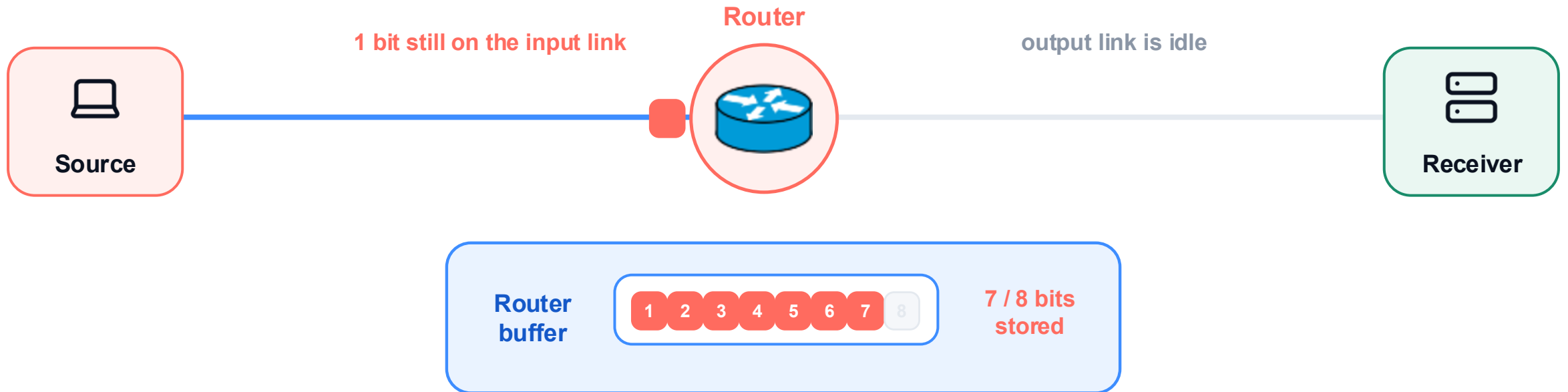
Bit 7 arrives and is stored in the router buffer

1 RECEIVE + STORE ALL 8 BITS



2 TRANSMIT

one square = one bit · eight bits = one packet



The output stays empty: store-and-forward is still waiting for bit 8.

Core structure

Routing + forwarding

Packet switching

Performance

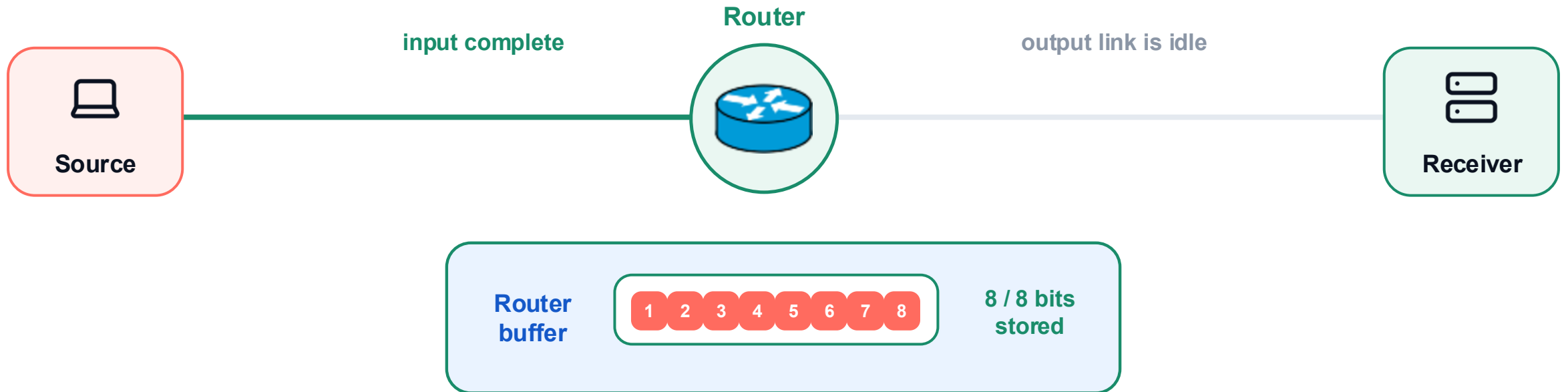
All 8 bits are buffered; transmission may now begin

1 RECEIVE + STORE ALL 8 BITS



2 TRANSMIT

one square = one bit · eight bits = one packet



Only after bit 8 arrives may the router start transmitting on the next link.

Core structure

Routing + forwarding

Packet switching

Performance

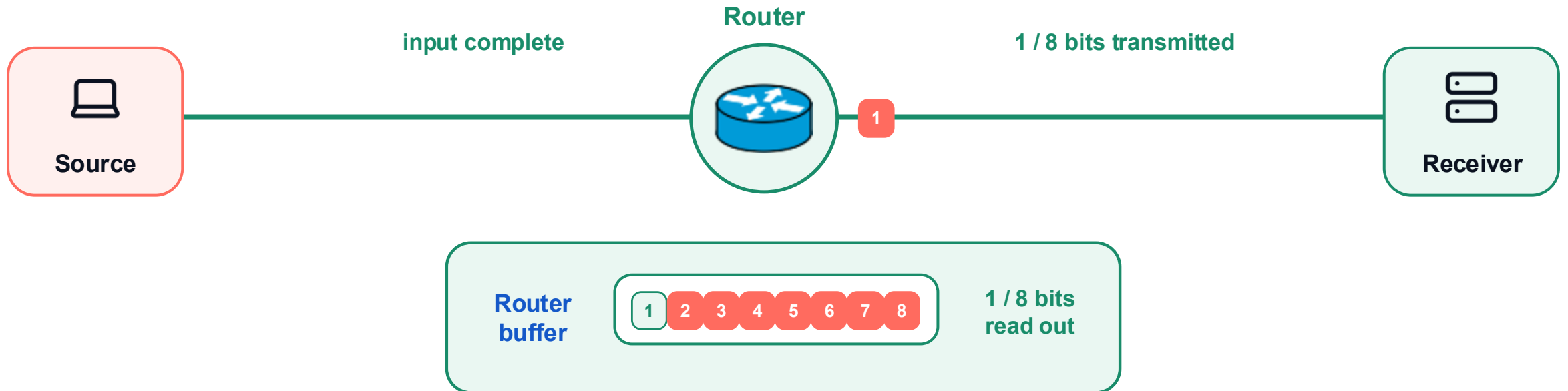
The router transmits bit 1 onto the next link

1 RECEIVE + STORE ALL 8 BITS



2 TRANSMIT

one square = one bit · eight bits = one packet



The complete packet was stored first; now the router sends one bit at a time.

Core structure

Routing + forwarding

Packet switching

Performance

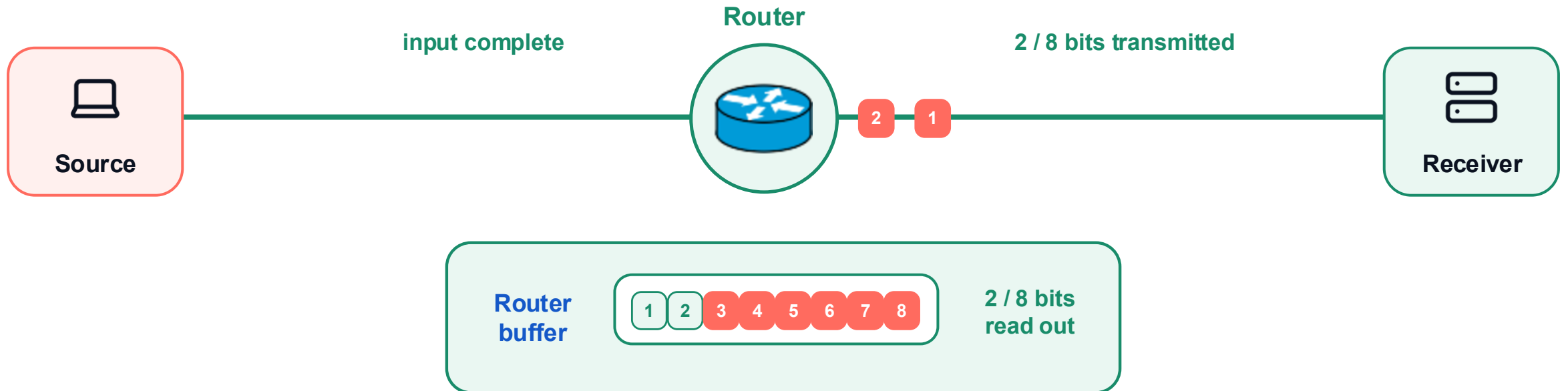
The router transmits bit 2 onto the next link

1 RECEIVE + STORE ALL 8 BITS



2 TRANSMIT

one square = one bit · eight bits = one packet



The complete packet was stored first; now the router sends one bit at a time.

Core structure

Routing + forwarding

Packet switching

Performance

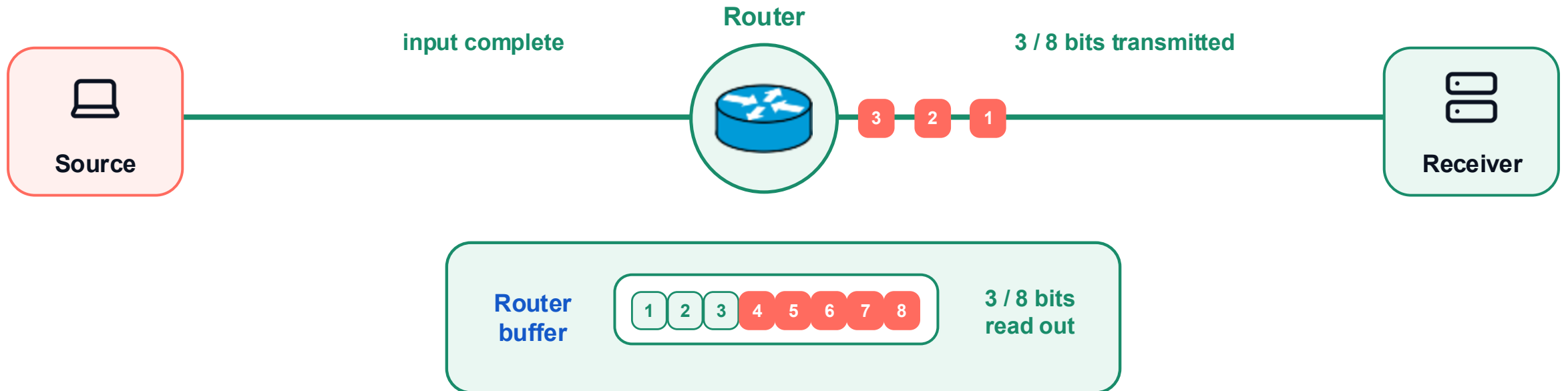
The router transmits bit 3 onto the next link

1 RECEIVE + STORE ALL 8 BITS



2 TRANSMIT

one square = one bit · eight bits = one packet



The complete packet was stored first; now the router sends one bit at a time.

Core structure

Routing + forwarding

Packet switching

Performance

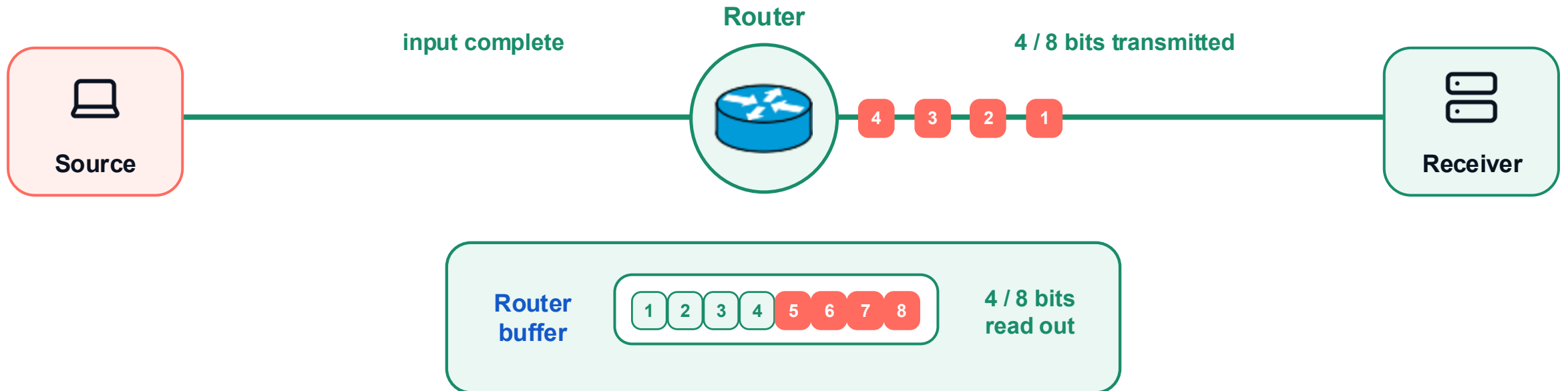
The router transmits bit 4 onto the next link

1 RECEIVE + STORE ALL 8 BITS



2 TRANSMIT

one square = one bit · eight bits = one packet



The complete packet was stored first; now the router sends one bit at a time.

Core structure

Routing + forwarding

Packet switching

Performance

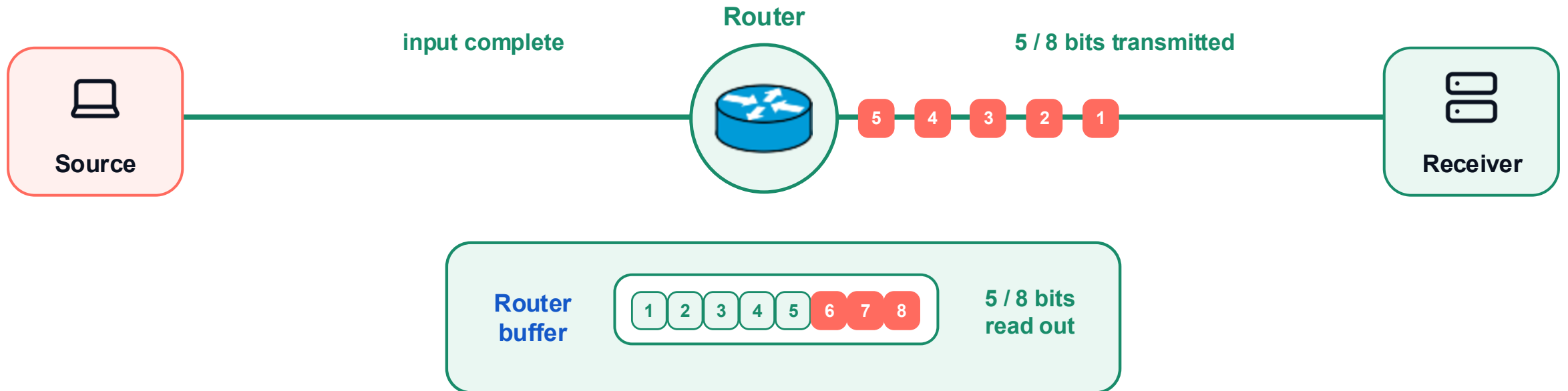
The router transmits bit 5 onto the next link

1 RECEIVE + STORE ALL 8 BITS



2 TRANSMIT

one square = one bit · eight bits = one packet



The complete packet was stored first; now the router sends one bit at a time.

Core structure

Routing + forwarding

Packet switching

Performance

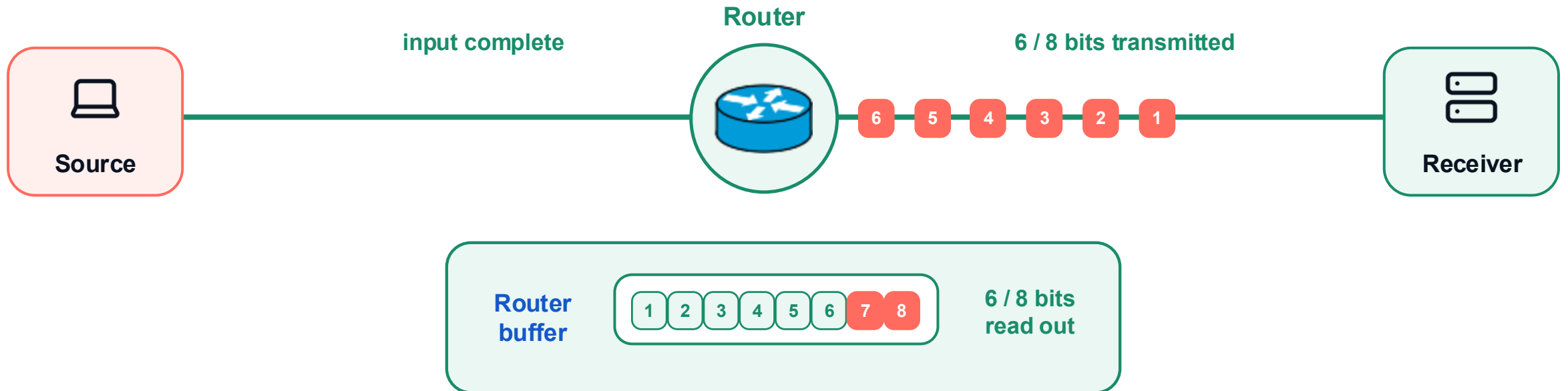
The router transmits bit 6 onto the next link

1 RECEIVE + STORE ALL 8 BITS



2 TRANSMIT

one square = one bit · eight bits = one packet



The complete packet was stored first; now the router sends one bit at a time.

Core structure

Routing + forwarding

Packet switching

Performance

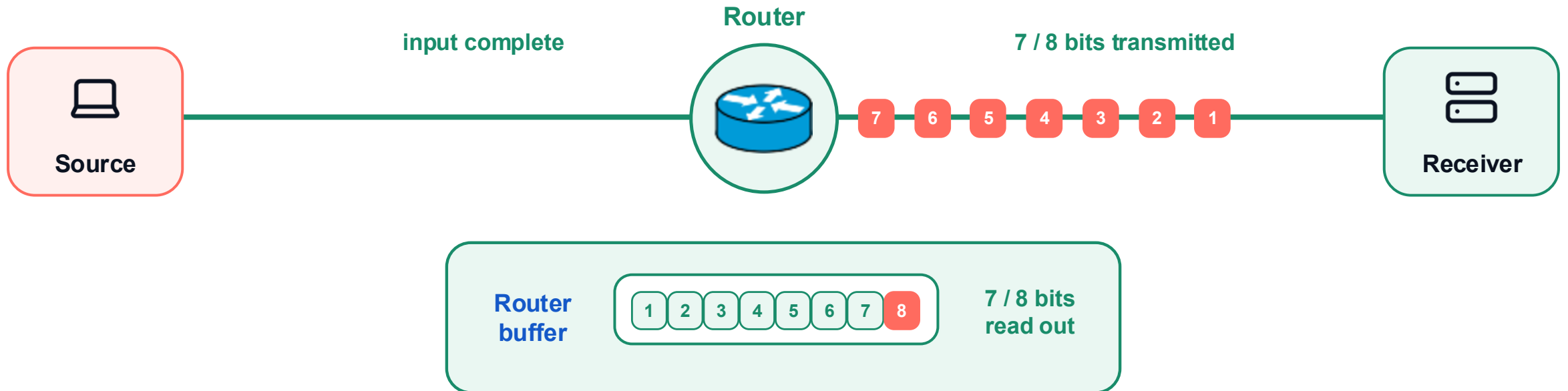
The router transmits bit 7 onto the next link

1 RECEIVE + STORE ALL 8 BITS



2 TRANSMIT

one square = one bit · eight bits = one packet



The complete packet was stored first; now the router sends one bit at a time.

Core structure

Routing + forwarding

Packet switching

Performance

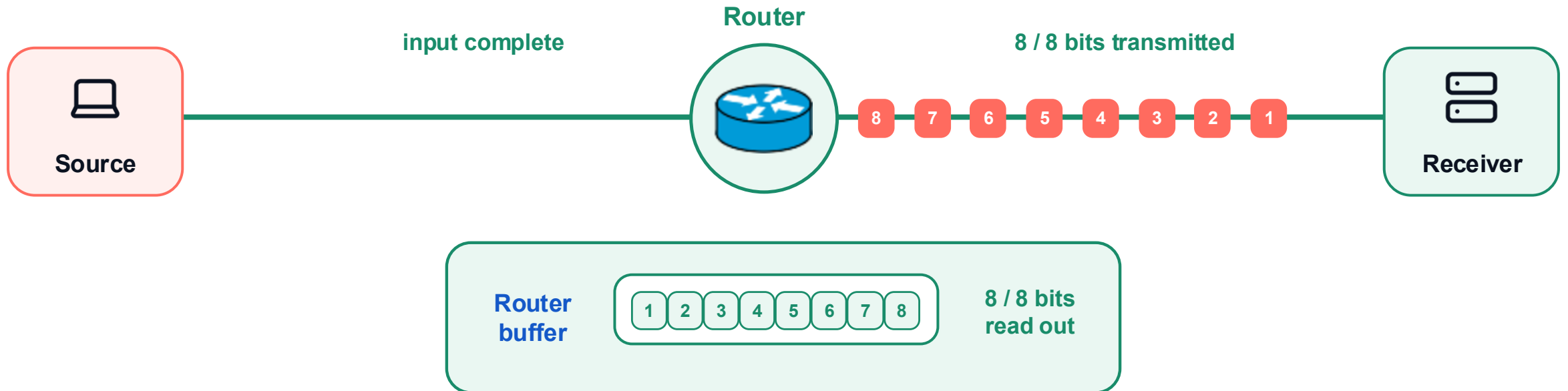
All 8 bits have been transmitted onto the next link

1 RECEIVE + STORE ALL 8 BITS



2 TRANSMIT

one square = one bit · eight bits = one packet



Store first. Then forward.

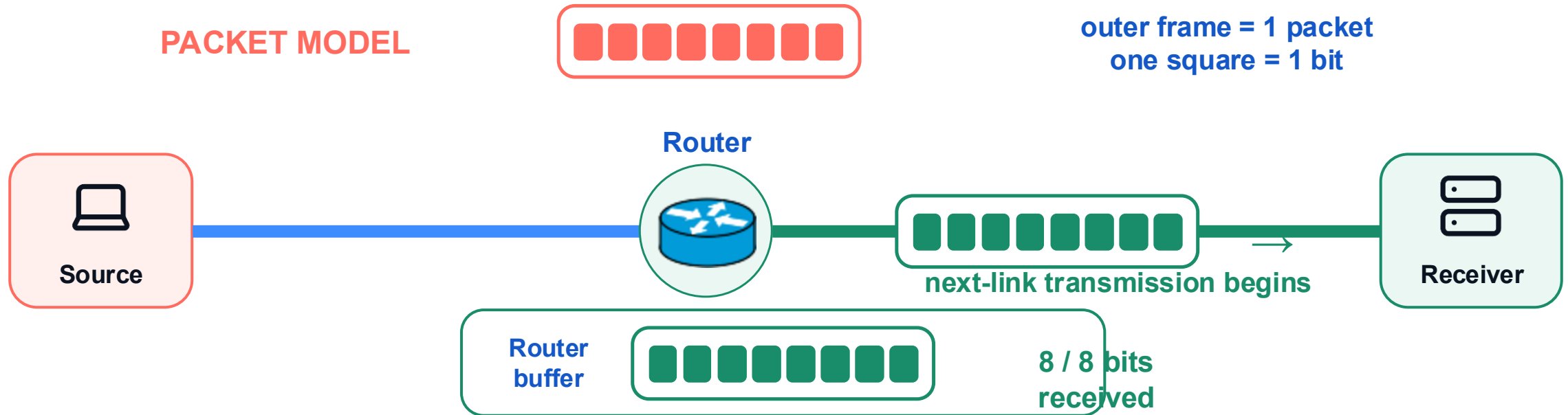
Core structure

Routing + forwarding

Packet switching

Performance

Store-and-forward waits for all 8 bits



Cut-through comparison

may start the next link before all 8 bits arrive

At 8/8 bits, the next-link transmission may begin.

Core structure

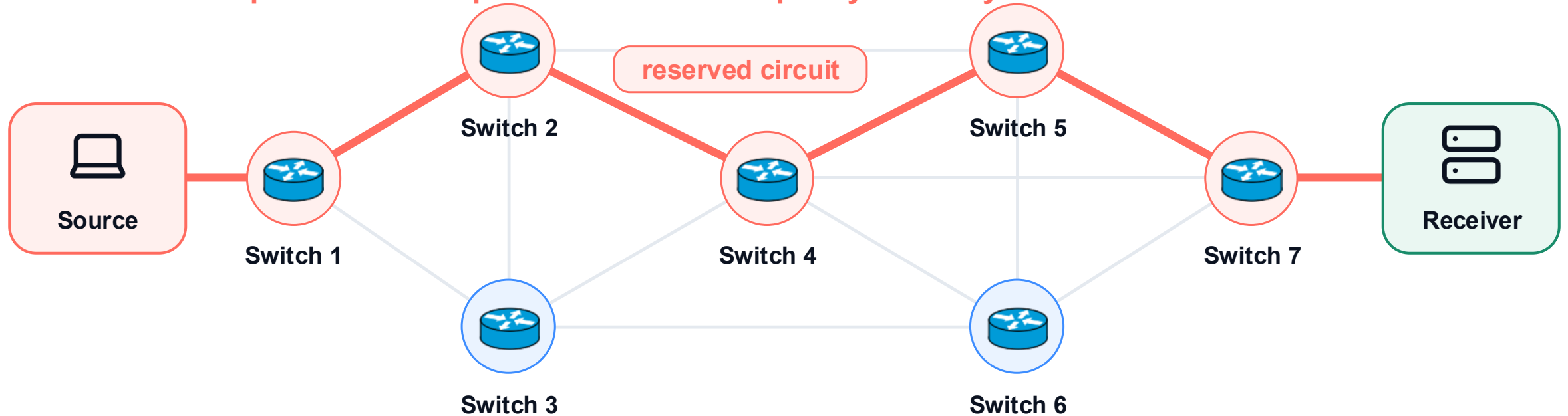
Routing + forwarding

Packet switching

Performance

Circuit switching reserves an end-to-end path

Setup chooses one path and reserves capacity on every red link before data flows.



1 set up the path → 2 reserve capacity on each link → 3 transmit

The circuit owns reserved capacity across several switches for the duration of the connection.

Core structure

Routing + forwarding

Packet switching

Performance

Packet switching shares link capacity on demand



Packet switching

- Shares link capacity packet by packet
- No setup; efficient for bursty traffic
- Queues can create variable delay and loss

Core structure

Routing + forwarding

Packet switching

Performance

Circuit switching reserves capacity before data flows



Packet switching

- Shares link capacity packet by packet
- No setup; efficient for bursty traffic
- Queues can create variable delay and loss



Circuit switching

- Reserves capacity on every link
- Requires setup before data flows
- Predictable service; idle capacity stays reserved

Core structure

Routing + forwarding

Packet switching

Performance

Packet and circuit switching make different tradeoffs



Packet switching

- Shares link capacity packet by packet
- No setup; efficient for bursty traffic
- Queues can create variable delay and loss



Circuit switching

- Reserves capacity on every link
- Requires setup before data flows
- Predictable service; idle capacity stays reserved

Packet switching favors statistical sharing; circuit switching favors reservation.

Core structure

Routing + forwarding

Packet switching

Performance

Why packet sharing is efficient



Why sharing works

- **Idle capacity can serve another user**
- Bursty users can share the same link
- No per-flow reservation is required

Core structure

Routing + forwarding

Packet switching

Performance

The same sharing can create congestion



Why sharing works

- Idle capacity can serve another user
- Bursty users can share the same link
- No per-flow reservation is required



When sharing breaks down

- Simultaneous bursts build queues
- Full buffers drop packets
- Congestion control must reduce the load

Core structure

Routing + forwarding

Packet switching

Performance

Packet switching is efficient, but congestion is real



Why sharing works

- Idle capacity can serve another user
- Bursty users can share the same link
- No per-flow reservation is required



When sharing breaks down

- Simultaneous bursts build queues
- Full buffers drop packets
- Congestion control must reduce the load

Packet switching is not a slam dunk; its strength and its risk both come from sharing.

Core structure

Routing + forwarding

Packet switching

Performance

Part 4 · Performance

04



**Follow one packet: delay,
loss, throughput.**

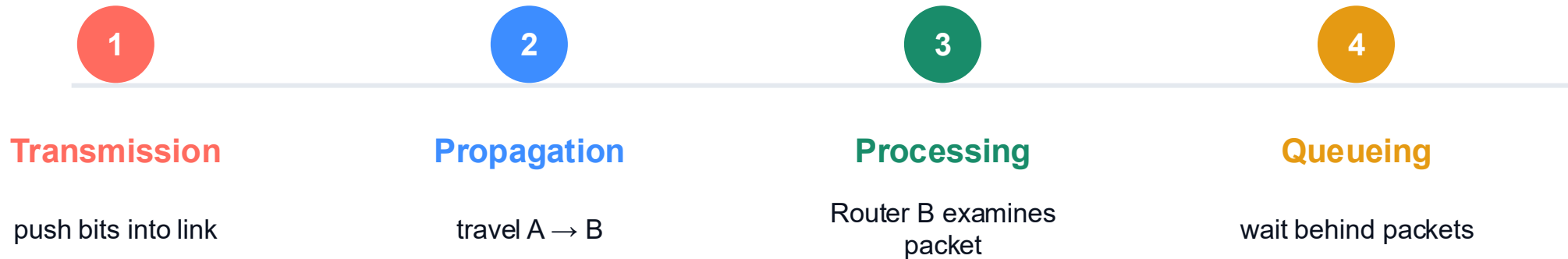
Core structure

Routing + forwarding

Packet switching

Performance

Performance follows one packet in order



Follow the packet: onto the link, across the link, through the router, then into the buffer.

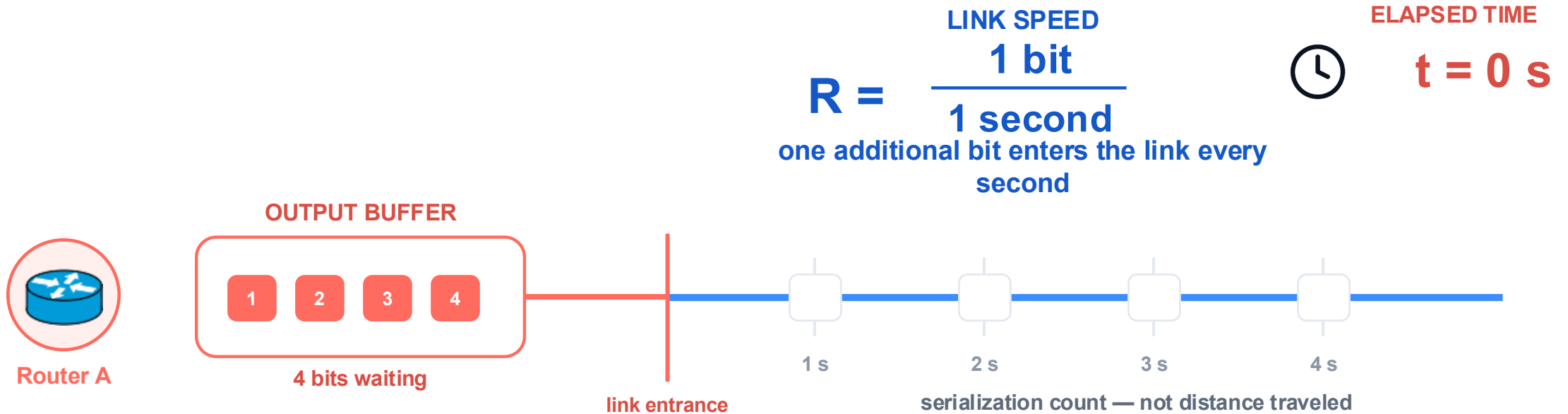
Core structure

Routing + forwarding

Packet switching

Performance

A 4-bit packet waits behind a 1 bit/s link



Transmission delay asks: how long until the last bit enters the link?

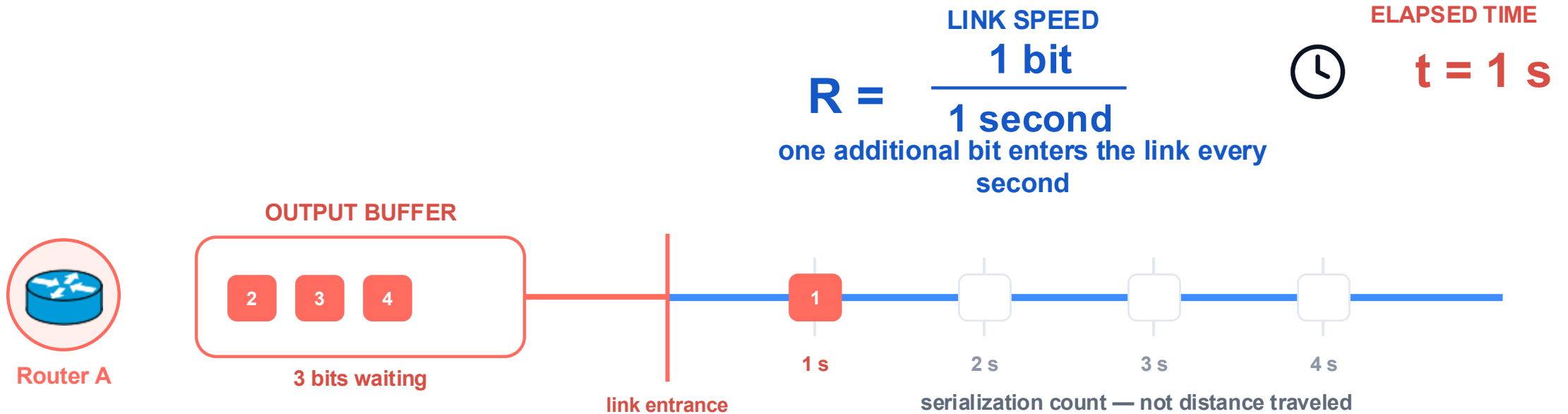
Core structure

Routing + forwarding

Packet switching

Performance

After 1 s, exactly one bit has entered the link



At 1 bit/s, elapsed seconds and transmitted bits advance together.

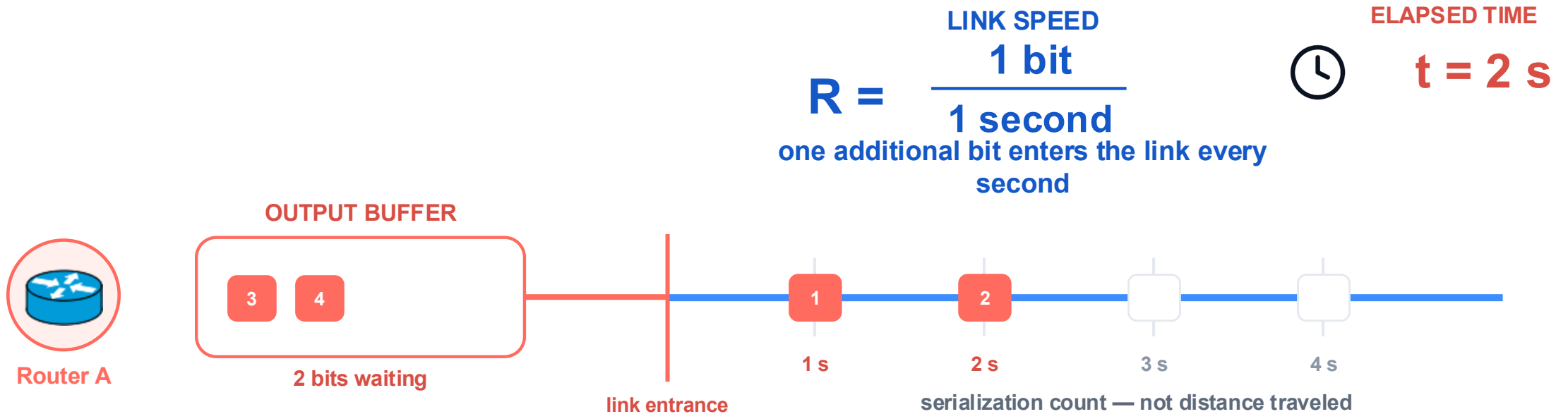
Core structure

Routing + forwarding

Packet switching

Performance

After 2 s, exactly two bits have entered the link



Two seconds elapsed → two of four bits have entered the link.

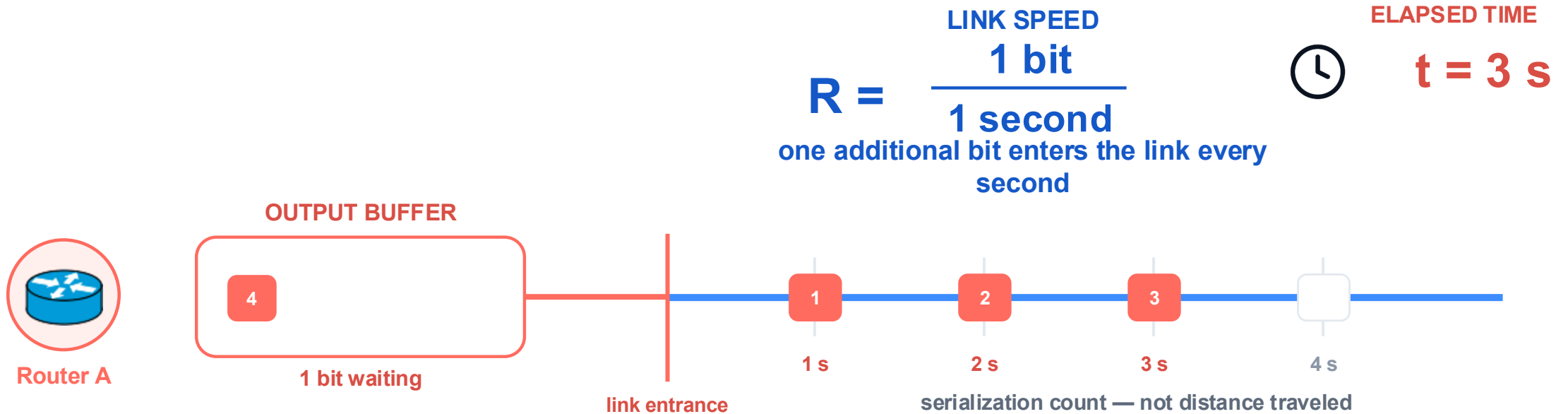
Core structure

Routing + forwarding

Packet switching

Performance

After 3 s, one bit remains in Router A's buffer



Three seconds elapsed → only the final bit is still waiting.

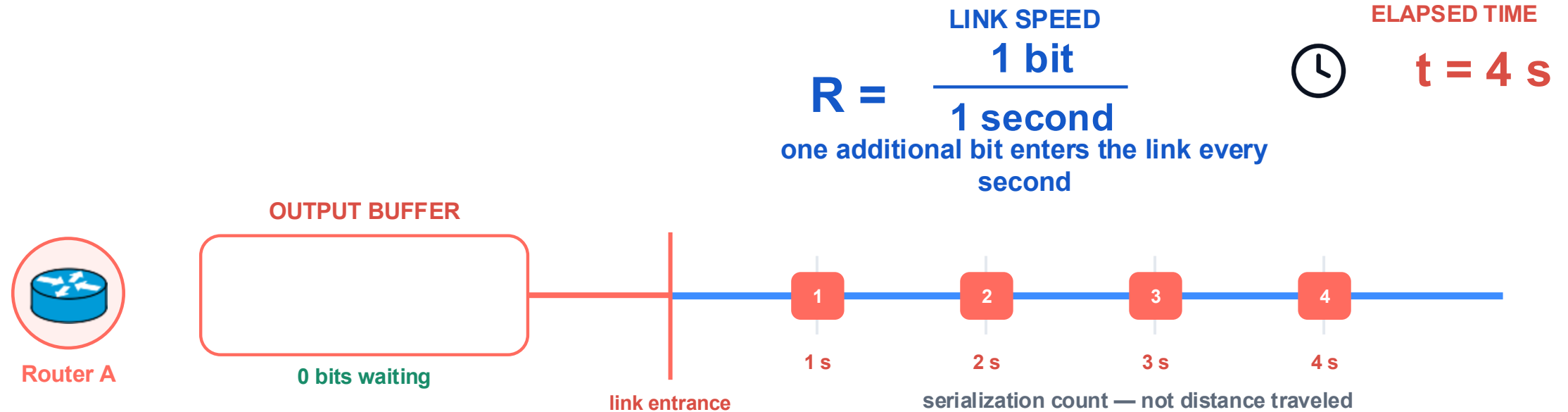
Core structure

Routing + forwarding

Packet switching

Performance

After 4 s, the last bit enters the link



$$d_{\text{trans}} = \frac{L}{R} = \frac{4 \text{ bits}}{1 \text{ bit/s}} = 4 \text{ s}$$

time until the last bit enters the link

Core structure

Routing + forwarding

Packet switching

Performance

A bit crosses 3,000 km of fiber at about 200,000 km/s

SIGNAL SPEED IN FIBER

$$s \approx 2 \times 10^8 \text{ m/s} \approx 200,000 \text{ km/s}$$

about two-thirds of the vacuum speed of light

ELAPSED TIME



t = 0 ms



Router A

1

0 km

1,500 km

3,000 km



Router B

distance $d = 3,000 \text{ km}$

Propagation follows distance and signal speed—not packet length.

Core structure

Routing + forwarding

Packet switching

Performance

After 7.5 ms, the bit is halfway to Router B

SIGNAL SPEED IN FIBER

$$s \approx 2 \times 10^8 \text{ m/s} \approx 200,000 \text{ km/s}$$

about two-thirds of the vacuum speed of light

ELAPSED TIME

🕒 $t = 7.5 \text{ ms}$



Half the distance takes half the propagation time.

Core structure

Routing + forwarding

Packet switching

Performance

After 15 ms, the bit reaches Router B

SIGNAL SPEED IN FIBER

$$s \approx 2 \times 10^8 \text{ m/s} \approx 200,000 \text{ km/s}$$

about two-thirds of the vacuum speed of light

ELAPSED TIME

🕒 $t = 15 \text{ ms}$



$$d_{\text{prop}} = \frac{d}{s} = \frac{3,000 \text{ km}}{200,000 \text{ km/s}} = 0.015 \text{ s} = 15 \text{ ms}$$

Core structure

Routing + forwarding

Packet switching

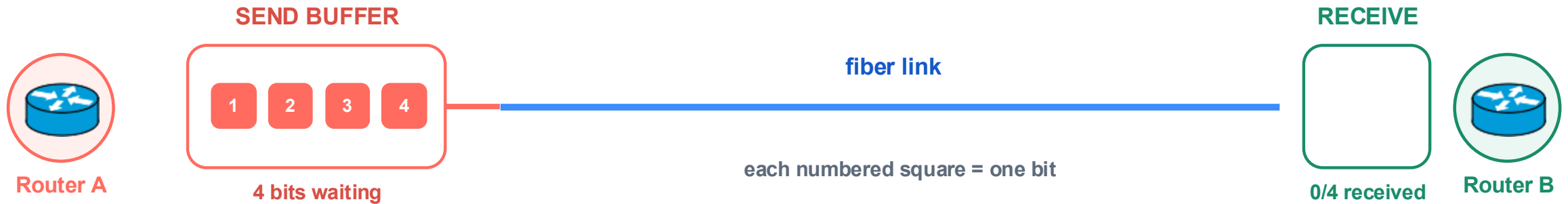
Performance

• — One-hop delay

$t = 0$ ms: four bits wait in Router A's buffer

4 bits • link rate: 1 bit/ms • 3,000 km fiber

$t = 0$ ms



Four numbered bits are ready to be transmitted.

Core structure

Routing + forwarding

Packet switching

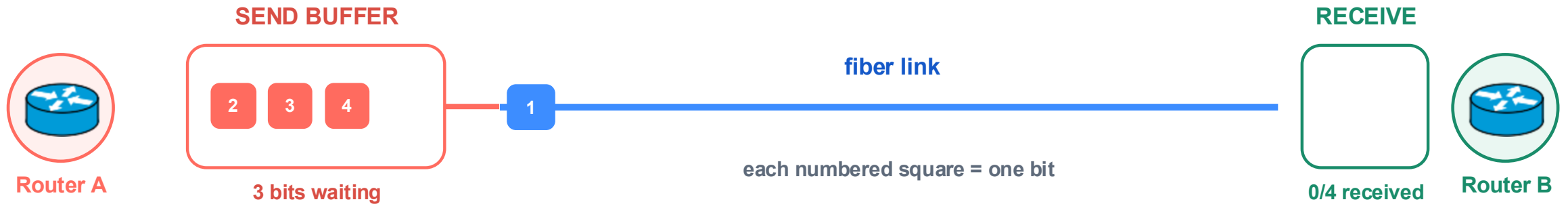
Performance

- — One-hop delay

t = 1 ms: bit 1 enters the link

4 bits • link rate: 1 bit/ms • 3,000 km fiber

t = 1 ms



At 1 bit/ms, exactly one bit enters the link each millisecond.

Core structure

Routing + forwarding

Packet switching

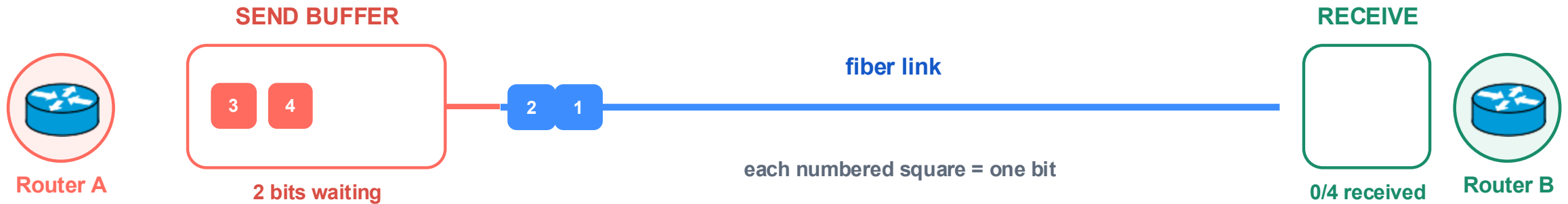
Performance

- — One-hop delay

t = 2 ms: bit 2 enters the link

4 bits • link rate: 1 bit/ms • 3,000 km fiber

t = 2 ms



Two bits are on the link; two bits still wait in the buffer.

Core structure

Routing + forwarding

Packet switching

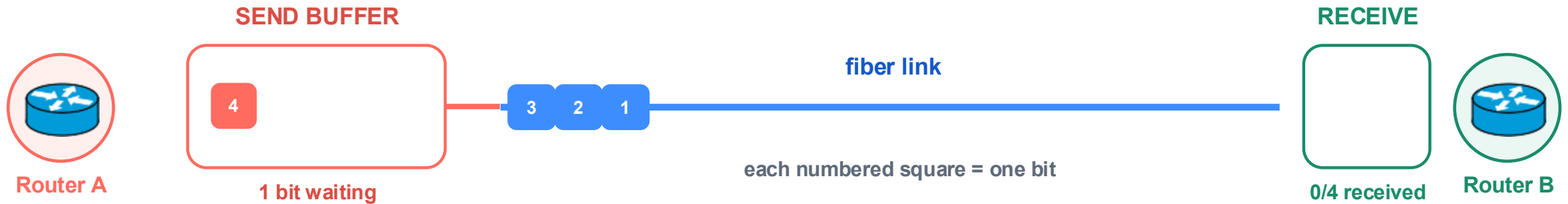
Performance

- — One-hop delay

t = 3 ms: bit 3 enters the link

4 bits • link rate: 1 bit/ms • 3,000 km fiber

t = 3 ms



Three bits are on the link; only bit 4 remains in the buffer.

Core structure

Routing + forwarding

Packet switching

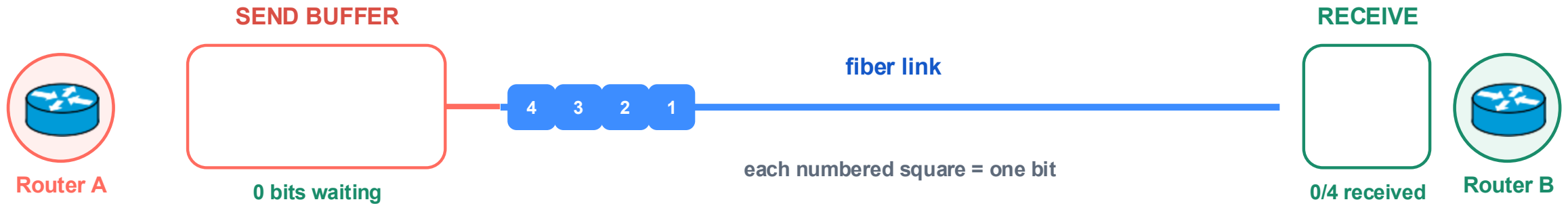
Performance

- — One-hop delay

t = 4 ms: all four bits are on the link

4 bits • link rate: 1 bit/ms • 3,000 km fiber

t = 4 ms



All four bits are on the link—and still far from Router B.

Core structure

Routing + forwarding

Packet switching

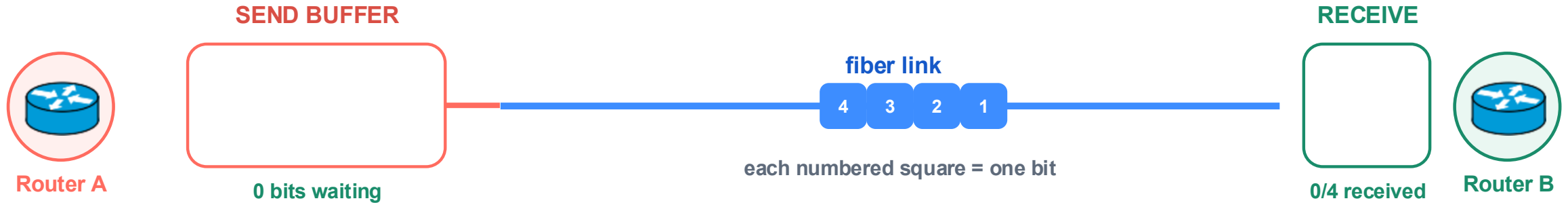
Performance

● — One-hop delay

$t = 10 \text{ ms}$: the four-bit train crosses the link

4 bits • link rate: 1 bit/ms • 3,000 km fiber

$t = 10 \text{ ms}$



The four bits propagate together across the empty distance.

Core structure

Routing + forwarding

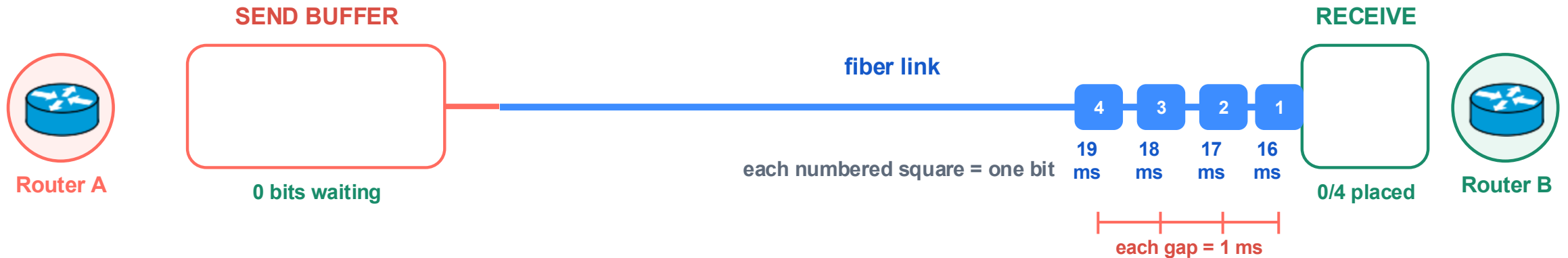
Packet switching

Performance

At Router B, bits arrive 1 ms apart

4 bits • link rate: 1 bit/ms • 3,000 km fiber

$t = 16\text{--}19\text{ ms}$



Adjacent completions: $1/R = 1\text{ ms}$

First → last completed bit: $3 \times 1\text{ ms} = 3\text{ ms}$

Receive all 4 bits: $L/R = 4/1 = 4\text{ ms}$ (15–19 ms)

Core structure

Routing + forwarding

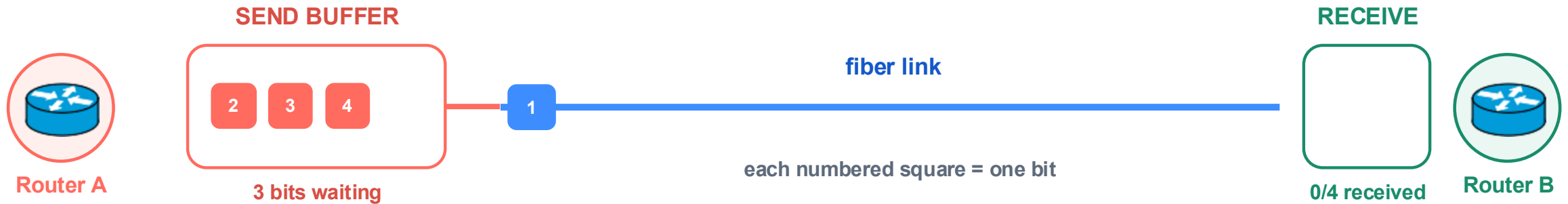
Packet switching

Performance

First, bit 1 takes 1 ms to enter the link

4 bits • link rate: 1 bit/ms • 3,000 km fiber

t = 1 ms



At t = 1 ms, the complete bit is on the fiber—and it keeps moving.

Core structure

Routing + forwarding

Packet switching

Performance

At $t = 1$ ms, bit 1 starts propagating

SIGNAL SPEED IN FIBER

$$s \approx 2 \times 10^8 \text{ m/s} \approx 200,000 \text{ km/s}$$

after entering the link, bit 1 keeps moving without a pause

COMPLETE BIT ARRIVAL

🕒 $t = 1 \text{ ms}$



$$d_{\text{prop}} = \frac{d}{s} = \frac{3,000 \text{ km}}{200,000 \text{ km/s}} = 0.015 \text{ s} = 15 \text{ ms}$$

Core structure

Routing + forwarding

Packet switching

Performance

At $t = 6$ ms, bit 1 has crossed 1,000 km

SIGNAL SPEED IN FIBER

$$s \approx 2 \times 10^8 \text{ m/s} \approx 200,000 \text{ km/s}$$

after entering the link, bit 1 keeps moving without a pause

COMPLETE BIT ARRIVAL

🕒 $t = 6 \text{ ms}$



$$d_{\text{prop}} = \frac{d}{s} = \frac{3,000 \text{ km}}{200,000 \text{ km/s}} = 0.015 \text{ s} = 15 \text{ ms}$$

Core structure

Routing + forwarding

Packet switching

Performance

At $t = 11$ ms, bit 1 has crossed 2,000 km

SIGNAL SPEED IN FIBER

$$s \approx 2 \times 10^8 \text{ m/s} \approx 200,000 \text{ km/s}$$

after entering the link, bit 1 keeps moving without a pause

COMPLETE BIT ARRIVAL

🕒 $t = 11 \text{ ms}$



$$d_{\text{prop}} = \frac{d}{s} = \frac{3,000 \text{ km}}{200,000 \text{ km/s}} = 0.015 \text{ s} = 15 \text{ ms}$$

Core structure

Routing + forwarding

Packet switching

Performance

• — One-hop delay

1 ms to enter + 15 ms to propagate = 16 ms

SIGNAL SPEED IN FIBER

$$s \approx 2 \times 10^8 \text{ m/s} \approx 200,000 \text{ km/s}$$

after entering the link, bit 1 keeps moving without a pause

COMPLETE BIT ARRIVAL

🕒 $t = 16 \text{ ms}$



$$d_{\text{prop}} = \frac{d}{s} = \frac{3,000 \text{ km}}{200,000 \text{ km/s}} = 0.015 \text{ s} = 15 \text{ ms}$$

Core structure

Routing + forwarding

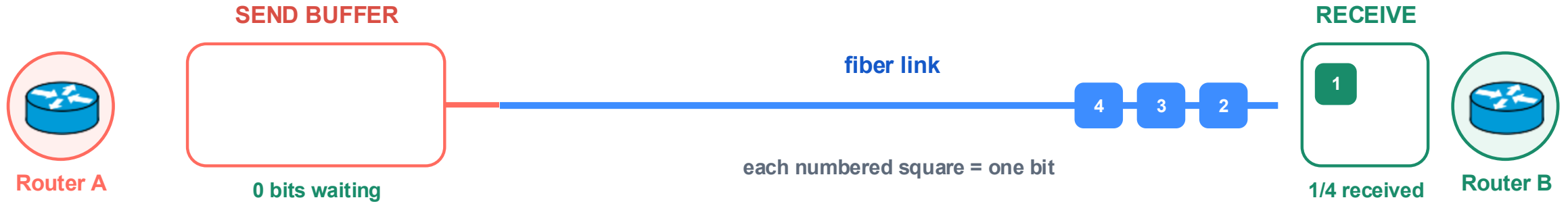
Packet switching

Performance

At $t = 16$ ms, bit 1 enters Router B's buffer

4 bits • link rate: 1 bit/ms • 3,000 km fiber

$t = 16$ ms



The end of ms 16 is the start of ms 17—the first complete bit is now buffered.

Core structure

Routing + forwarding

Packet switching

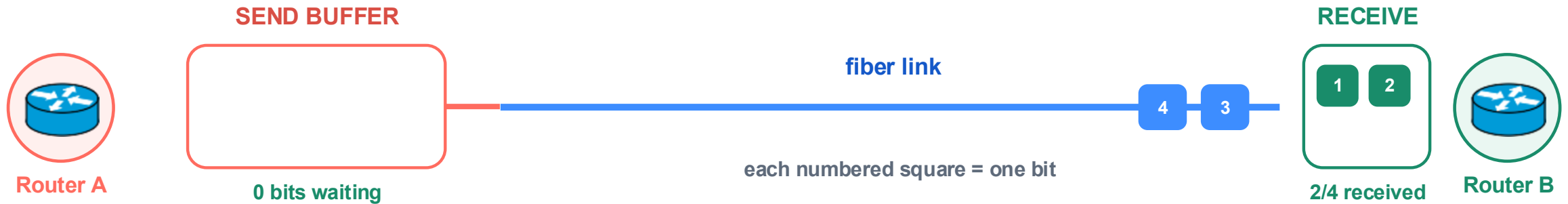
Performance

• — One-hop delay

$t = 17 \text{ ms}$: Router B receives bit 2

4 bits • link rate: 1 bit/ms • 3,000 km fiber

$t = 17 \text{ ms}$



Router B now has two bits; two bits are still on the link.

Core structure

Routing + forwarding

Packet switching

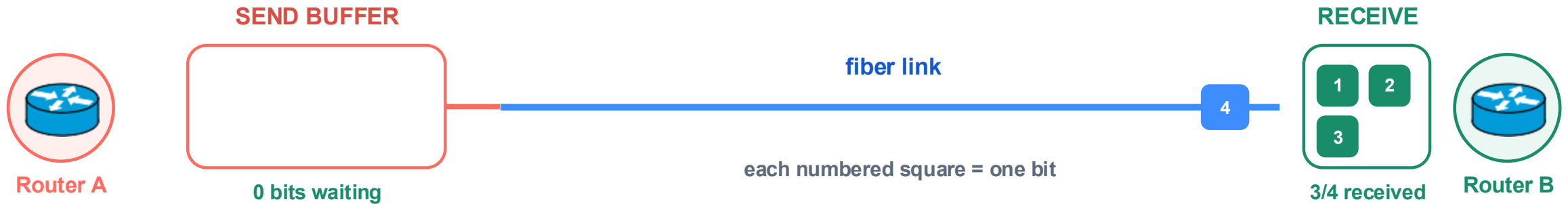
Performance

• — One-hop delay

$t = 18 \text{ ms}$: Router B receives bit 3

4 bits • link rate: 1 bit/ms • 3,000 km fiber

$t = 18 \text{ ms}$



Only the last bit remains in flight.

Core structure

Routing + forwarding

Packet switching

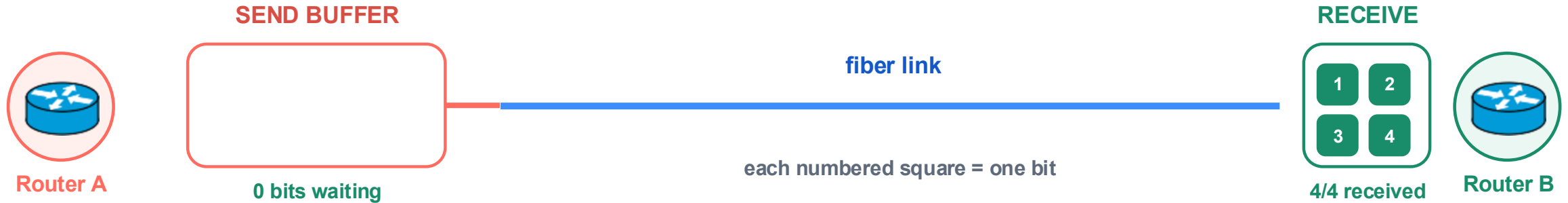
Performance

• — One-hop delay

$t = 19 \text{ ms}$: Router B receives the last bit

4 bits • link rate: 1 bit/ms • 3,000 km fiber

$t = 19 \text{ ms}$



transmission delay + propagation delay = total completion time

$$4 \text{ ms} + 15 \text{ ms} = 19 \text{ ms}$$

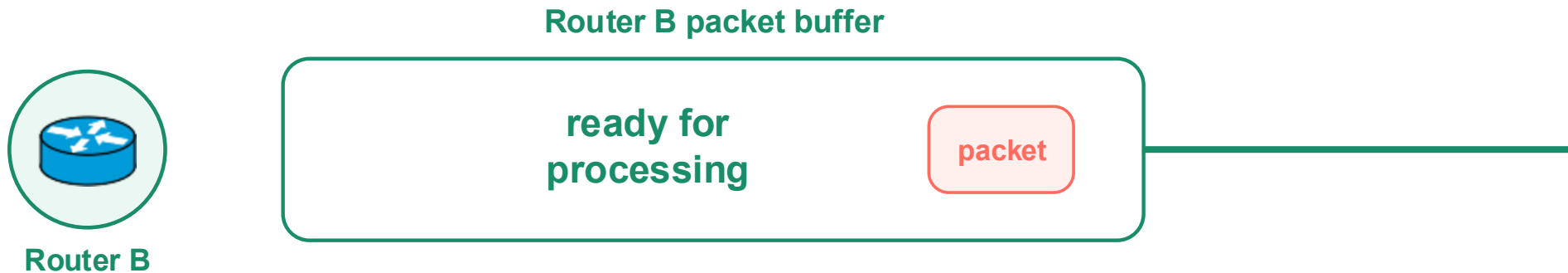
Core structure

Routing + forwarding

Packet switching

Performance

The complete packet is now stored in Router B's buffer

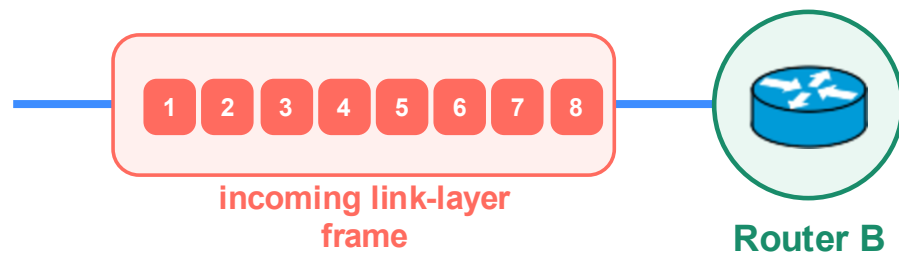


What work happens
before forwarding?

Now follow this packet through Router B's forwarding path.



1 · Verify and decapsulate the incoming frame



Check frame integrity (for example, CRC)

Remove the old link-layer header + trailer

LINK-LAYER INPUT WORK

Router B exposes the IP packet before making its forwarding decision.

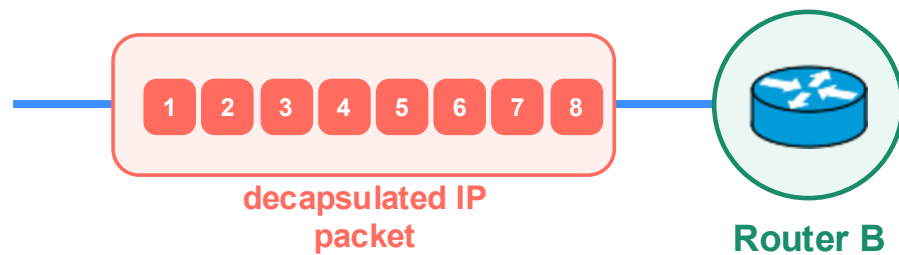
Core structure

Routing + forwarding

Packet switching

Performance

2 · Read and update the network-layer header



Read destination; validate the IP header

Decrement TTL / Hop Limit; update fields

NETWORK-LAYER HEADER WORK

Header fields guide delivery and prevent packets from circulating forever.

Core structure

Routing + forwarding

Packet switching

Performance

3 · Use the forwarding table to choose an output



FORWARDING LOOKUP

The forwarding table maps a destination prefix to the router's next action.

Core structure

Routing + forwarding

Packet switching

Performance

4 · Re-encapsulate for the next link



Resolve the next-hop link address

Add a new link-layer header + trailer

OUTPUT-LINK ENCAPSULATION

Link-layer framing changes hop by hop; the IP packet continues onward.

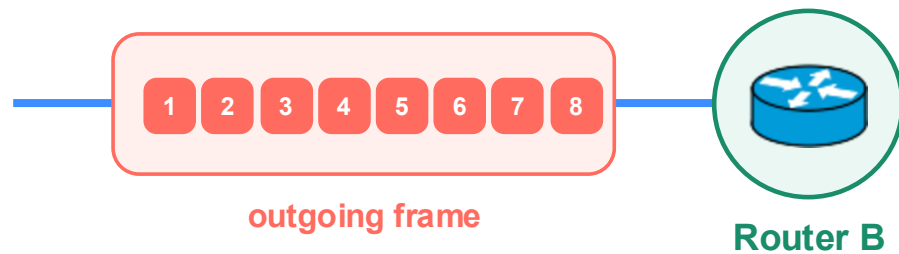
Core structure

Routing + forwarding

Packet switching

Performance

5 · Hand the frame to the selected output queue



Place the frame in the output queue

The scheduler decides when
transmission starts

PROCESSING → QUEUEING BOUNDARY

Processing ends here; any wait that follows is queueing delay.

Core structure

Routing + forwarding

Packet switching

Performance

Can every arriving packet start service immediately?



What if Router B or its output link is already busy?

The single-packet story assumed no contention.

Core structure

Routing + forwarding

Packet switching

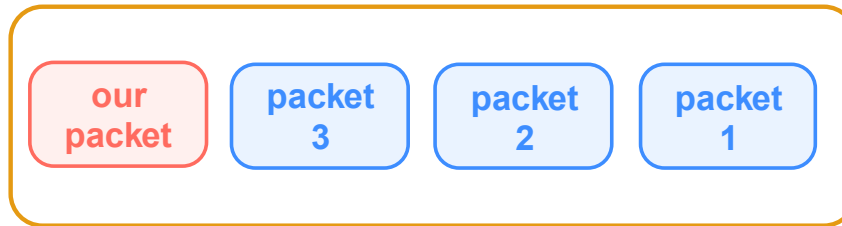
Performance

No—the packet may find an existing output queue



Router B

FIFO output queue (simple model)



waits behind 3 packets

Queue = packets waiting for service

In this simple FIFO story, earlier arrivals leave first.

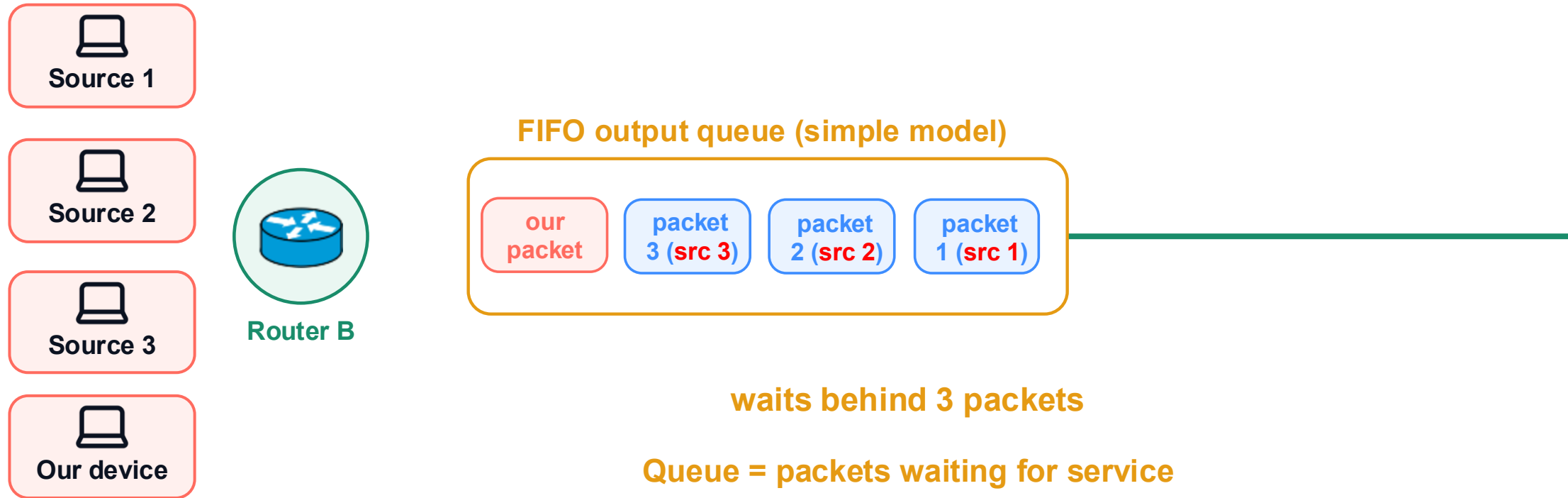
Core structure

Routing + forwarding

Packet switching

Performance

No—the packet may find an existing output queue



In this simple FIFO story, earlier arrivals leave first.

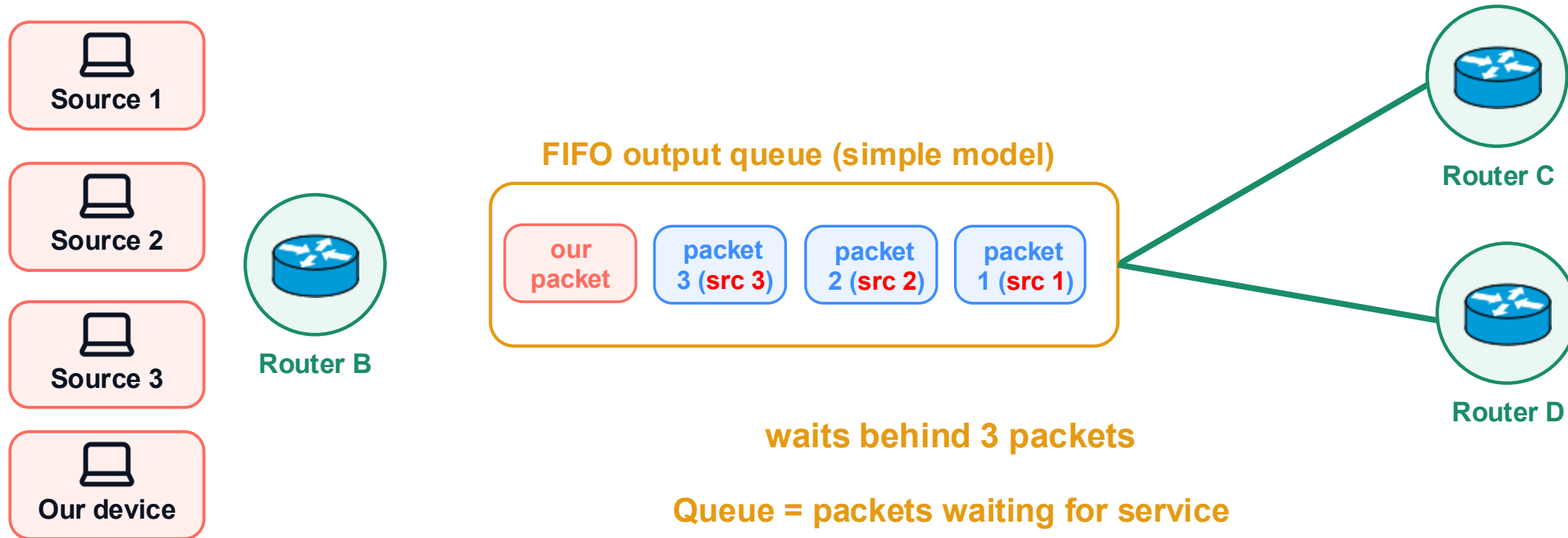
Core structure

Routing + forwarding

Packet switching

Performance

No—the packet may find an existing output queue



In this simple FIFO story, earlier arrivals leave first.

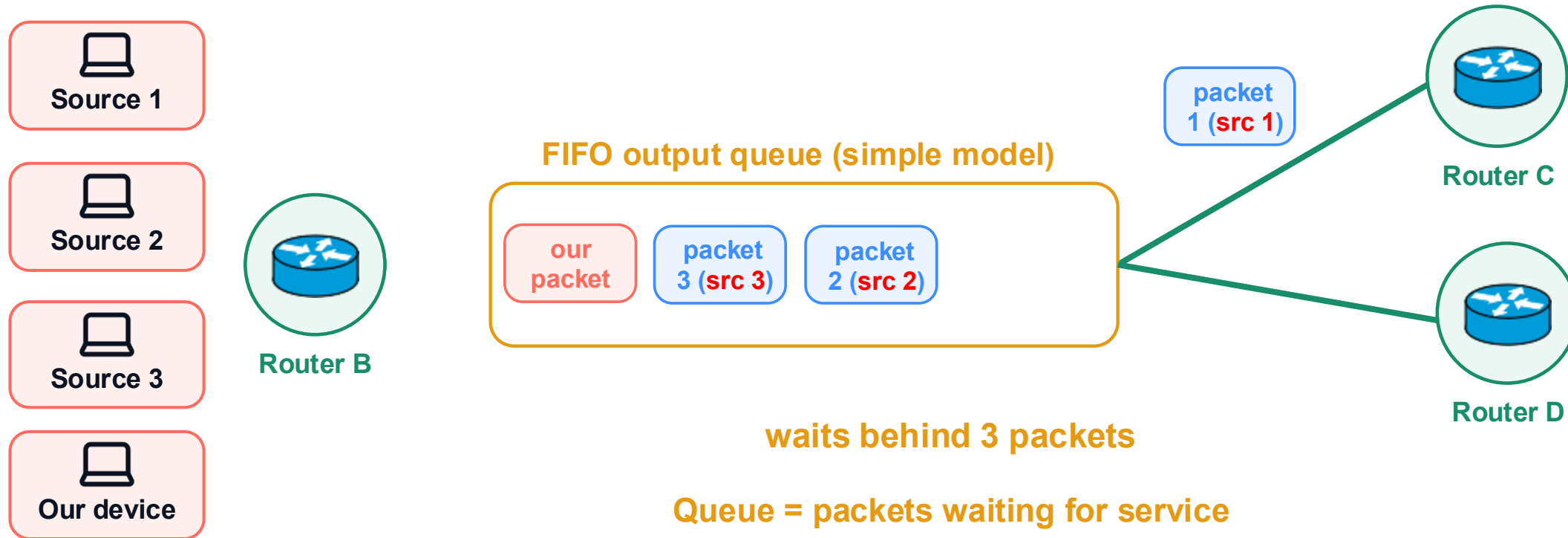
Core structure

Routing + forwarding

Packet switching

Performance

No—the packet may find an existing output queue



In this simple FIFO story, earlier arrivals leave first.

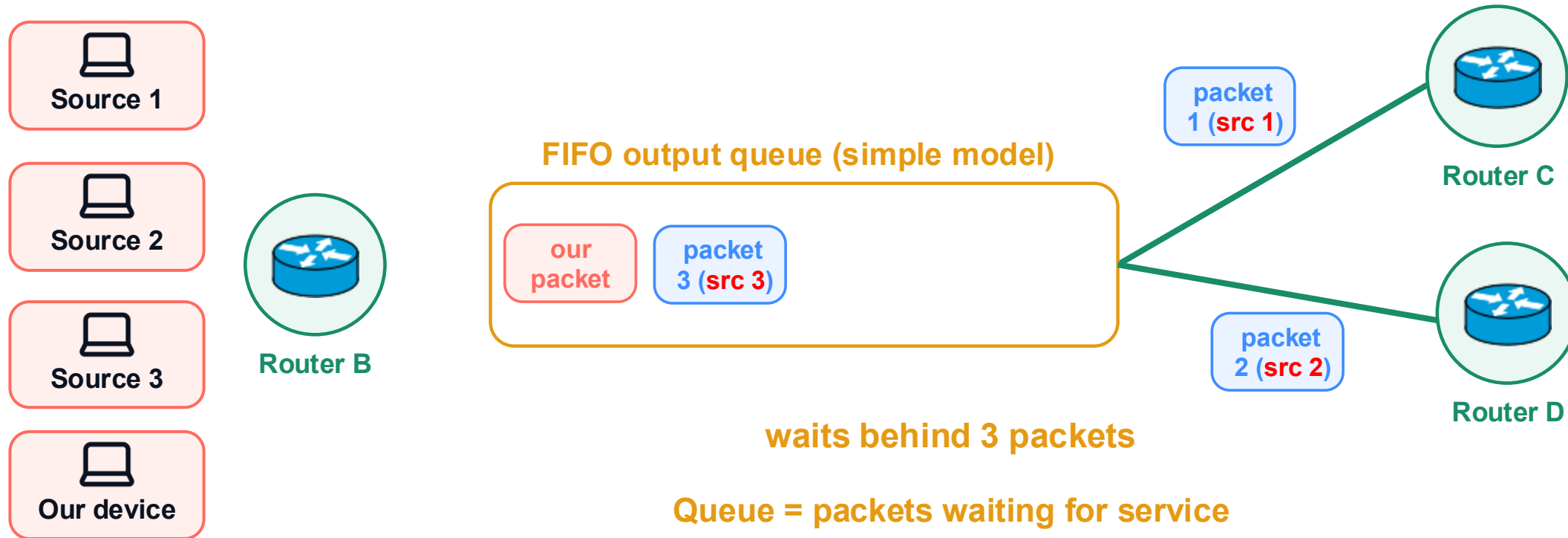
Core structure

Routing + forwarding

Packet switching

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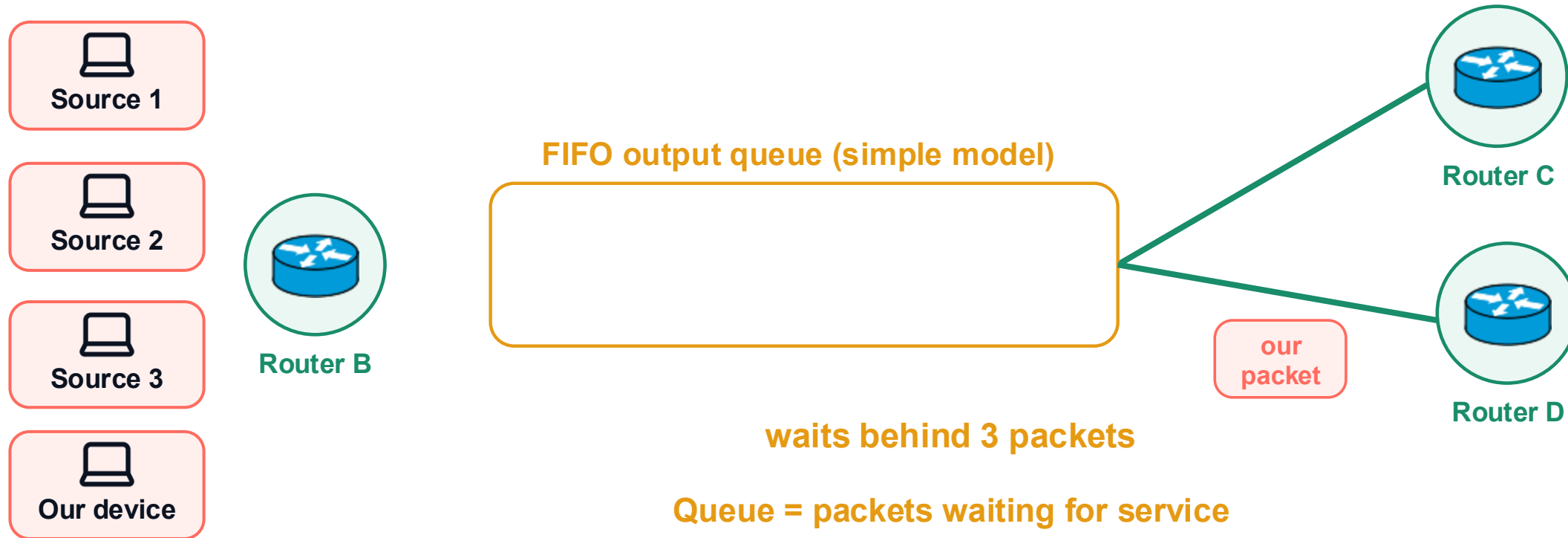
Core structure

Routing + forwarding

Packet switching

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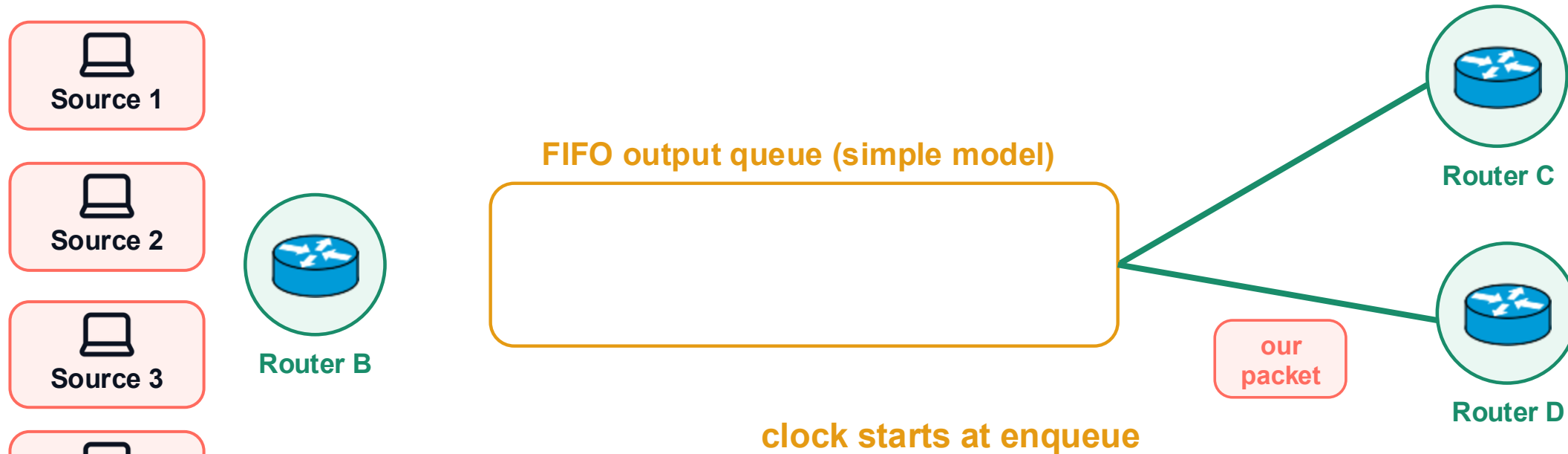
Core structure

Routing + forwarding

Packet switching

Performance

No—the packet may find an existing output queue



$$d_{\text{queue}} = t_{\text{dequeue}} - t_{\text{enqueue}}$$

It ends when the scheduler selects the packet for transmission.

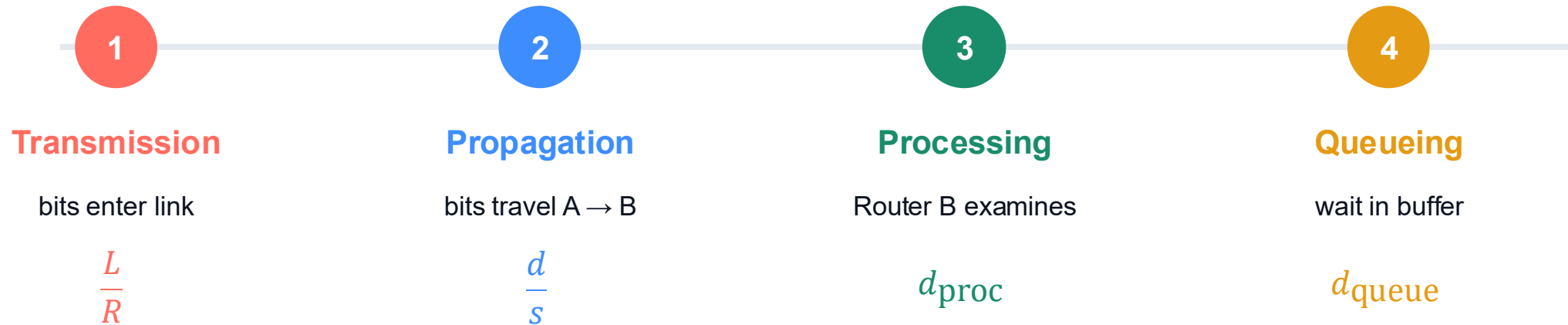
Core structure

Routing + forwarding

Packet switching

Performance

The four delays occur in the packet's travel order



One packet, one journey, four places where time is spent.

Core structure

Routing + forwarding

Packet switching

Performance

One-hop delay is the sum of the four components

$$d_{\text{nodal}} = d_{\text{trans}} + d_{\text{prop}} + d_{\text{proc}} + d_{\text{queue}}$$

Faster link rate R	→	shorter transmission delay
Shorter distance d	→	shorter propagation delay
Less router work	→	shorter processing delay
Fewer packets ahead	→	shorter queueing delay

Each delay has a different physical cause—and a different way to reduce it.

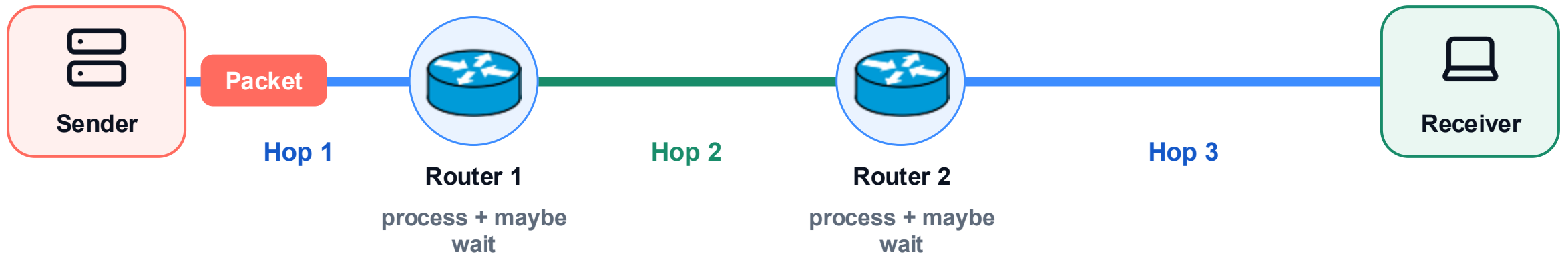
Core structure

Routing + forwarding

Packet switching

Performance

The same four steps repeat at every hop



At every router: process → queue → transmit → propagate.

Keep the route fixed and watch one packet advance one stage at a time.

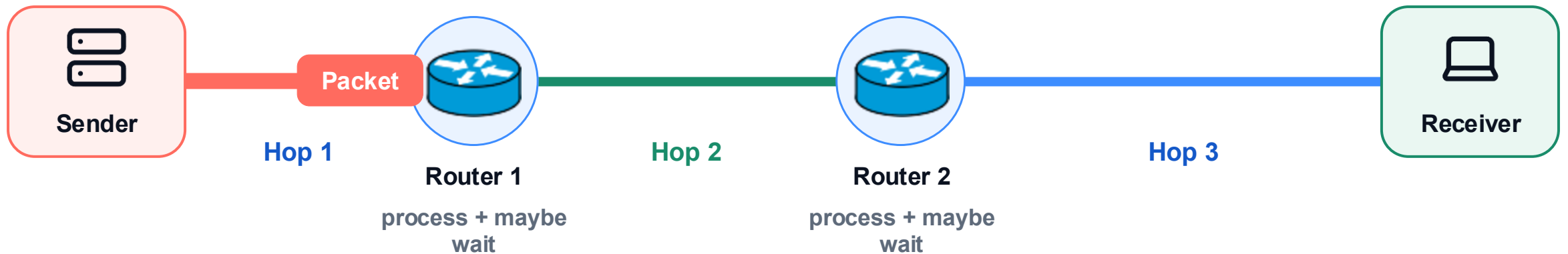
Core structure

Routing + forwarding

Packet switching

Performance

The packet crosses the first link



The packet pays transmission and propagation delay on Hop 1.

Keep the route fixed and watch one packet advance one stage at a time.

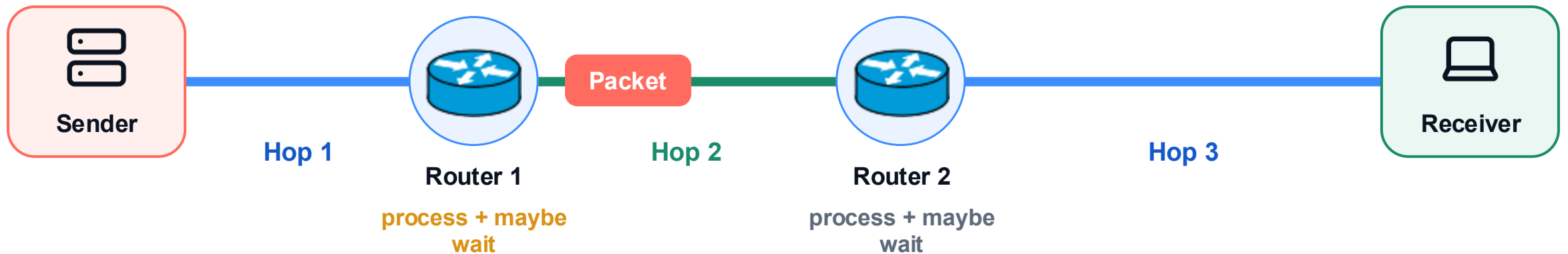
Core structure

Routing + forwarding

Packet switching

Performance

The packet may wait again at the next router



Router 1 processes the packet and may queue it again.

Keep the route fixed and watch one packet advance one stage at a time.

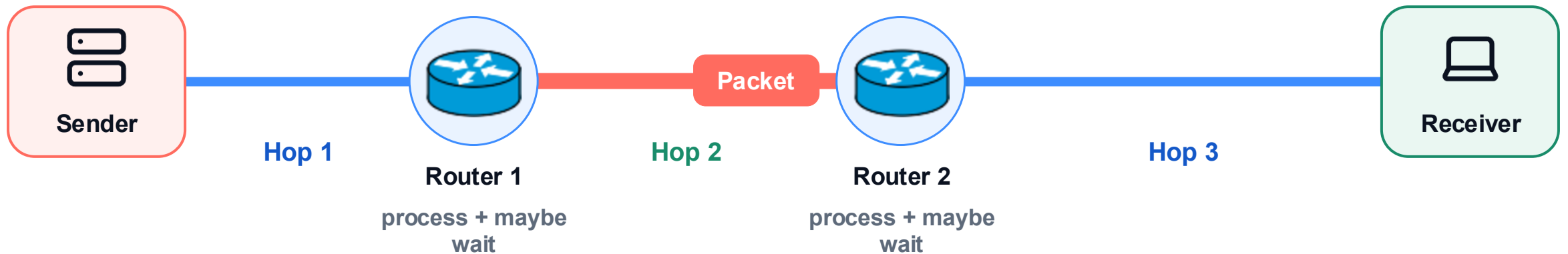
Core structure

Routing + forwarding

Packet switching

Performance

The packet crosses the next link



The same four-stage cycle begins again on Hop 2.

Keep the route fixed and watch one packet advance one stage at a time.

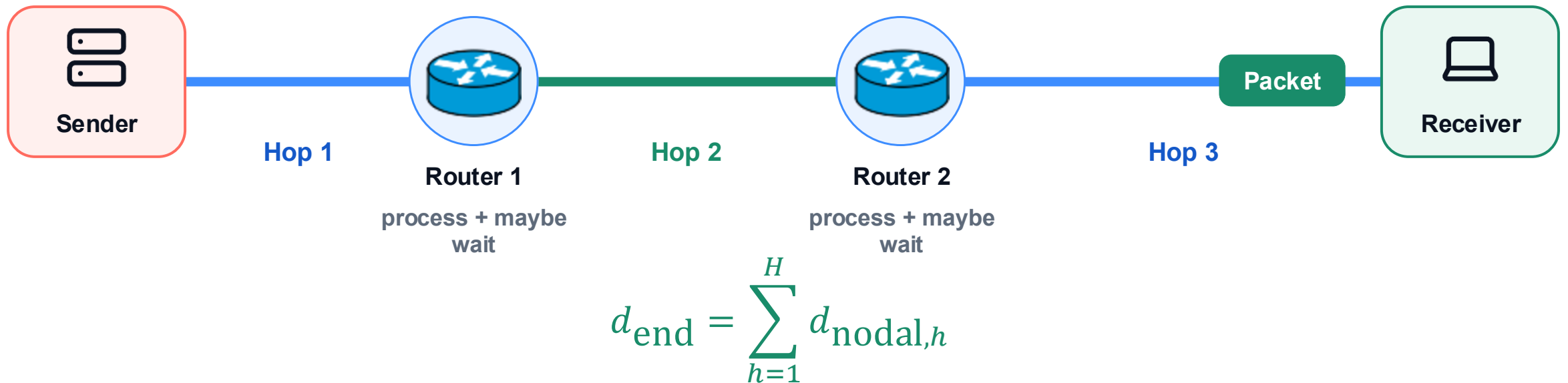
Core structure

Routing + forwarding

Packet switching

Performance

End-to-end delay adds the delay of every hop



End-to-end delay grows by repeating the same local process across the selected route.

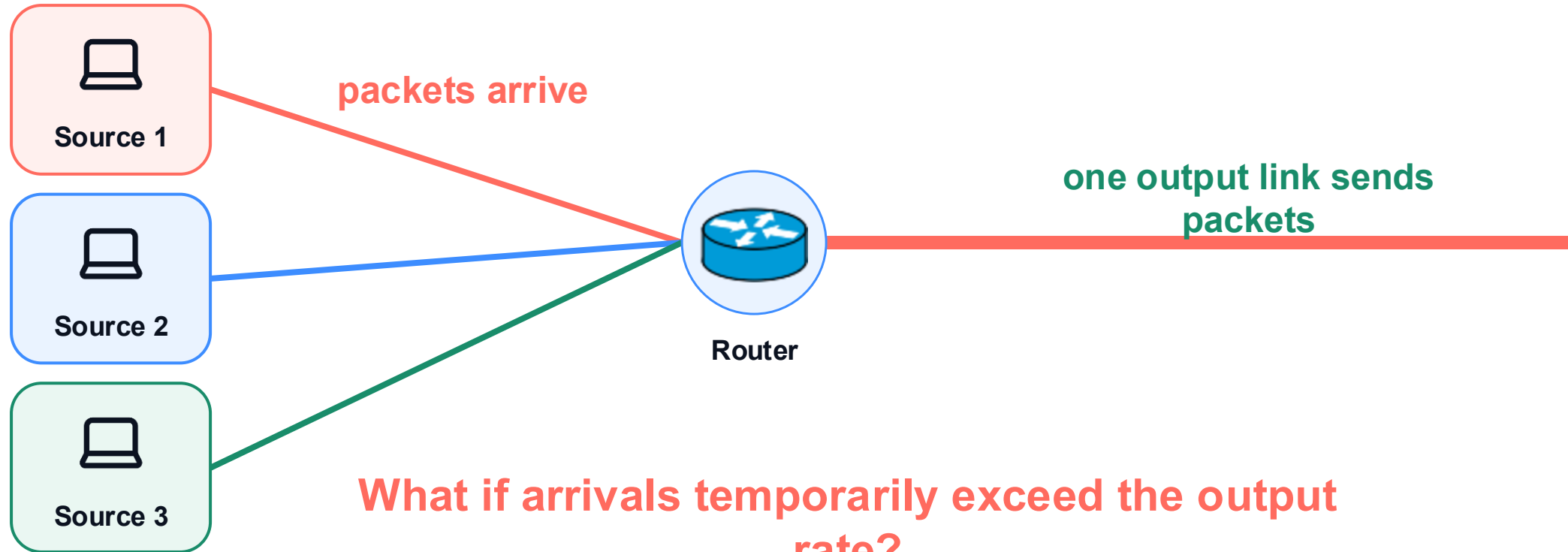
Core structure

Routing + forwarding

Packet switching

Performance

What changes when packets arrive too quickly?



Keep the output link fixed; now increase the arriving traffic.

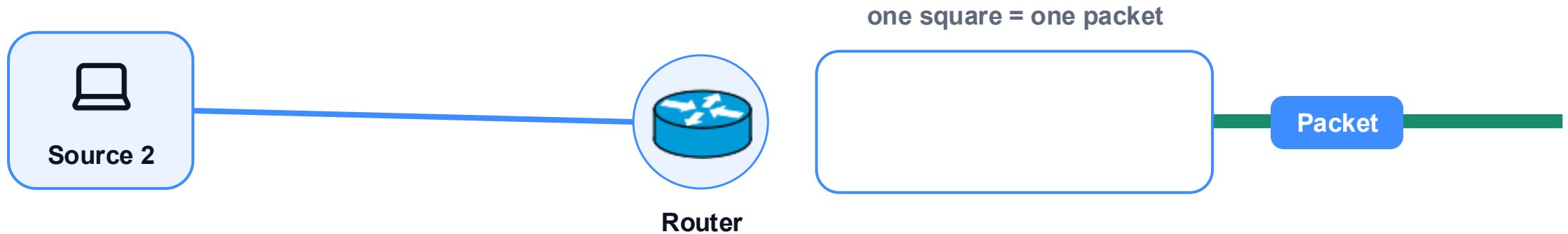
Core structure

Routing + forwarding

Packet switching

Performance

Below capacity, the output link keeps up



arrival rate $\leq$ output rate

Packets leave as fast as they arrive, so the queue stays short.

Queueing is small when service keeps up with arrivals.

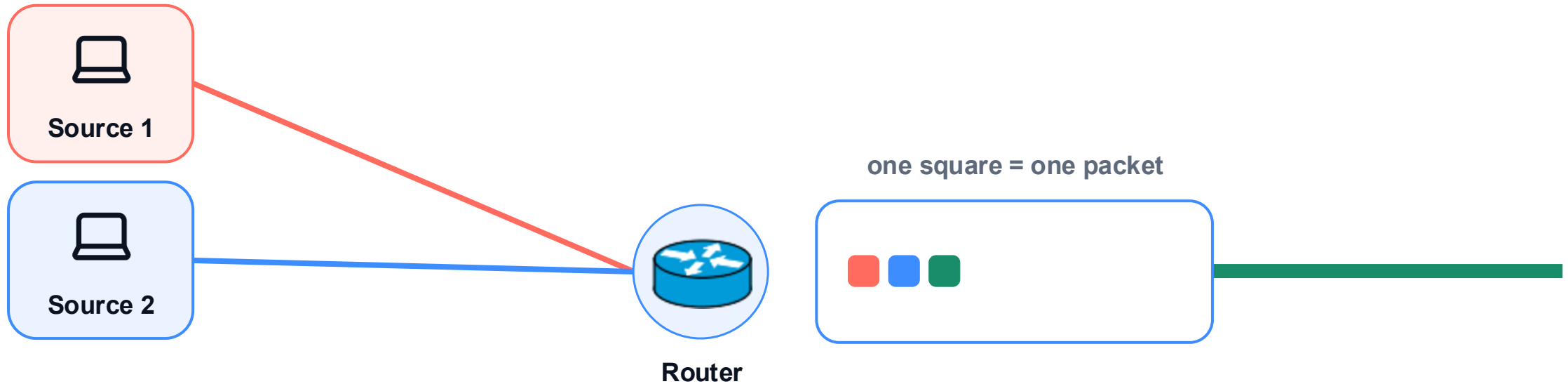
Core structure

Routing + forwarding

Packet switching

Performance

A burst makes packets wait in the buffer



arrival rate > output rate for a short burst

Later packets wait behind packets that arrived first.

Queueing delay appears before packet loss.

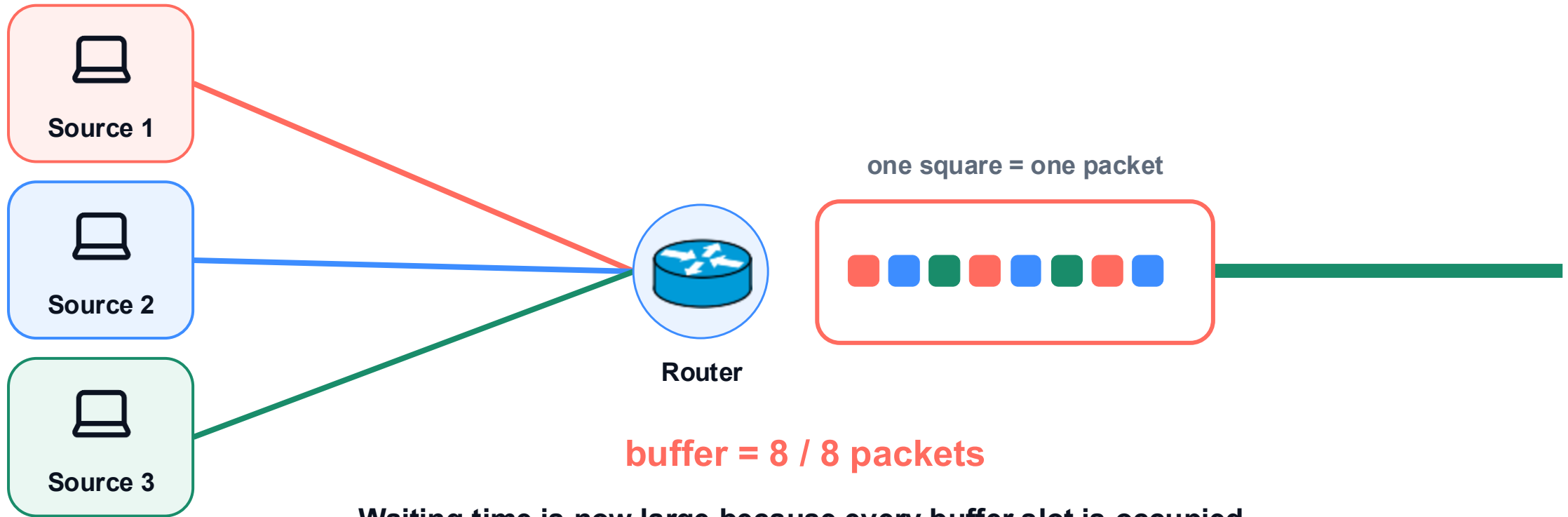
Core structure

Routing + forwarding

Packet switching

Performance

A full buffer cannot accept another packet



Waiting time is now large because every buffer slot is occupied.

A finite buffer can absorb a burst, but only temporarily.

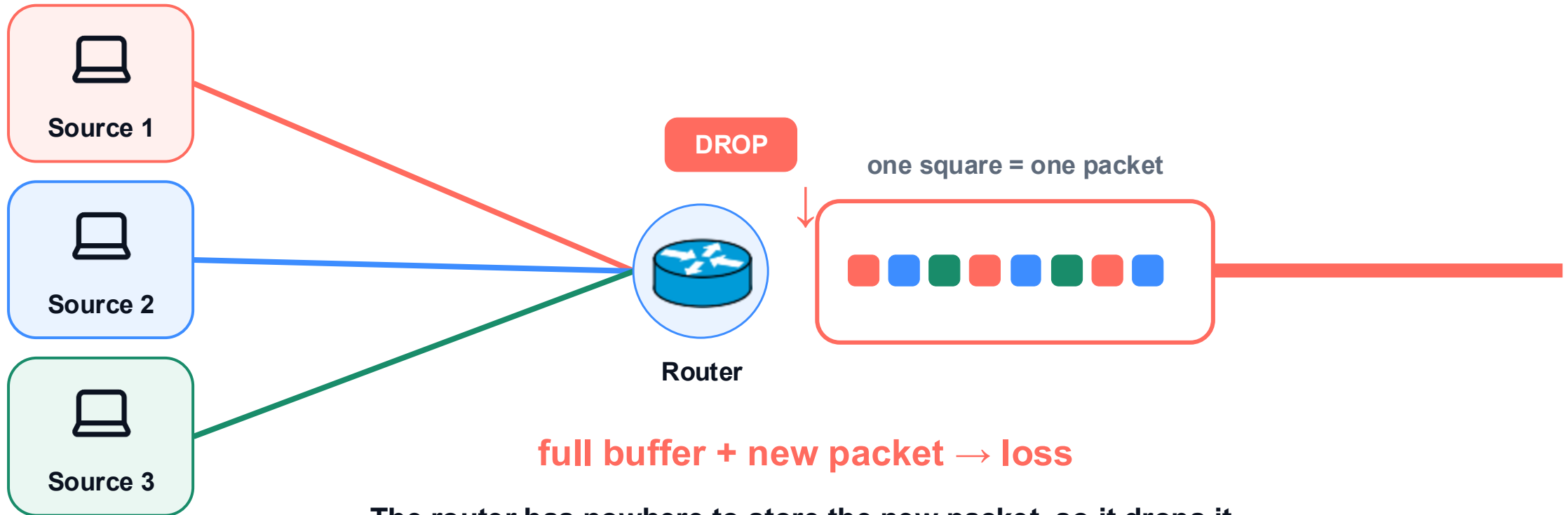
Core structure

Routing + forwarding

Packet switching

Performance

The next arriving packet is dropped



The router has nowhere to store the new packet, so it drops it.

Loss is the next step after a finite buffer becomes full.

Core structure

Routing + forwarding

Packet switching

Performance

Throughput measures useful delivery



$$T = \frac{N_{\text{delivered}}}{t_{\text{elapsed}}}$$

Throughput counts useful bits that actually reach the receiver.

Core structure

Routing + forwarding

Packet switching

Performance

Throughput measures useful delivery



$$T = \frac{N_{\text{delivered}}}{t_{\text{elapsed}}}$$

$$\frac{10 \text{ bits}}{10 \text{ seconds}} = 10 \text{ bits/second}$$

Throughput counts useful bits that actually reach the receiver.

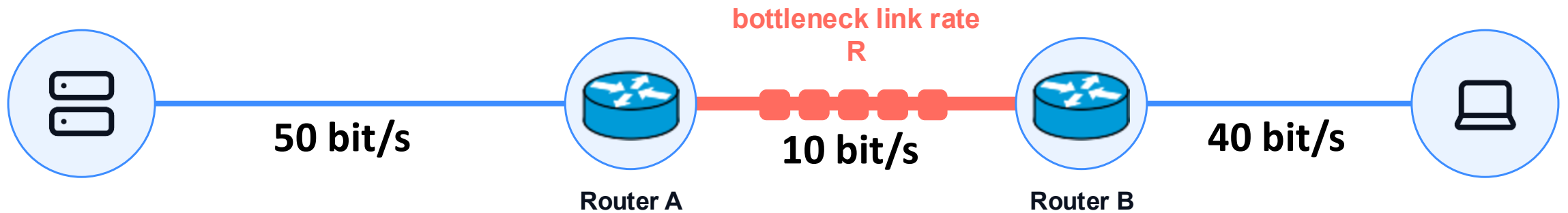
Core structure

Routing + forwarding

Packet switching

Performance

The slowest link limits one connection



$$T = \min(R_{\text{sender}}, R_{\text{bottleneck}}, R_{\text{receiver}})$$

One slow link can limit the entire end-to-end connection.

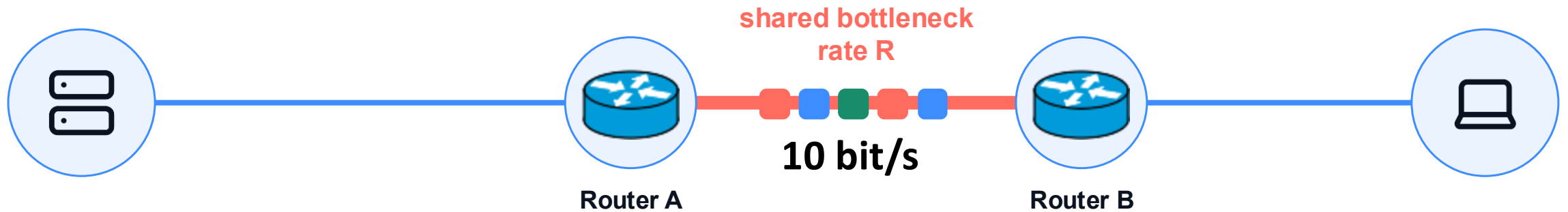
Core structure

Routing + forwarding

Packet switching

Performance

Sharing one bottleneck divides capacity



$$T_{\text{flow}} \approx \min(R_{\text{sender}}, R_{\text{receiver}}, \frac{R}{3})$$

When three flows share one bottleneck, each receives only part of its capacity.

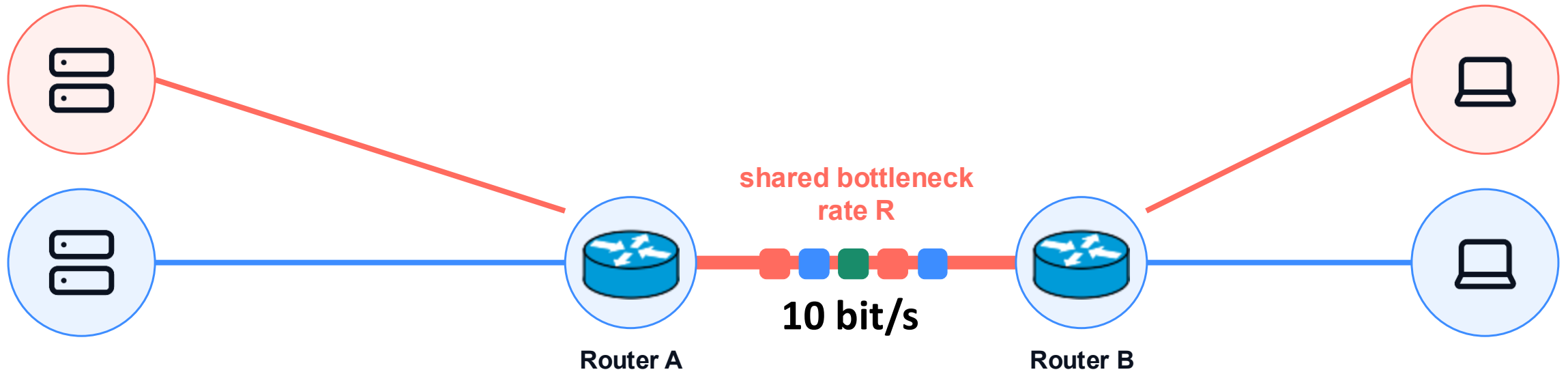
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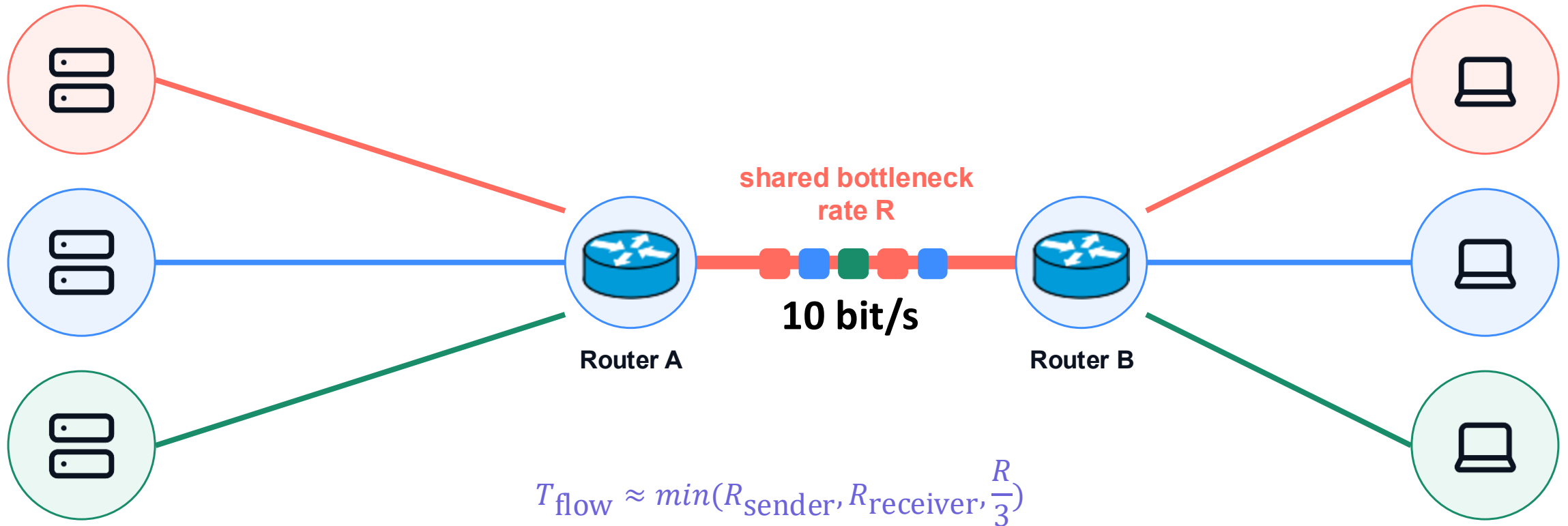
Core structure

Routing + forwarding

Packet switching

Performance

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When three flows share one bottleneck, each receives only part of its capacity.

Core structure

Routing + forwarding

Packet switching

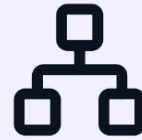
Performance

The Internet core is shared, distributed, and constrained



Shared

Packets from many users reuse links and routers.



Distributed

Independent networks coordinate without one owner.



Constrained

Rates, distance, buffers, and contention shape performance.

The Internet core works because local actions compose into global communication.

Core structure

Routing + forwarding

Packet switching

Performance